

# Peru — Ley 29733 de Protección de Datos Personales

Discovery and masking of Peruvian personal data: DNI and RUC with a valid check digit, quasi-identifiers, the sensitive data of article 2, 5 — including economic income —, health data, financial data and criminal records.

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85

DOMAINS

144

CLASSIFIERS

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ALGORITHMS

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The first sections explain what the set is for and how it works. The domain chapters then describe every domain: what it is, how its data is found and how it is masked, with examples. The appendices list every algorithm, file and classifier the set contains.

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|       |                    |                          |
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# About this profile set

This profile set finds the personal data in a Peruvian database and masks it, so that a non-production copy — for testing, development, training or analysis — no longer identifies anyone.

Masking names and DNIs alone is not enough: a birth date, a sex and a small distrito already point at one person. The set works in the layers where the risk is:

- **Direct identifiers** — DNI and RUC, foreigner card, passport, civil registry records, EsSalud and pension numbers, student code, professional registration, names, contact, address, accounts, cards, wallets, devices, cookies, plates, property, free text.
- **Quasi-identifiers** — birth date, age, sex, distrito and provincia with their UBIGEO, postal code, urbanización, centro poblado, marital status, occupation, employer, education.
- **Sensitive data** — those of article 2, numeral 5: biometric data that identifies on its own, racial and ethnic origin, **economic income**, political, religious, philosophical and moral opinions or convictions, trade union membership, and information on health or sexual life; plus genetic data, gender identity, indigenous people and language, nationality and victim status.
- **Health data** — sensitive without qualification, in a group of its own: clinical record, insurance, ICD-10, procedures, medicines, HIV, blood group, disability, occupational, reproductive and mental health.
- **Financial data** — the income the law itself calls sensitive, plus credit history, debt, assets, pensions, social programmes, the SISFOH classification and housing tenure.
- **Criminal records** — record certificates, procedural situation and court or prosecution file numbers.

In a Masking Engine, the set is a profile set with its classifiers, the domains they vote for, the algorithms those domains name — and the algorithms those rely on — and the lookup files they read. The appendices list each of them with its full configuration.

The assignment threshold is 60%. An unambiguous column name reaches it on its own; an ambiguous one — NOMBRE , DOCUMENTO , EXPEDIENTE , CARGO — needs the values to agree. A column with a meaningless name ( CAMPO1 , OBS ) is still found when its values have a recognizable shape or vocabulary: a DNI, an eleven-digit RUC, a nine-digit mobile, an address with urbanización, manzana and lote, a plate, a distrito.

Column names are recognized in Spanish, in the abbreviations of Peruvian systems ( NUM\_DOC , FEC\_NAC , UBIGEO , RUC , ESSALUD , CODIGO\_MODULAR ) and in English.

Before masking, inventory the binary columns (BLOB, images, PDF). Onboarding processes often keep a photo of the DNI — with the fingerprint and the signature — and a validation selfie. Report those columns as residual risk.

# Ley 29733 in brief

The **Ley 29733 de Protección de Datos Personales** was published on 3 July 2011 and developed by a regulation of 2013. On **30 November 2024** a new regulation replaced it, **Decreto Supremo 016-2024-JUS**, in force since **30 March 2025**. Applying and supervising both falls to the **Autoridad Nacional de Protección de Datos Personales (ANPD)**, within the Ministry of Justice and Human Rights.

What matters for a copy of production data:

- **Personal data** is any information about a natural person that identifies them or makes them identifiable through means that can reasonably be used (art. 2, 4). The data subject is always a natural person (art. 2,

12).

- **Sensitive data** is a short and blunt list (art. 2, 5): biometric data that by itself can identify the subject; data on racial and ethnic origin; **economic income**; political, religious, philosophical or moral opinions or convictions; trade union membership; and information on health or sexual life. Income appears in no neighbouring law: in Peru a salary column carries the weight of a diagnosis.
- **Anonymization** makes the subject unidentifiable in a way that **cannot be reversed** (art. 2, 14). **Dissociation** is the procedure that prevents identification, and article 41.1 of the 2024 regulation states outright that **this procedure is reversible**. Article 41.2 asks the controller to choose the technique that fits the data, the number of subjects and the risk.
- **Where they appear**. Using data for a purpose other than the one collected is a serious infringement — *unless an anonymization or dissociation procedure has taken place* (art. 133.7 of the regulation). That sentence is the legal basis for a masked non-production copy.
- **Security**. The controller keeps security measures in place (arts. 39 and 40 of the regulation ask for an impact assessment where the risk is high), appoints a **Personal Data Protection Officer** where articles 37 to 39 require one, and notifies a breach to the ANPD within **48 hours** of becoming aware of it (art. 34).
- **Minors**. Data of a child under fourteen needs the consent of the parents; between fourteen and eighteen the adolescent consents themselves (art. 25 of the regulation). A copy that keeps exact birth dates keeps that distinction visible.
- **Sanctions**. Fines are measured in UIT and reach the tens of unidades impositivas tributarias for very serious infringements, with the ANPD able to order the processing to stop.

**What the law leaves out.** The law does not apply to data held by natural persons for domestic purposes, nor to data processed for national defence, public security or the investigation of crimes, nor — as the regulation makes explicit — to anonymized data.

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## Dissociation, anonymization and non-production copies

Peru is one of the few countries that writes the difference down in the law itself. **Anonymization** is irreversible (art. 2, 14). **Dissociation** is the procedure that prevents identification, and article 41.1 of the 2024 regulation says in so many words that it **is reversible**.

The algorithms of this set are deterministic for the Masking Engine's key: the same DNI masks the same way in every table, which keeps joins working. That key is what makes the procedure reversible. The masked copy is **dissociated** in the sense of article 41, not **anonymized**, and remains personal data for whoever holds the key and production.

- Keep the engine key apart from the copy, and treat whoever holds both as processing personal data.
- Article 133.7 of the regulation excuses the change of purpose when a dissociation procedure has taken place; article 41.2 asks for the technique to fit the data, the number of subjects and the risk. Record which technique this set applies to each column — the appendices list them one by one.
- If what you need is anonymized data — irreversible, out of the law altogether —, measure the re-identification risk of the quasi-identifiers that remain; article 2, 14 is a higher standard than this set meets.

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# How sensitive data is discovered

A profiling job looks at every column of its rule set and asks each classifier of the profile set how confident it is that the column belongs to its domain. The set uses the four classifier frameworks of the Masking Engine:

- **Column name** (Path) — regular expressions on the column name. Each rule has a match strength: the confidence it gives when it matches.
- **Data type** (Type) — accepts the column type and length, or rules the domain out. It never assigns a domain on its own.
- **Value format** (Regex) — tests a sample of the values against patterns, some with a check digit. The confidence follows the share of values that match.
- **Known values** (List) — looks the sampled values up in lists; for names, word by word.

Evidence from the name and from the values is combined per domain, and the column is assigned to the domain with the highest confidence when it reaches the assignment threshold of the profile set — **60%** here. Values that do not match a format or a list count against their domain by the classifier's reject strength.

Column-name rules are shown as their alternatives. They ignore case and must not be glued to other letters:

`rut` finds `RUT_CLIENTE` and `CLIENTE_RUT` but not `RUTA`. In them, `_?` is an optional underscore, `(a|b)` either form, `s?` an optional letter and `[a-z_]*` any continuation.

Profiling proposes; it does not decide. Review the domains assigned in the inventory before running a masking job, and look for columns the set left unassigned.

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## How data is masked

Each domain names one algorithm, which a masking job applies to every column the inventory assigns to the domain. Algorithms can be composed: a **Regex Decompose** splits a value into parts and gives each part its own action; a **String Algorithm Chain** applies algorithms one after another. The tree under each domain shows everything its algorithm uses; Appendix A has every configuration in full.

The algorithms are deterministic for their key: the same value masks the same way in every table, database and job, so joins between masked systems keep working — as long as every engine uses the same keys.

The masked examples in this document were produced with one particular key. On an engine with other keys the masked values differ; their shape does not.

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## Essential and extended packs

The set can be adopted whole or in part. The **extended pack** is everything this document describes. The **essential pack** is the minimum for the law — identity documents, names, contact details, address and birth date. To adopt only the essential pack, create on the engine just its domains, the classifiers that vote for them and the algorithms and files those use, and put only those classifiers in the profile set.

| PACK      | DOMAINS | CLASSIFIERS | ALGORITHMS | FILES |
|-----------|---------|-------------|------------|-------|
| Essential | 12      | 25          | 28         | 5     |

| PACK     | DOMAINS | CLASSIFIERS | ALGORITHMS | FILES |
|----------|---------|-------------|------------|-------|
| Extended | 85      | 144         | 122        | 79    |

The essential pack holds these domains, marked **Essential** in the tables and chapters:

- DNI — PE\_L1\_DNI
- RUC — PE\_L1\_RUC
- Foreigner documents — PE\_L1\_CARNE\_EXTRANJERIA
- Passport — PE\_L1\_PASAPORTE
- Given name — PE\_L1\_NOMBRE
- Surname — PE\_L1\_APELLIDO
- Full name — PE\_L1\_NOMBRE\_COMPLETO
- E-mail — PE\_L1\_EMAIL
- Phone — PE\_L1\_TELEFONO
- Address — PE\_L1\_DIRECCION
- Address complement — PE\_L1\_DIRECCION\_COMPLEMENTO
- Birth date — PE\_L2\_FECHA\_NACIMIENTO

With fewer domains competing, a column that a domain of the extended pack would claim by its name can be proposed for an essential domain instead — a numeric code, for instance, can look like a telephone number. Review the profiling result.

# Domains at a glance

Every domain of the set, how it is found and the algorithm that masks it.

| DOMAIN                                                                         | FOUND BY    |              |              | ALGORITHM            |
|--------------------------------------------------------------------------------|-------------|--------------|--------------|----------------------|
| DIRECT IDENTIFIERS                                                             |             |              |              |                      |
| DNI <div>ESSENTIAL</div> <div>PE_L1_DNI</div>                                  | COLUMN NAME | DATA TYPE    | VALUE FORMAT | PE_DOCUMENTO         |
| RUC <div>ESSENTIAL</div> <div>PE_L1_RUC</div>                                  | COLUMN NAME | VALUE FORMAT |              | PE_RUC               |
| Verification character <div>PE_L1_DIGITO_VERIFICACION</div>                    | COLUMN NAME | DATA TYPE    |              | PE_CM_DIGITOS_LETRAS |
| Foreigner documents <div>ESSENTIAL</div> <div>PE_L1_CARNE_EXTRANJERIA</div>    | COLUMN NAME |              |              | PE_CM_ALFANUM        |
| Passport <div>ESSENTIAL</div> <div>PE_L1_PASAPORTE</div>                       | COLUMN NAME | VALUE FORMAT |              | PE_CM_ALFANUM        |
| Civil registry records <div>PE_L1_ACTA</div>                                   | COLUMN NAME |              |              | PE_CM_ALFANUM        |
| EsSalud, AFP and ONP <div>PE_L1_SEGURO_SOCIAL</div>                            | COLUMN NAME |              |              | PE_CM_ALFANUM        |
| Student code <div>PE_L1_CODIGO_ESTUDIANTE</div>                                | COLUMN NAME |              |              | PE_CM_ALFANUM        |
| Professional registration <div>PE_L1_COLEGIATURA</div>                         | COLUMN NAME |              |              | PE_CM_ALFANUM        |
| Given name <div>ESSENTIAL</div> <div>PE_L1_NOMBRE</div>                        | COLUMN NAME | KNOWN VALUES |              | PE_NOMBRES           |
| Surname <div>ESSENTIAL</div> <div>PE_L1_APELLIDO</div>                         | COLUMN NAME | KNOWN VALUES |              | PE_APELLIDOS         |
| Full name <div>ESSENTIAL</div> <div>PE_L1_NOMBRE_COMPLETO</div>                | COLUMN NAME | KNOWN VALUES |              | PE_NOMBRE_COMPLETO   |
| Business name <div>PE_L1_RAZON_SOCIAL</div>                                    | COLUMN NAME |              |              | PE_RAZON_SOCIAL      |
| E-mail <div>ESSENTIAL</div> <div>PE_L1_EMAIL</div>                             | COLUMN NAME | DATA TYPE    | VALUE FORMAT | PE_EMAIL             |
| Phone <div>ESSENTIAL</div> <div>PE_L1_TELEFONO</div>                           | COLUMN NAME | VALUE FORMAT |              | PE_TELEFONO          |
| Address <div>ESSENTIAL</div> <div>PE_L1_DIRECCION</div>                        | COLUMN NAME | DATA TYPE    | VALUE FORMAT | PE_DIRECCION         |
| Address complement <div>ESSENTIAL</div> <div>PE_L1_DIRECCION_COMPLEMENTO</div> | COLUMN NAME |              |              | PE_CM_ALFANUM        |
| Bank account <div>PE_L1_CUENTA_BANCARIA</div>                                  | COLUMN NAME |              |              | PE_CM_DIGITOS        |
| Payment card <div>PE_L1_TARJETA</div>                                          | COLUMN NAME | VALUE FORMAT |              | PE_TARJETA           |

| DOMAIN                                                            | FOUND BY    |              |              | ALGORITHM            |
|-------------------------------------------------------------------|-------------|--------------|--------------|----------------------|
| <b>Digital wallet</b><br>PE_L1_BILLETERA                          | COLUMN NAME | VALUE FORMAT |              | PE_CM_ALFANUM        |
| <b>IP address</b><br>PE_L1_IP                                     | COLUMN NAME | VALUE FORMAT |              | PE_IP                |
| <b>Device identifier</b><br>PE_L1_DISPOSITIVO                     | COLUMN NAME | VALUE FORMAT |              | PE_CM_HEX            |
| <b>Cookies and tracking identifiers</b><br>PE_L1_COOKIE           | COLUMN NAME |              |              | PE_CM_ALFANUM        |
| <b>Vehicle plate</b><br>PE_L1_PLACA                               | COLUMN NAME | VALUE FORMAT |              | PE_CM_DIGITOS_LETRAS |
| <b>Vehicle and licence</b><br>PE_L1_VEHICULO                      | COLUMN NAME | VALUE FORMAT |              | PE_CM_ALFANUM        |
| <b>Property</b><br>PE_L1_INMUEBLE                                 | COLUMN NAME |              |              | PE_CM_ALFANUM        |
| <b>Customer and contract numbers</b><br>PE_L1_CONTRATO            | COLUMN NAME |              |              | PE_CM_ALFANUM        |
| <b>User and social media</b><br>PE_L1_USUARIO                     | COLUMN NAME |              |              | PE_CM_ALFANUM        |
| <b>Credentials</b><br>PE_L1_CREDENCIAL                            | COLUMN NAME |              |              | PE_REDACTAR          |
| <b>Geolocation</b><br>PE_L1_GEOLOCALIZACION                       | COLUMN NAME | VALUE FORMAT |              | PE_COORDENADA        |
| <b>Free text</b><br>PE_L1_TEXTO_LIBRE                             | COLUMN NAME | DATA TYPE    | VALUE FORMAT | PE_TEXTO_LIBRE       |
| QUASI-IDENTIFIERS                                                 |             |              |              |                      |
| <b>Birth date</b> ESSENTIAL<br>PE_L2_FECHA_NACIMIENTO             | COLUMN NAME | DATA TYPE    |              | PE_FECHA_NACIMIENTO  |
| <b>Birth year</b><br>PE_L2_ANIO_NACIMIENTO                        | COLUMN NAME | DATA TYPE    |              | PE_ANIO              |
| <b>Age</b><br>PE_L2_EDAD                                          | COLUMN NAME | DATA TYPE    |              | PE_EDAD              |
| <b>Personal event dates</b><br>PE_L2_FECHA_EVENTO                 | COLUMN NAME | DATA TYPE    |              | PE_FECHA_EVENTO      |
| <b>Sex</b><br>PE_L2_SEXO                                          | COLUMN NAME | KNOWN VALUES |              | PE_SEXO              |
| <b>Distrito and provincia</b><br>PE_L2_DISTRITO                   | COLUMN NAME | KNOWN VALUES |              | PE_DISTRITO          |
| <b>UBIGEO</b><br>PE_L2_UBIGEO                                     | COLUMN NAME | VALUE FORMAT |              | PE_UBIGEO            |
| <b>Centro poblado and caserío</b><br>PE_L2_CENTRO_POBLADO         | COLUMN NAME |              |              | PE_ZONA_RURAL        |
| <b>Urbanización and asentamiento humano</b><br>PE_L2_URBANIZACION | COLUMN NAME |              |              | PE_URBANIZACION      |
| <b>Postal code</b><br>PE_L2_CODIGO_POSTAL                         | COLUMN NAME | DATA TYPE    |              | PE_CODIGO_POSTAL     |

| DOMAIN                                                                    | FOUND BY    |              | ALGORITHM                |
|---------------------------------------------------------------------------|-------------|--------------|--------------------------|
| <b>Marital status</b><br>PE_L2_ESTADO_CIVIL                               | COLUMN NAME | KNOWN VALUES | PE_ESTADO_CIVIL          |
| <b>Occupation</b><br>PE_L2_OCUPACION                                      | COLUMN NAME |              | PE_OCUPACION_O_CODIGO    |
| <b>Employer</b><br>PE_L2_EMPLEADOR                                        | COLUMN NAME |              | PE_EMPLEADOR             |
| <b>Education</b><br>PE_L2_NIVEL_EDUCATIVO                                 | COLUMN NAME | KNOWN VALUES | PE_NIVEL_EDUCATIVO       |
| <b>School</b><br>PE_L2_INSTITUCION_EDUCATIVA                              | COLUMN NAME |              | PE_INSTITUCION_EDUCATIVA |
| <b>Dependants and children</b><br>PE_L2_PERSONAS_A_CARGO                  | COLUMN NAME | DATA TYPE    | PE_CONTEO_LIMITADO       |
| SENSITIVE DATA (ART. 2, 5)                                                |             |              |                          |
| <b>Ethnic origin</b><br>PE_L3_ORIGEN_ETNICO                               | COLUMN NAME | KNOWN VALUES | PE_ORIGEN_ETNICO         |
| <b>Indigenous people</b><br>PE_L3_PUEBLO_INDIGENA                         | COLUMN NAME | KNOWN VALUES | PE_PUEBLO_INDIGENA       |
| <b>Language</b><br>PE_L3_LENGUA                                           | COLUMN NAME | KNOWN VALUES | PE_LENGUA                |
| <b>Nationality</b><br>PE_L3_NACIONALIDAD                                  | COLUMN NAME | KNOWN VALUES | PE_NACIONALIDAD          |
| <b>Religion</b><br>PE_L3_RELIGION                                         | COLUMN NAME | KNOWN VALUES | PE_RELIGION              |
| <b>Philosophical and moral convictions</b><br>PE_L3_CONVICCION_FILOSOFICA | COLUMN NAME |              | PE_CATEGORIA_SUPRIMIDA   |
| <b>Political opinion</b><br>PE_L3_OPINION_POLITICA                        | COLUMN NAME | KNOWN VALUES | PE_IDEOLOGIA_POLITICA    |
| <b>Party membership</b><br>PE_L3_AFILIACION_PARTIDARIA                    | COLUMN NAME | KNOWN VALUES | PE_PARTIDO               |
| <b>Trade union membership</b><br>PE_L3_AFILIACION_SINDICAL                | COLUMN NAME | KNOWN VALUES | PE_CATEGORIA_SUPRIMIDA   |
| <b>Biometric data</b><br>PE_L3_BIOMETRICO                                 | COLUMN NAME | DATA TYPE    | PE_REDACTAR              |
| <b>Genetic data</b><br>PE_L3_GENETICO                                     | COLUMN NAME |              | PE_REDACTAR              |
| <b>Sexual orientation</b><br>PE_L3_ORIENTACION_SEXUAL                     | COLUMN NAME | KNOWN VALUES | PE_ORIENTACION_SEXUAL    |
| <b>Sexual life</b><br>PE_L3_VIDA_SEXUAL                                   | COLUMN NAME |              | PE_CATEGORIA_SUPRIMIDA   |
| <b>Gender identity</b><br>PE_L3_IDENTIDAD_GENERO                          | COLUMN NAME | KNOWN VALUES | PE_IDENTIDAD_GENERO      |
| <b>Victims</b><br>PE_L3_VICTIMA                                           | COLUMN NAME |              | PE_CATEGORIA_SUPRIMIDA   |
| HEALTH DATA                                                               |             |              |                          |

| DOMAIN                                                                  | FOUND BY    |              | ALGORITHM                           |
|-------------------------------------------------------------------------|-------------|--------------|-------------------------------------|
| <b>Health identifiers</b><br>PE_SALUD_IDENTIFICADOR                     | COLUMN NAME |              | PE_CM_ALFANUM                       |
| <b>Health insurance</b><br>PE_SALUD_COBERTURA                           | COLUMN NAME | KNOWN VALUES | PE_COBERTURA_SALUD                  |
| <b>Diagnosis and ICD-10</b><br>PE_SALUD_DIAGNOSTICO                     | COLUMN NAME | VALUE FORMAT | PE_DIAGNOSTICO                      |
| <b>Procedures</b><br>PE_SALUD_PROCEDIMIENTO                             | COLUMN NAME |              | PE_PROCEDIMIENTO                    |
| <b>Medicines</b><br>PE_SALUD_MEDICAMENTO                                | COLUMN NAME | KNOWN VALUES | PE_MEDICAMENTO                      |
| <b>Clinical text</b><br>PE_SALUD_TEXTO_CLINICO                          | COLUMN NAME | DATA TYPE    | PE_TEXTO_LIBRE                      |
| <b>Test results</b><br>PE_SALUD_RESULTADO_EXAMEN                        | COLUMN NAME |              | PE_CM_ALFANUM                       |
| <b>HIV and sexually transmitted infections</b><br>PE_SALUD_VIH          | COLUMN NAME |              | PE_CATEGORIA_SUPRIMIDA              |
| <b>Blood group and Rh</b><br>PE_SALUD_GRUPO_SANGUINEO                   | COLUMN NAME | KNOWN VALUES | PE_GRUPO_SANGUINEO                  |
| <b>Disability</b><br>PE_SALUD_DISCAPACIDAD                              | COLUMN NAME | KNOWN VALUES | PE_DISCAPACIDAD                     |
| <b>Occupational health</b><br>PE_SALUD_OCUPACIONAL                      | COLUMN NAME | KNOWN VALUES | PE_APTITUD_LABORAL                  |
| <b>Sexual and reproductive health</b><br>PE_SALUD_REPRODUCTIVA          | COLUMN NAME |              | PE_CATEGORIA_SUPRIMIDA              |
| <b>Mental health</b><br>PE_SALUD_MENTAL                                 | COLUMN NAME |              | PE_CATEGORIA_SUPRIMIDA              |
| FINANCIAL AND SOCIOECONOMIC DATA                                        |             |              |                                     |
| <b>Income and pay</b><br>PE_FIN_INGRESOS                                | COLUMN NAME | DATA TYPE    | PE_INGRESOS                         |
| <b>Credit history</b><br>PE_FIN_HISTORIAL_CREDITICIO                    | COLUMN NAME |              | PE_HISTORIAL_CREDITICIO             |
| <b>Debt</b><br>PE_FIN_DEUDA                                             | COLUMN NAME | DATA TYPE    | PE_VALOR_DEUDA                      |
| <b>Assets</b><br>PE_FIN_PATRIMONIO                                      | COLUMN NAME | DATA TYPE    | PE_PATRIMONIO                       |
| <b>Pension and benefits</b><br>PE_FIN_PENSION                           | COLUMN NAME | DATA TYPE    | PE_VALOR_PENSION                    |
| <b>Social programmes</b><br>PE_FIN_PROGRAMA_SOCIAL                      | COLUMN NAME | KNOWN VALUES | PE_PROGRAMA_SOCIAL                  |
| <b>SISFOH classification</b><br>PE_FIN_CLASIFICACION_SOCIOECONOMIC<br>A | COLUMN NAME | KNOWN VALUES | PE_CLASIFICACION_SOCIOECONOMIC<br>A |
| <b>Housing tenure</b><br>PE_FIN_VIVIENDA                                | COLUMN NAME | KNOWN VALUES | PE_TENENCIA_VIVIENDA                |
| CRIMINAL RECORDS                                                        |             |              |                                     |

| DOMAIN                           | FOUND BY    |              | ALGORITHM              |
|----------------------------------|-------------|--------------|------------------------|
| <b>Criminal records</b>          | COLUMN NAME |              | PE_ANTECEDENTES        |
| PE_PENAL_ANTECEDENTES            |             |              |                        |
| <b>Court or prosecution file</b> | COLUMN NAME | VALUE FORMAT | PE_EXPEDIENTE_JUDICIAL |
| PE_PENAL_EXPEDIENTE              |             |              |                        |

# Direct identifiers

Data that identifies a person on its own. The DNI is the key of almost every Peruvian record, and the RUC of a natural person is the number 10 followed by that same DNI. Both come out with a valid check digit, because applications validate it, and both mask the same way everywhere.

## 8.1 DNI

PE\_L1\_DNI

ESSENTIAL COLUMN NAME DATA TYPE VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The number of the Documento Nacional de Identidad issued by RENIEC: eight digits, 45678901 , the key of almost every Peruvian record. The card prints a **verification character** beside it, and systems store it glued to the number ( 45678901-7 ) or in a column of its own.

The same domain takes RUCs: a document column often holds both.

HOW IT IS FOUND

COLUMN NAME PE\_L1\_DNI - Path

90% dni d\_n\_i documento\_?nacional(\_?de?identidad)? n(o|ro|um|umero)?  
(dni|documento|doc|identificaci[oó]n|id(ent)?) doc\_?identidad nro\_?doc num\_?doc cui  
c[oó]digo\_?[uú]nico\_?de?identificaci[oó]n  
(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|  
trabajador|colaborador|alumno|estudiante|deudor|codeudor|fiador|avalista|solicitante|propietar  
io|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elec  
tor|votante|contribuyente|proveedor|socio|conyuge|padre|madre|hijo|familiar|jubilado|pensionis  
ta|aportante|derechohabiente)\_?(dni|doc|documento) dni\_  
(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|t  
rabajador|colaborador|alumno|estudiante|deudor|codeudor|fiador|avalista|solicitante|propietari  
o|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elect  
or|votante|contribuyente|proveedor|socio|conyuge|padre|madre|hijo|familiar|jubilado|pensionist  
a|aportante|derechohabiente)

60% documento doc identificaci[oó]n

DATA TYPE PE\_L1\_DNI - Type

Accepts text (length 8 or more) or number (length 8 or more); any other type rules the domain out.

VALUE FORMAT PE\_L1\_DNI - Regex

| PATTERN               | STRENGTH | NOTES |
|-----------------------|----------|-------|
| \d{8}[\s\~] [0-9A-JK] |          | 90%   |
| \d{8}                 |          | 50%   |

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Digit by digit, with the verification character recomputed — the digit series or the letter series, whichever the value carries. A value shaped like a RUC is masked as a RUC.

PE\_DOCUMENTO REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PE\_CM\_ALFANUM .

PE\_RUC STRING ALGORITHM CHAIN

Applies PE\_RUC\_CD → PE\_RUC\_ESPACIO → PE\_RUC\_AUX → PE\_RUC\_VERIFICADOR , in this order.

PE\_RUC\_CD CHECK DIGIT

Masks the digits with PE\_RUC\_CUERPO and recomputes a modulus 11 check digit with weights 2 3 4 5 6 7 2 3 4 5 from the right.

PE\_RUC\_CUERPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches:

PE\_CM\_DIGITOS .

PE\_CM\_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

PE\_RUC\_ESPACIO REGEX DECOMPOSE

Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_RUC\_AUX CHECK DIGIT

Masks the digits with PE\_IDENTIDAD and recomputes a modulus 11 check digit with weights 0 4 6 8 10 1 3 4 6 8 10 from the right.

PE\_IDENTIDAD REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

PE\_RUC\_VERIFICADOR REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_DNI\_CON\_DIGITO STRING ALGORITHM CHAIN

Applies PE\_DNI\_CD → PE\_DNI\_ESPACIO → PE\_DNI\_AUX → PE\_DNI\_VERIFICADOR , in this order.

PE\_DNI\_CD CHECK DIGIT

Masks the digits with PE\_CM\_DIGITOS and recomputes a modulus 11 check digit with weights 2 3 4 5 6 7 2 3 from the right.

PE\_DNI\_ESPACIO REGEX DECOMPOSE

Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_DNI\_AUX CHECK DIGIT

Masks the digits with PE\_IDENTIDAD and recomputes a modulus 11 check digit with weights 0 4 6 8 10 1 3 4 6 from the right.

PE\_DNI\_VERIFICADOR REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_DNI\_CON\_LETRA STRING ALGORITHM CHAIN

Applies PE\_DNI\_CD → PE\_DNI\_ESPACIO → PE\_DNI\_AUX → PE\_DNI\_LETRA , in this order.

PE\_DNI\_LETRA REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE       |   | MASKED      |
|-------------|---|-------------|
| 45678901    | → | 32016288    |
| 45678901-7  | → | 32016288-8  |
| 20131378972 | → | 20817541918 |

ESSENTIAL

COLUMN NAME

VALUE FORMAT

STRING ALGORITHM CHAIN

WHAT IT IS

The eleven-digit taxpayer number of SUNAT, 20131378972 : two digits for the kind of taxpayer — **10** and 15 for a natural person, 17 for an estate, 20 for a legal person —, eight for the number and one check digit. For a natural person with a 10, those eight digits **are the DNI**.

HOW IT IS FOUND

COLUMN NAME

PE\_L1\_RUC – Path

90%  
ruc[a-z0-9]\* n(o|ro|um|umero)?ruc registro?[uú]nico?(de\_)?contribuyente  
(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|trabajador|colaborador|alumno|estudiante|deudor|codeudor|fiador|avalista|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elector|votante|contribuyente|proveedor|socio|conyuge|padre|madre|hijo|familiar|jubilado|pensionista|aportante|derechohabiente)?ruc ruc\_  
(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|trabajador|colaborador|alumno|estudiante|deudor|codeudor|fiador|avalista|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elector|votante|contribuyente|proveedor|socio|conyuge|padre|madre|hijo|familiar|jubilado|pensionista|aportante|derechohabiente) id?tributario tax?id

VALUE FORMAT

PE\_L1\_RUC – Regex

| PATTERN            | STRENGTH | NOTES |
|--------------------|----------|-------|
| (10 15 17 20)\d{9} |          | 90%   |
| \d{11}             |          | 50%   |

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The two digits of the kind of taxpayer stay, the eight of the number are mapped and the check digit is recomputed, so DNI and RUC of the same person keep agreeing.

PE\_RUC STRING ALGORITHM CHAIN

Applies PE\_RUC\_CD → PE\_RUC\_ESPACIO → PE\_RUC\_AUX → PE\_RUC\_VERIFICADOR , in this order.

PE\_RUC\_CD CHECK DIGIT

Masks the digits with PE\_RUC\_CUERPO and recomputes a modulus 11 check digit with weights 2 3 4 5 6 7 2 3 4 5 from the right.

PE\_RUC\_CUERPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PE\_CM\_DIGITOS .

PE\_CM\_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

PE\_RUC\_ESPACIO REGEX DECOMPOSE

Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_RUC\_AUX CHECK DIGIT

Masks the digits with PE\_IDENTIDAD and recomputes a modulus 11 check digit with weights 0 4 6 8 10 1 3 4 6 8 10 from the right.

PE\_IDENTIDAD REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

PE\_RUC\_VERIFICADOR REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

#### EXAMPLES

| VALUE       |   | MASKED      |
|-------------|---|-------------|
| 20131378972 | → | 20817541918 |
| 10456789017 | → | 10320162888 |

## 8.3 Verification character

PE\_L1\_DIGITO\_VERIFICACION

COLUMN NAME DATA TYPE CHARACTER MAPPING

#### WHAT IT IS

The DNI's verification digit or letter kept in a column of its own.

#### HOW IT IS FOUND

COLUMN NAME PE\_L1\_DIGITO\_VERIFICACION - Path

80% d[íi]gito?(de\_)?verificaci[oó]n digito?verificador dig?ver dv?dni dv c[oó]digo?  
(de\_)?verificaci[oó]n?(dni|documento) check?digit

DATA TYPE PE\_L1\_DIGITO\_VERIFICACION - Type

Accepts text (length up to 1) or number (length up to 1); any other type rules the domain out.

#### HOW IT IS MASKED

Digits to digits and letters to letters. It cannot be recomputed there: the number it belongs to is not in the value.

PE\_CM\_DIGITOS\_LETRAS CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

#### EXAMPLES

| VALUE |   | MASKED |
|-------|---|--------|
| 7     | → | 5      |
| K     | → | 0      |

## 8.4 Foreigner documents

PE\_L1\_CARNE\_EXTRANJERIA

ESSENTIAL COLUMN NAME CHARACTER MAPPING

#### WHAT IT IS

The carné de extranjería, the PTP and CPP permits of Migraciones, the migratory status and the visa number — the documents of the Venezuelan and other foreign communities.

#### HOW IT IS FOUND

COLUMN NAME PE\_L1\_CARNE\_EXTRANJERIA - Path

85% carn[eé]?(de\_)?extranjer[ií]a carnet?extranjeria c[\.s]?e[\.s]??(n(o|ro|um|umero))?  
n(o|ro|um|umero)?(carn[eé]|extranjer[ií]a) permiso?temporal?(de\_)?permanencia ptp  
carn[eé]?(de\_)?permiso?temporal cpp calidad?migratoria condici[oó]n?migratoria  
n(o|ro|um|umero)?(de\_)?(residencia|migraciones) visa(\_?n(o|ro|um|umero))?

#### HOW IT IS MASKED

Character by character, keeping length and format.

PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE     |   | MASKED    |
|-----------|---|-----------|
| 001234567 | → | 003065281 |

## 8.5 Passport

PE\_L1\_PASAPORTE

ESSENTIAL COLUMN NAME VALUE FORMAT CHARACTER MAPPING

#### WHAT IT IS

Peruvian passport numbers and foreign ones.

#### HOW IT IS FOUND

COLUMN NAME PE\_L1\_PASAPORTE - Path

90% pasaporte[a-z0-9]\* n(o|ro|um|umero)?pasaporte passport[a-z0-9]\*

VALUE FORMAT PE\_L1\_PASAPORTE - Regex

| PATTERN           | STRENGTH | NOTES |
|-------------------|----------|-------|
| [A-Z]{1,2}\d{6,8} |          | 40%   |

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Character by character, keeping length and format.

PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE    |   | MASKED   |
|----------|---|----------|
| A1234567 | → | U2438404 |

## 8.6 Civil registry records

PE\_L1\_ACTA

COLUMN NAME CHARACTER MAPPING

#### WHAT IT IS

The acta, book and folio of the birth, marriage and death records RENIEC and the municipal registry offices keep, and the CUA that numbers them.

#### HOW IT IS FOUND

COLUMN NAME PE\_L1\_ACTA - Path

85% acta?(de\_)?(nacimiento|matrimonio|defunci[oó]n)(\_(n(o|ro|um|umero)|folio|libro))?  
n(o|ro|um|umero)?acta partida?(de\_)?(nacimiento|matrimonio|defunci[oó]n) folio?  
(acta|libro|partida) libro?(acta|registro) asiento?registral?civil c[oó]digo?  
[uú]nico?de?acta cua

#### HOW IT IS MASKED

Character by character, keeping length and format.

#### PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE    |   | MASKED   |
|----------|---|----------|
| 123/1985 | → | 091/7273 |

## 8.7 EsSalud, AFP and ONP

PE\_L1\_SEGURO\_SOCIAL

#### COLUMN NAME CHARACTER MAPPING

#### WHAT IT IS

The EsSalud *autogenerado*, the CUSPP of the private pension system and the ONP and individual account numbers.

#### HOW IT IS FOUND

##### COLUMN NAME PE\_L1\_SEGURO\_SOCIAL - Path

85% autogenerado c[oó]digo\_(de\_)?(asegurado|autogenerado|essalud) n(o|ro|um|umero)\_?  
(essalud|asegurado|afiliado|autogenerado) essalud[a-z0-9]\* cuspp c[oó]digo\_[uú]nico\_  
(de\_)?identificaci[oó]n\_(de\_l\_)?spp n(o|ro|um|umero)\_?(afp|onp|cuenta\_?individual|cic)  
afp[a-z0-9]\* onp[a-z0-9]\* n(o|ro|um|umero)\_?(de\_)?aportante

#### HOW IT IS MASKED

Character by character, keeping length and format.

#### PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE        |   | MASKED       |
|--------------|---|--------------|
| JUQU12345678 | → | SAGI03506074 |

## 8.8 Student code

PE\_L1\_CODIGO\_ESTUDIANTE

#### COLUMN NAME CHARACTER MAPPING

#### WHAT IT IS

The student code, the enrolment number and the university card, and the SIAGIE identifier of school pupils.

#### HOW IT IS FOUND

##### COLUMN NAME PE\_L1\_CODIGO\_ESTUDIANTE - Path

80% c[oó]digo\_(de\_)?(estudiante|alumno|matr[ií]cula|educando) cod\_(alumno|estudiante)  
n(o|ro|um|umero)\_?(de\_)?(matr[ií]cula|carn[eé]\_?universitario) carn[eé]\_(de\_)?  
(estudiante|universitario) id\_(alumno|estudiante) c[oó]digo\_modular\_(de\_l\_)?estudiante  
siagie

#### HOW IT IS MASKED

Character by character, keeping length and format.

#### PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE    |   | MASKED   |
|----------|---|----------|
| 20201234 | → | 10992819 |

## 8.9 Professional registration

PE\_L1\_COLEGIATURA

#### COLUMN NAME CHARACTER MAPPING

#### WHAT IT IS

The number of the professional college — CMP for physicians, CIP for engineers, CAL for lawyers —, published together with the name.

#### HOW IT IS FOUND

##### COLUMN NAME PE\_L1\_COLEGIATURA – Path

85% `colegiatura n(o|ro|um|umero)?(de_)?(colegiatura|colegiado|registro_?profesional) cmp`  
`c[oó]digo?(m[eé]dico|profesional) registro?(m[eé]dico|profesional|cip|cal) cip`  
`matr[ií]cula_?profesional habilitaci[oó]n_?profesional`

#### HOW IT IS MASKED

Character by character, keeping length and format.

#### PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE     |   | MASKED    |
|-----------|---|-----------|
| CMP-12345 | → | DRF-17346 |

## 8.10 Given name

PE\_L1\_NOMBRE

#### ESSENTIAL COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

#### WHAT IT IS

The given name, one or two: Milagros , Juan Carlos , María del Rosario .

#### HOW IT IS FOUND

##### COLUMN NAME PE\_L1\_NOMBRE – Path

85% `primer_?nombre segundo_?nombre nombres?_(de_)?pila pri(mer)?_?nom seg(undo)?_?nom`  
`nombre_?1 nombre_?2 prenombr(es)? first_?names? given_?names? fname middle_?names?`  
`nombres (except when followed by completos? , y_?apellidos? , apellidos? )`

50% `nombre (except when followed by list E1) nom (except when followed by list E2) name (except when followed by list E3)`

**KNOWN VALUES** PE\_L1\_NOMBRE – List

| LIST                    | VALUES | STRENGTH | FIRST VALUES                           |
|-------------------------|--------|----------|----------------------------------------|
| pe-detectar-nombres.txt | 456    | 70%      | José, Jose, María, Maria, Juan, Carlos |

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Word by word, replacing each name with one from the list. `de`, `del` and `la` stay in place. More than four words are masked character by character.

**PE\_NOMBRES** **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: **PE\_CM\_ALFANUM**.

**PE\_PALABRA\_NOMBRE** **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: **PE\_NOMBRE**.

**PE\_NOMBRE** **NAME**

Replaces the name with one of the 395 names in `pe-nombres.txt`.

**PE\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE             |   | MASKED                    |
|-------------------|---|---------------------------|
| María del Rosario | → | <b>Paulino del Javier</b> |
| JUAN CARLOS       | → | <b>FILOMENO BRENDA</b>    |

## 8.11 Surname

PE\_L1\_APELLIDO

**ESSENTIAL** **COLUMN NAME** **KNOWN VALUES** **REGEX DECOMPOSE**

#### WHAT IT IS

The surnames — the father's and the mother's —, in separate columns or together: `Quispe Huamán`. `Quispe` and `Mamani`, the two most common, are Quechua and Aymara.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L1\_APELLIDO – Path

**85%** `apellidos?` `primer_?apellido` `segundo_?apellido` `apellido_?(1|2|paterno|materno|casada)`  
`ape_?(1|2|pat|mat)` `ap_?(paterno|materno)` `last_?names?` `surnames?` `family_?names?` `lname`

**KNOWN VALUES** PE\_L1\_APELLIDO – List

| LIST                      | VALUES | STRENGTH | FIRST VALUES                                    |
|---------------------------|--------|----------|-------------------------------------------------|
| pe-detectar-apellidos.txt | 519    | 70%      | Quispe, Mamani, Flores, Huamán, Huaman, Sánchez |

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Word by word, replacing each surname with one from the list; particles stay in place.

**PE\_APELLIDOS** **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: **PE\_CM\_ALFANUM**.

**PE\_PALABRA\_APELLIDO** **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: **PE\_APELLIDO**.

**PE\_APELLIDO** **NAME**

Replaces the name with one of the 422 names in `pe-apellidos.txt`.

#### PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

##### VALUE

##### MASKED

Quispe Huamán

→

**Ortiz Zambrano**

## 8.12 Full name

PE\_L1\_NOMBRE\_COMPLETO

ESSENTIAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

#### WHAT IT IS

Given names and surnames in one column: Juan Carlos Quispe Huamán . Also those of the mother, father, guardian, spouse, partner, emergency contact and reference.

RENIEC lists write `APELLIDOS, NOMBRES` with a comma; the set reads that form too.

HOW IT IS FOUND

COLUMN NAME PE\_L1\_NOMBRE\_COMPLETO – Path

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 90% | <code>nombres?_?(completos? y_apellidos? apellidos?)</code> <code>apellidos?_?(y_)?nombres?</code> <code>nombre_?(del_ de_la_)?</code><br><code>(cliente cli paciente pac usuario afiliado asegurado beneficiario benef titular empleado trabajador colaborador alumno estudiante deudor codeudor fiador avalista solicitante propietario arrendatario inquilino conductor responsable representante apoderado persona ciudadano contribuyente madre padre conyuge conviviente contacto referencia victima denunciante imputado procesado testigo medico docente socio heredero donante tutor jubilado pensionista derechohabiente customer patient employee member holder mother father spouse contact person student driver owner)</code><br><code>(cliente cli paciente pac usuario afiliado asegurado beneficiario benef titular empleado trabajador colaborador alumno estudiante deudor codeudor fiador avalista solicitante propietario arrendatario inquilino conductor responsable representante apoderado persona ciudadano contribuyente madre padre conyuge conviviente contacto referencia victima denunciante imputado procesado testigo medico docente socio heredero donante tutor jubilado pensionista derechohabiente customer patient employee member holder mother father spouse contact person student driver owner)_?(nombre nom)</code> <code>nom_?</code><br><code>(cliente cli paciente pac usuario afiliado asegurado beneficiario benef titular empleado trabajador colaborador alumno estudiante deudor codeudor fiador avalista solicitante propietario arrendatario inquilino conductor responsable representante apoderado persona ciudadano contribuyente madre padre conyuge conviviente contacto referencia victima denunciante imputado procesado testigo medico docente socio heredero donante tutor jubilado pensionista derechohabiente customer patient employee member holder mother father spouse contact person student driver owner)</code> <code>nombre_?(madre padre tutor conyuge conviviente contacto_emergencia referencia)</code> <code>full_?</code><br><code>name</code> <code>person_name</code> <code>razon_social_persona_natural</code> |
| 60% | <code>madre</code> <code>padre</code> <code>tutor</code> <code>conyuge</code> <code>conviviente</code> <code>representante_legal</code> <code>apoderado</code> <code>fiador</code><br><code>beneficiario</code> <code>heredero</code> <code>titular</code> <code>referencia_(personal familiar)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 50% | <code>nombre</code> (except when followed by <a href="#">list E1</a> ) <code>nom</code> (except when followed by <a href="#">list E2</a> ) <code>name</code> (except when followed by <a href="#">list E3</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

KNOWN VALUES PE\_L1\_NOMBRE\_COMPLETO – List

| LIST                      | VALUES | STRENGTH | FIRST VALUES                                    |
|---------------------------|--------|----------|-------------------------------------------------|
| pe-detectar-nombres.txt   | 456    | 65%      | José, Jose, María, Maria, Juan, Carlos          |
| pe-detectar-apellidos.txt | 519    | 65%      | Quispe, Mamani, Flores, Huamán, Huaman, Sánchez |

Values are split into words, and each word is looked up.  
A value that matches none counts 30% against the domain.

HOW IT IS MASKED

In Peruvian order: two words are a given name and a surname; three, a given name and two surnames; four, two and two. The comma form is masked keeping the comma.

PE\_NOMBRE\_COMPLETO REGEX DECOMPOSE

Tries 7 patterns in order and gives each part of the value its own action; when none matches: PE\_CM\_ALFANUM .

PE\_APELLIDOS REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM` .

`PE_PALABRA_APELLIDO` **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: `PE_APELLIDO` .

`PE_APELLIDO` **NAME**

Replaces the name with one of the 422 names in `pe-apellidos.txt` .

`PE_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

`PE_NOMBRES` **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM` .

`PE_PALABRA_NOMBRE` **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: `PE_NOMBRE` .

`PE_NOMBRE` **NAME**

Replaces the name with one of the 395 names in `pe-nombres.txt` .

#### EXAMPLES

| VALUE                      |   | MASKED                          |
|----------------------------|---|---------------------------------|
| Juan Carlos Quispe Huamán  | → | Filomeno Brenda Ortiz Zambrano  |
| Milagros Quispe            | → | Pablo Ortiz                     |
| QUISPE HUAMÁN, JUAN CARLOS | → | ORTIZ ZAMBRANO, FILOMENO BRENDA |

## 8.13 Business name

PE\_L1\_RAZON\_SOCIAL

COLUMN NAME

SECURE LOOKUP

#### WHAT IT IS

The business name of a sole trader, beside the RUC that names them. The law protects the natural person behind it; a company as such is not a data subject.

#### HOW IT IS FOUND

COLUMN NAME

PE\_L1\_RAZON\_SOCIAL – Path

85%

`raz[oó]n_?social` `razonsocial` `nombre_?(comercial|de_?la_?empresa|empresa)`

`denominaci[oó]n_?social` `nombre_?legal` `business_?name` `company_?name` `legal_?name`

#### HOW IT IS MASKED

Replaced by a fictitious business name.

`PE_RAZON_SOCIAL` **SECURE LOOKUP**

Replaces the value with one of the 20 values in `pe-razones-sociales.txt` , chosen by a keyed hash.

#### EXAMPLES

| VALUE                                |   | MASKED                            |
|--------------------------------------|---|-----------------------------------|
| Corporación Comercial Peruana S.A.C. | → | Café de Altura Chanchamayo S.A.C. |

## 8.14 E-mail

PE\_L1\_EMAIL

ESSENTIAL

COLUMN NAME

DATA TYPE

VALUE FORMAT

REGEX DECOMPOSE

#### WHAT IT IS

Personal and corporate e-mail addresses.

HOW IT IS FOUND

**COLUMN NAME** PE\_L1\_EMAIL - Path

**85%** e\_?mail[a-z0-9\_]\* mail[a-z0-9\_]\* correo(\_?electr[oó]nico)?[a-z0-9\_]\* dir\_?correo  
direcci[oó]n\_?(de\_)?correo direcci[oó]n\_?electr[oó]nica

**DATA TYPE** PE\_L1\_EMAIL - Type

Accepts text (length 5 or more); any other type rules the domain out.

**VALUE FORMAT** PE\_L1\_EMAIL - Regex

| PATTERN                                                 | STRENGTH | NOTES |
|---------------------------------------------------------|----------|-------|
| [A-Za-z0-9._%+'\-\-]+@[A-Za-z0-9.\-\-]+\.[A-Za-z]{2,24} | 100%     |       |

A value that matches none counts 20% against the domain.

HOW IT IS MASKED

User and domain are mapped character by character, and the top-level domain — .com , .com.pe — becomes .test , reserved for tests: a masked address can never deliver.

**PE\_EMAIL** **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: PE\_CM\_ALFANUM .

**PE\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE                     | MASKED                       |
|---------------------------|------------------------------|
| milagros.quispe@gmail.com | → gevevjew.cuahxa@pgoeb.test |

8.15 Phone

PE\_L1\_TELEFONO

**ESSENTIAL** **COLUMN NAME** **VALUE FORMAT** **REGEX DECOMPOSE**

WHAT IT IS

Nine-digit mobiles, all starting with 9, and landlines with an area code — 1 for Lima and Callao, two digits elsewhere — and six or seven digits.

HOW IT IS FOUND

**COLUMN NAME** PE\_L1\_TELEFONO - Path

**85%** tel tel[eé]fono[a-z0-9\_]\* tel\_?(fijo|casa|oficina|trabajo|contacto|celular|m[oó]vil)  
celular[a-z0-9\_]\* cel m[oó]vil whats\_?app anexo\_?telef[oó]nico n(o|ro|um|umero)\_?  
(tel|tel[eé]fono|celular|cel|contacto) fax phone[a-z0-9\_]\* mobile[a-z0-9\_]\*

**VALUE FORMAT** PE\_L1\_TELEFONO - Regex

| PATTERN                                          | STRENGTH | NOTES |
|--------------------------------------------------|----------|-------|
| (\+?51[\s\-\-])?9\d{2}[\s\-\-]?d{3}[\s\-\-]?d{3} | 90%      |       |
| \( ?0?1\)?[\s\-\-]?d{3}[\s\-\-]?d{4}             | 50%      |       |

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

+51 , the leading 9 of a mobile and the area code of a landline stay; the rest of the digits are mapped, so the number keeps saying mobile or landline and which region.

## PE\_TELEFONO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: PE\_CM\_DIGITOS .

### PE\_CM\_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

#### EXAMPLES

| VALUE           |   | MASKED          |
|-----------------|---|-----------------|
| +51 987 654 321 | → | +51 912 410 723 |
| (01) 234-5678   | → | (01) 158-7565   |

## 8.16 Address

PE\_L1\_DIRECCION

### ESSENTIAL

### COLUMN NAME

### DATA TYPE

### VALUE FORMAT

### SECURE LOOKUP

#### WHAT IT IS

The address with the kind of street and the number — Av. Arequipa 1234, Urb. Santa Patricia —, and the Mz. B Lt. 12 of an asentamiento humano.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L1\_DIRECCION - Path

80%

direcci[oó]n[a-z0-9]\* (except when preceded by *mac* , *ip* , *e\_mail* , *web* , *url* , *correo* ; except when followed by *list* E4) domicilio[a-z0-9]\* dom\_?(residencia|particular|laboral|cliente|fiscal) residencia lugar\_?(de\_?)?residencia direccion\_?fiscal calle\_(?y\_?n[uú]mero)? address[a-z0-9]\* (except when preceded by *mac* , *ip* , *e\_mail* , *web* , *url* ; except when followed by *list* E5)

**DATA TYPE** PE\_L1\_DIRECCION - Type

Accepts text (length 6 or more); any other type rules the domain out.

**VALUE FORMAT** PE\_L1\_DIRECCION - Regex

| PATTERN                                                                                                                                                        | STRENGTH | NOTES |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|
| (?i)                                                                                                                                                           | 80%      |       |
| (av avenida jr jir[oó]n calle ca pasaje psje prolongaci[oó]n alameda malec[oó]n carretera)\.?\s+[a-záéíóúñ0-9.' ]{2,40}\s*(n° nro\.? #)?\s*d{1,5}.*            |          |       |
| (?i).*\b(mz manzana)\.?\s*[a-z0-9]{1,3}\s*(lt lote)\.?\s*d{1,3}.*                                                                                              | 80%      |       |
| (?i).*\b(urb urbanizaci[oó]n aa\.?hh\.? asentamiento_?humano pueblo_?joven coop cooperativa residencial conjunto_?habitacional)\.?\s+[a-záéíóúñ0-9.' ]{2,40}.* | 70%      |       |
| (?i)k(m il[oó]metro)\.?\s*d{1,3}\s+(carretera autopista v[íí]a).*                                                                                              | 70%      |       |

A value that matches none counts 20% against the domain.

#### HOW IT IS MASKED

The whole line is replaced by a fictitious Peruvian address of the same form. Urbanización, centro poblado, distrito and postal code have domains of their own for when they live in separate columns.

### PE\_DIRECCION SECURE LOOKUP

Replaces the value with one of the 4,000 values in `pe-direcciones.txt` , chosen by a keyed hash.

## EXAMPLES

### VALUE

### MASKED

Av. Arequipa 1234, Urb. Santa Patricia

→

Res. Santa Cruz, Carretera Independencia 446

## 8.17 Address complement

PE\_L1\_DIRECCION\_COMPLEMENTO

### ESSENTIAL

### COLUMN NAME

### CHARACTER MAPPING

#### WHAT IT IS

Manzana, lote, interior, flat, floor, block or stage in columns of their own.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L1\_DIRECCION\_COMPLEMENTO - Path

70%

```
manzana mz lote lt interior int?(direccion|domicilio) dpto?(n(o|ro|um|umero))?
departamento?(n(o|ro|um|umero)|interior) piso block bloque etapa sector?
(vivienda|domicilio) n(o|ro|um|umero)?(casa|puerta|dpto|interior) complemento(_?
direcci[oó]n)? referencia?(de_)?(direcci[oó]n|domicilio)
```

#### HOW IT IS MASKED

Character by character, keeping length and format.

PE\_CM\_ALFANUM

### CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

## EXAMPLES

### VALUE

### MASKED

Mz. B Lt. 12

→

Rt. Z Kj. 69

## 8.18 Bank account

PE\_L1\_CUENTA\_BANCARIA

### COLUMN NAME

### CHARACTER MAPPING

#### WHAT IT IS

Savings, current, payroll, CTS and deduction account numbers, and the twenty-digit CCI. The bank code is not personal data and has no domain.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L1\_CUENTA\_BANCARIA - Path

85%

```
cuenta_?
(bancaria|banco|corriente|ahorro|sueldo|haberes|cts|abono|dep[oó]sito|destino|origen|detracci[
oó]n) n(o|ro|um|umero)?(de_)?(cuenta|cta) nro?cta num?cta cta?
(cte|corriente|ahorro|bancaria) cci c[oó]digo?(de_)?cuenta?interbancario iban
account?(no|num|number) bank?account
```

#### HOW IT IS MASKED

Digits are mapped; separators stay.

PE\_CM\_DIGITOS

### CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

## EXAMPLES

VALUE

MASKED

0011-0123-0100123456

→ 4708-9581-6701632534

## 8.19 Payment card

PE\_L1\_TARJETA

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

### WHAT IT IS

Credit and debit card numbers, recognized by the Luhn algorithm.

### HOW IT IS FOUND

COLUMN NAME

PE\_L1\_TARJETA - Path

85%

tarjeta(\_?(cr[eé]dito|d[eé]bito|n(o|ro|um|umero)))? n(o|ro|um|umero)?tarjeta pan card\_?  
(no|num|number) credit\_?card tc\_?(numero|num|nro)

VALUE FORMAT

PE\_L1\_TARJETA - Regex

PATTERN

STRENGTH

NOTES

\d{13,19}

90%

LUHN check digit; after removing [\s-]

A value that matches none counts 30% against the domain.

### HOW IT IS MASKED

Payment Card: another number that passes the Luhn algorithm. Only card-shaped values reach it.

PE\_TARJETA

REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

PE\_TARJETA\_LUHN PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

## EXAMPLES

VALUE

MASKED

4557 8800 0000 0001

→ 1941 2536 7434 1799

## 8.20 Digital wallet

PE\_L1\_BILLETERA

COLUMN NAME

VALUE FORMAT

CHARACTER MAPPING

### WHAT IT IS

Yape, Plin and other mobile wallet identifiers — keyed by the phone number — and the bitcoin addresses some systems keep.

### HOW IT IS FOUND

COLUMN NAME

PE\_L1\_BILLETERA - Path

85%

billetera(\_?(digital|electr[oó]nica|m[oó]vil))? wallet(\_?(id|address|direccion))? yape[a-z0-9\_]\*  
plin[a-z0-9\_]\* tunki lukita agora\_?pay c[oó]digo\_?qr qr\_?(id|codigo)  
direcci[oó]n\_?(bitcoin|btc|wallet) btc\_?(address|direccion)

**VALUE FORMAT** PE\_L1\_BILLETERA – Regex

| PATTERN                            | STRENGTH | NOTES |
|------------------------------------|----------|-------|
| (bc1 [13])[a-zA-HJ-NP-Z0-9]{25,59} | 70%      |       |

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Character by character, keeping the shape. A masked bitcoin address no longer passes its own checksum, which is what a test copy wants.

**PE\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE                                      | MASKED                                      |
|--------------------------------------------|---------------------------------------------|
| bc1qar0srrr7xfkvy5l643lydnw9re59gtzwwf5mdq | → mh2wis8tsdm0pdjdv8v511yqxqg4ze53qkjbp7d9f |

## 8.21 IP address

PE\_L1\_IP

**COLUMN NAME** **VALUE FORMAT** **REGEX DECOMPOSE**

#### WHAT IT IS

IPv4 and IPv6 addresses from access logs and transactions.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L1\_IP – Path

**85%** ip ip?(address|addr|origen|destino|cliente|usuario|acceso|login|remota|publica)  
direcci[oó]n?ip dir?ip ipv[46] remote?addr(ess)? client?ip host?ip

**VALUE FORMAT** PE\_L1\_IP – Regex

| PATTERN                                                     | STRENGTH | NOTES |
|-------------------------------------------------------------|----------|-------|
| ((25[0-5] 2[0-4]\d 1?\d?\d)\.){3}(25[0-5] 2[0-4]\d 1?\d?\d) | 100%     |       |
| [0-9A-Fa-f]{1,4}(:[0-9A-Fa-f]{0,4}){2,7}                    | 90%      |       |

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Each IPv4 octet is mapped to 1–254; an IPv6 address is mapped keeping its hex characters.

**PE\_IP** **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: **PE\_CM\_HEX**.

**PE\_OCTETO** **NUMERIC MAPPING**

Maps the number to another between 1 and 254.

**PE\_CM\_HEX** **CHARACTER MAPPING**

Maps each character to another of its group (5 groups), keeping length and format.

#### EXAMPLES

| VALUE         | MASKED          |
|---------------|-----------------|
| 190.117.45.12 | → 164.108.51.94 |

## 8.22 Device identifier

PE\_L1\_DISPOSITIVO

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

### WHAT IT IS

IMEI, IMSI, ICCID, MAC address, advertising and session identifiers.

### HOW IT IS FOUND

COLUMN NAME PE\_L1\_DISPOSITIVO - Path

80% imei imsi iccid mac mac\_address direccion\_mac device\_(id|uuid|serial) id\_(dispositivo|equipo|celular) serial\_(equipo|celular|dispositivo) advertising\_id idfa gaid session\_id id\_sesion

VALUE FORMAT PE\_L1\_DISPOSITIVO - Regex

| PATTERN                                | STRENGTH | NOTES            |
|----------------------------------------|----------|------------------|
| ([0-9A-Fa-f]{2}[:\-]){5}[0-9A-Fa-f]{2} | 100%     |                  |
| \d{15}                                 | 80%      | LUHN check digit |

A value that matches none counts 30% against the domain.

### HOW IT IS MASKED

Character by character, with hex digits kept hex: a MAC stays a MAC.

PE\_CM\_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

### EXAMPLES

| VALUE             | MASKED              |
|-------------------|---------------------|
| 3c:22:fb:9a:10:e4 | → 5e:83:bb:3c:31:f9 |

## 8.23 Cookies and tracking identifiers

PE\_L1\_COOKIE

COLUMN NAME CHARACTER MAPPING

### WHAT IT IS

The identifier a site stores in the browser and the campaign parameters that travel with it. It is personal data when it singles out a person through means that can reasonably be used.

### HOW IT IS FOUND

COLUMN NAME PE\_L1\_COOKIE - Path

80% cookie\_(? (id|value|valor|nombre)) ? cookies \_?ga \_?gid fbp fbclid utm\_(source|medium|campaign|term|content) id\_(navegador|browser|visitante) visitor\_id tracking\_id client\_id

### HOW IT IS MASKED

Character by character, keeping length and format.

PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

GA1.2.123456789.1700000000

→ ZE0.5.891552838.7889916184

8.24 Vehicle plate

PE\_L1\_PLACA

COLUMN NAME

VALUE FORMAT

CHARACTER MAPPING

WHAT IT IS

The Placa Única Nacional de Rodaje: three letters and three digits since 2011 ( ABC-123 ), a variant that mixes them ( A1B-234 ), and two letters and four digits on a motorcycle.

HOW IT IS FOUND

COLUMN NAME

PE\_L1\_PLACA - Path

85%

placa(\_?(veh[ií]culo|carro|moto|automotor|rodaje))? n(o|ro|um|umero)?placa placa\_  
[uú]nica?nacional?(de\_)?rodaje punr matr[ií]cula?(veh[ií]culo|automotor) license\_  
plate plate(\_?(no|number))?

VALUE FORMAT

PE\_L1\_PLACA - Regex

PATTERN

STRENGTH

NOTES

[A-Z] [0-9A-Z] [A-Z] [\s-]? \d{3}

80%

[A-Z]{2} [\s-]? \d{4}

50%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Letters to letters and digits to digits, position by position: the plate format stays.

PE\_CM\_DIGITOS\_LETRAS

CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

ABC-123

→ AFJ-775

A1B-234

→ V9D-496

8.25 Vehicle and licence

PE\_L1\_VEHICULO

COLUMN NAME

VALUE FORMAT

CHARACTER MAPPING

WHAT IT IS

The VIN or chassis number, the engine number, the vehicle identification card and the driving licence — the *brevete*.

HOW IT IS FOUND

COLUMN NAME

PE\_L1\_VEHICULO - Path

85%

vin chasis n(o|ro|um|umero)?(chasis|motor|serie|vin) motor?n(o|ro|um|umero) tarjeta\_  
(de\_)?(identificaci[oó]n?vehicular|propiedad|circulaci[oó]n) licencia?(de\_)?conducir(\_?  
(n(o|ro|um|umero)))? n(o|ro|um|umero)?licencia brevete

**VALUE FORMAT** PE\_L1\_VEHICULO - Regex

| PATTERN             | STRENGTH | NOTES |
|---------------------|----------|-------|
| [A-HJ-NPR-Z0-9]{17} |          | 70%   |

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

Character by character, keeping length and format.

**PE\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE             | MASKED              |
|-------------------|---------------------|
| 8AJHA8CD3J1234567 | → 8EWKE1QN1T0772894 |

## 8.26 Property

PE\_L1\_INMUEBLE

**COLUMN NAME** **CHARACTER MAPPING**

#### WHAT IT IS

The partida registral and ficha of SUNARP, the cadastral code and the self-assessment record: they lead to the home and its owner.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L1\_INMUEBLE - Path

**80%** partida\_?(registral|electr[oó]nica) n(o|ro|um|umero)?partida?registral ficha?registral  
tomo\_?(y\_)?folio n(o|ro|um|umero)?(de\_)?(predio|inmueble|catastral) c[oó]digo?  
(catastral|predial|de?predio) catastro cup asiento?registral sunarp autoval[uú]o?  
(n(o|ro|um|umero)|c[oó]digo) hr?pu

#### HOW IT IS MASKED

Character by character, keeping length and format.

**PE\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE     | MASKED      |
|-----------|-------------|
| P12345678 | → T33398422 |

## 8.27 Customer and contract numbers

PE\_L1\_CONTRATO

**COLUMN NAME** **CHARACTER MAPPING**

#### WHAT IT IS

Contract, policy, loan, application, member, subscriber, supply, meter, case or administrative file numbers, and customer or employee codes.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L1\_CONTRATO – Path

**70%** n(o|ro|um|umero)?(contrato|p[ro]liza|cr[eé]dito|pr[eé]stamo|solicitud|expediente?  
(administrativo|interno)|tr[aá]mite|referencia|cliente|socio|suscriptor|suministro|medidor|ser  
vicio|caso|ticket|beneficio) contrato?(n(o|ro|um|umero)) c[ó]digo?  
(cliente|socio|empleado|afiliado|suministro) id?(cliente|socio|empleado|afiliado)  
n(o|ro)?empleado c[ó]digo?(de\_)?(empleado|trabajador|planilla) n(o|ro|um|umero)?  
expediente?sig

#### HOW IT IS MASKED

Character by character, keeping length and format. The mapping is deterministic, so joins keep working.

**PE\_CM\_ALFANUM CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

**VALUE**

**MASKED**

CT-123456

→ NG-557411

## 8.28 User and social media

PE\_L1\_USUARIO

**COLUMN NAME** **CHARACTER MAPPING**

#### WHAT IT IS

Login, alias and social media profiles.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L1\_USUARIO – Path

**70%** usuario user?name username login alias apodo nick(\_?name)? perfil?  
(instagram|facebook|twitter|tiktok|linkedin|x) instagram facebook twitter tiktok  
linkedin red?social handle

#### HOW IT IS MASKED

Character by character, keeping length and format.

**PE\_CM\_ALFANUM CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

**VALUE**

**MASKED**

@milagros\_quispe

→ @gevevjew\_cuahxa

## 8.29 Credentials

PE\_L1\_CREDENCIAL

**COLUMN NAME** **REGEX DECOMPOSE**

#### WHAT IT IS

Passwords and their hashes, PIN, security and OTP codes, CVV, tokens, API keys, the SUNAT *clave SOL*, private keys and seed phrases.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L1\_CREDENCIAL – Path

**90%** `contrase[nñ]a` `clave` (except when followed by `catastral`, `primaria`, `for[aá]nea`, `valor`, `producto`)  
`password` `passwd` `pwd` `pass` `hash_?(clave|contrasena|password)` `pin` `c[oo]digo_?`  
`(seguridad|verificaci[oó]n|acceso|otp)` `cvv` `cvc` `token(_?(acceso|refresh|api))?` `access_?`  
`token` `refresh_?token` `api_?key` `llave_?(api|privada)` `secreto` `secret` `otp` `clave_?sol`  
`clave_?(de_?)?internet` `pregunta_?secreta` `respuesta_?secreta` `frase_?semilla` `seed_?`  
`phrase`

#### HOW IT IS MASKED

Every character becomes X. A credential has no use in a test copy, and a hash masked character by character could still be cracked.

**PE\_REDACTAR** **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

#### EXAMPLES

| VALUE      | MASKED       |
|------------|--------------|
| Clave#2024 | → XXXXXXXXXX |

## 8.30 Geolocation

PE\_L1\_GEOLOCALIZACION

**COLUMN NAME** **VALUE FORMAT** **REGEX DECOMPOSE**

#### WHAT IT IS

Latitude and longitude, together or in separate columns, from apps, visits, addresses and deliveries.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L1\_GEOLOCALIZACION – Path

**85%** `lat` `latitud` `lon` `lng` `longitud` `coordenadas?` `coord_?[xy]` `geo_?`  
`(lat|lon|lng|loc|localizacion|posicion)` `georreferenciaci[oó]n` `geolocalizaci[oó]n` `gps(_?`  
`(ubicaci[oó]n|posici[oó]n|coordenadas))?` `ubicaci[oó]n_?gps`

**50%** `long`

**VALUE FORMAT** PE\_L1\_GEOLOCALIZACION – Regex

| PATTERN                                              | STRENGTH | NOTES |
|------------------------------------------------------|----------|-------|
| <code>-\d{1,2}\.\d{3,}\s*,\s*-(7 8)\d\.\d{3,}</code> |          | 90%   |
| <code>-(7[0-9] 8[01])\.\d{4,}</code>                 |          | 50%   |
| <code>-\d{1,2}\.\d{4,}</code>                        |          | 40%   |

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

The integer part and first decimal stay — about 11 km — and the other decimals are mapped: the place is generalized, not moved.

**PE\_COORDENADA** **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

**PE\_CM\_DIGITOS** **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

## EXAMPLES

### VALUE

-12.046374,-77.042793

→

### MASKED

-12.061019,-77.036229

## 8.31 Free text

PE\_L1\_TEXT0\_LIBRE

### COLUMN NAME

### DATA TYPE

### VALUE FORMAT

### FREE TEXT REDACTION

#### WHAT IT IS

Remarks, complaint descriptions, accounts of events and messages where anything gets written — DNI, RUC, phone and names in the middle of a sentence.

#### HOW IT IS FOUND

##### COLUMN NAME

PE\_L1\_TEXT0\_LIBRE - Path

60%

observaci[oó]n(es)? obs comentarios? notas? anotaci[oó]n(es)? glosa sustento  
descripci[oó]n\_?(queja|reclamo|solicitud|caso|hechos|novedad|atenci[oó]n|denuncia) detalle\_  
(reclamo|solicitud|caso|atenci[oó]n) hechos relato justificaci[oó]n mensaje *(except when  
preceded by list E6; except when followed by list E7)* texto(\_?libre)? resumen\_?(caso|atenci[oó]n) queja  
reclamo denuncia\_?texto remarks? comments? notes? free\_?text memo

##### DATA TYPE

PE\_L1\_TEXT0\_LIBRE - Type

Accepts text (length 30 or more); any other type rules the domain out.

##### VALUE FORMAT

PE\_L1\_TEXT0\_LIBRE - Regex

##### PATTERN

##### STRENGTH

##### NOTES

\d{8}-[0-9A-JK]

70% anywhere in the value

[\w.+\\-]+@[\\w\\-]+\\. [A-Za-z]{2,}

70% anywhere in the value

9\d{2}[\s\\-]?\\d{3}[\s\\-]?\\d{3}

60% anywhere in the value

A value that matches none does not count against the domain.

#### HOW IT IS MASKED

Free Text Redaction: DNIs, RUCs, e-mail, mobiles, cards, plates and IPs become [DAT0], and the names of the lists [NOMBRE].

##### PE\_TEXT0\_LIBRE FREE TEXT REDACTION

Replaces the text matching 8 patterns with [DAT0], and the 1,906 terms of pe-texto-nombres.txt with [NOMBRE].

## EXAMPLES

### VALUE

Cliente Quispe, DNI 45678901, correo  
m.quispe@gmail.com, cel 987654321.

→

### MASKED

Cliente [NOMBRE], DNI [DAT0], correo [DAT0],  
cel [DAT0].

# Quasi-identifiers

Data that identifies nobody on its own, but does in combination. Peru is divided into 1,892 distritos, and they are small: 1,633 of them had fewer than 20,000 inhabitants in the 2017 census, and hundreds fewer than a thousand. A birth date, a sex and one of those distritos point at a handful of people. These attributes are generalized or shifted rather than replaced, and the data keeps its analytical value.

## 9.1 Birth date

PE\_L2\_FECHA\_NACIMIENTO

ESSENTIAL COLUMN NAME DATA TYPE DATE SHIFT

WHAT IT IS

The birth date, as a date, timestamp or text.

HOW IT IS FOUND

COLUMN NAME PE\_L2\_FECHA\_NACIMIENTO – Path

90% ( fecha|fec|fch|f )?( de\_? )?nac( imiento)? fnac fechanac( imiento)? cumplea[ñ]os dob  
date\_of\_birth birth\_date birthday

DATA TYPE PE\_L2\_FECHA\_NACIMIENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the domain out.

HOW IT IS MASKED

Shifted up to 60 days either way. Ages and school years change for few people; the date stays plausible and the same birth date always moves the same way.

PE\_FECHA\_NACIMIENTO DATE SHIFT

Moves the date by -60 to 60 days.

EXAMPLES

| VALUE      | MASKED             |
|------------|--------------------|
| 1985-07-20 | → 1985-08-05T00:00 |

## 9.2 Birth year

PE\_L2\_ANIO\_NACIMIENTO

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

The birth year in a column of its own.

HOW IT IS FOUND

COLUMN NAME PE\_L2\_ANIO\_NACIMIENTO – Path

90% a[ñ]o\_?( de\_? )?nac( imiento)? anio\_?nac( imiento)? year\_of\_birth birth\_year yob

DATA TYPE PE\_L2\_ANIO\_NACIMIENTO – Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the domain out.

HOW IT IS MASKED

The decade stays and the last digit is mapped: 1985 becomes a year from 1980 to 1989.

PE\_ANIO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

**PE\_CM\_DIGITOS** **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

#### EXAMPLES

VALUE

MASKED

1985

→

**1980**

### 9.3

## Age

PE\_L2\_EDAD

**COLUMN NAME**

**DATA TYPE**

**REGEX DECOMPOSE**

#### WHAT IT IS

Age in years, also age group or five-year band.

#### HOW IT IS FOUND

**COLUMN NAME**

PE\_L2\_EDAD – Path

85%

edad(\_?(actual|a[nños|anios|paciente|afiliado|cliente|al\_?

(ingreso|diagn[ó]stico|fallecimiento)))? grupo\_?etario rango\_?(de\_?)?edad quinquenio

age(\_?(years|group)))?

**DATA TYPE**

PE\_L2\_EDAD – Type

Accepts number (length up to 6) or text (length up to 6); any other type rules the domain out.

#### HOW IT IS MASKED

The tens digit stays and the unit is mapped (37 becomes 30–39). A hundred or more becomes 90: rare ages identify.

**PE\_EDAD**

**REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

**PE\_CM\_DIGITOS** **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

#### EXAMPLES

VALUE

MASKED

37

→

**35**

104

→

**90**

### 9.4

## Personal event dates

PE\_L2\_FECHA\_EVENTO

**COLUMN NAME**

**DATA TYPE**

**DATE SHIFT**

#### WHAT IT IS

Dates that say something about the person: death, marriage, hiring and leaving, retirement, admission, diagnosis, arrest, sentence, sick leave, DNI issue.

HOW IT IS FOUND

**COLUMN NAME** PE\_L2\_FECHA\_EVENTO – Path

**80%** (fecha|fec|fch|f)?(de\_)?  
(defunci[oó]n|fallecimiento|muerte|matrimonio|divorcio|ingreso|cese|egreso|alta|baja|retiro|con-  
ntrataci[oó]n|despido|jubilaci[oó]n|hospitalizaci[oó]n|internamiento|diagn[oó]stico|atenci[oó]  
n|consulta|parto|captura|detenci[oó]n|sentencia|condena|emisi[oó]n(\_?(dni|documento))?  
|caducidad\_?dni|vacunaci[oó]n|cirug[ií]a|accidente|descanso\_?m[eé]dico) fec\_  
(emi|emision|defuncion|ingreso|cese|baja) date\_?of\_  
(death|marriage|hire|admission|discharge)

**DATA TYPE** PE\_L2\_FECHA\_EVENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the domain out.

HOW IT IS MASKED

Shifted up to 30 days either way.

**PE\_FECHA\_EVENTO DATE SHIFT**

Moves the date by -30 to 30 days.

EXAMPLES

| VALUE      |   | MASKED           |
|------------|---|------------------|
| 2021-03-15 | → | 2021-03-05T00:00 |

9.5

Sex

PE\_L2\_SEX0

**COLUMN NAME** **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

Sex as the system records it: F / M , Femenino / Masculino , Mujer / Hombre or the codes 1 and 2.

HOW IT IS FOUND

**COLUMN NAME** PE\_L2\_SEX0 – Path

**85%** sexo(\_?(biol[oó]gico|al\_?nacer|paciente|afiliado|cliente))? (tp|tipo|cod|cd|id)?sexo  
g[eé]nero (except when followed by *identidad* , *autopercebido* ) sex gender (except when followed by  
*identity* )

**KNOWN VALUES** PE\_L2\_SEX0 – List

| LIST                 | VALUES | STRENGTH | FIRST VALUES                        |
|----------------------|--------|----------|-------------------------------------|
| pe-detectar-sexo.txt | 9      | 50%      | f, m, h, femenino, masculino, mujer |

A value that matches none counts 50% against the domain.

HOW IT IS MASKED

Replaced within its own vocabulary: F stays F or M , 1 stays 1 or 2 .

**PE\_SEX0 REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

**PE\_SEX0\_FEMENINO\_MASCULINO SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-sexo-femenino-masculino.txt , chosen by a keyed hash.

**PE\_SEX0\_MUJER\_HOMBRE SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-sexo-mujer-hombre.txt , chosen by a keyed hash.

#### PE\_SEX0\_FEMALE\_MALE SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-female-male.txt`, chosen by a keyed hash.

#### PE\_SEX0\_F\_M SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-f-m.txt`, chosen by a keyed hash.

#### PE\_SEX0\_H\_M SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-h-m.txt`, chosen by a keyed hash.

#### PE\_SEX0\_1\_2 SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-1-2.txt`, chosen by a keyed hash.

#### EXAMPLES

| VALUE    |   | MASKED   |
|----------|---|----------|
| F        | → | M        |
| Femenino | → | Femenino |

## 9.6 Distrito and provincia

PE\_L2\_DISTRITO

### COLUMN NAME

### KNOWN VALUES

### DATA CLEANSING

#### WHAT IT IS

The distrito or provincia of residence, birth or care, by name — alone or with the provincia or the departamento, Balsas – Amazonas .

#### HOW IT IS FOUND

### COLUMN NAME

PE\_L2\_DISTRITO – Path

85%

`distrito(_?(residencia|nacimiento|domicilio))?` (except when followed by [list E8](#)) `dist` (except when followed by `cod`, `codigo`, `id`) `provincia(_?(residencia|nacimiento|domicilio))?` (except when followed by `cod`, `codigo`, `id`) `prov` (except when followed by `cod`, `codigo`, `id`) `ciudad(_?(residencia|nacimiento|domicilio|cliente))?` `(nom|nombre|desc)_?(distrito|provincia|ciudad)` `lugar_(de_)?nacimiento` `city` `town`

### KNOWN VALUES

PE\_L2\_DISTRITO – List

### LIST

### VALUES

### STRENGTH

### FIRST VALUES

|                           |       |     |                                                         |
|---------------------------|-------|-----|---------------------------------------------------------|
| pe-detectar-distritos.txt | 1,981 | 80% | Chachapoyas, Asunción, Asuncion, Balsas, Cheto, Chiquín |
|---------------------------|-------|-----|---------------------------------------------------------|

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

A distrito with fewer than 20,000 inhabitants becomes the nearest distrito of its provincia with at least 20,000 — or of its departamento when the provincia has none. The thirty-three provincias under 20,000 follow the same rule inside their departamento. The 25 departamentos stay.

### PE\_DISTRITO DATA CLEANSING

Replaces each value found in `pe-distritos-generalizados.txt` (44,952 pairs) by its counterpart; values not in the file stay as they are.

## EXAMPLES

| VALUE             |   | MASKED                 |
|-------------------|---|------------------------|
| Asunción          | → | Los Baños del Inca     |
| Lima              | → | Lima                   |
| Balsas – Amazonas | → | Chachapoyas – Amazonas |

## 9.7 UBIGEO

PE\_L2\_UBIGEO

COLUMN NAME VALUE FORMAT DATA CLEANSING

### WHAT IT IS

The INEI code: two digits for the departamento, four for the provincia and six for the distrito.

### HOW IT IS FOUND

COLUMN NAME PE\_L2\_UBIGEO – Path

85% `ubigeo[a-z0-9_]* (cod|codigo|cd|id)?(ubigeo|distrito|dist|provincia|prov|ciudad|lugar)(_?(inei|reniec|res|residencia|nac|nacimiento))?(cod|codigo)?(inei|reniec|geogr[aá]fico)`  
`coddist codprov`

VALUE FORMAT PE\_L2\_UBIGEO – Regex

| PATTERN                                       | STRENGTH | NOTES |
|-----------------------------------------------|----------|-------|
| <code>(0[1-9] 1\d 2[0-5])\d{2}(\d{2})?</code> |          | 40%   |

A value that matches none counts 30% against the domain.

### HOW IT IS MASKED

The same generalization as the name: a small distrito's UBIGEO becomes the UBIGEO of the distrito that replaces its name. The code of a large one does not change.

PE\_UBIGEO DATA CLEANSING

Replaces each value found in `pe-ubigeo-generalizado.txt` (1,666 pairs) by its counterpart; values not in the file stay as they are.

## EXAMPLES

| VALUE  |   | MASKED |
|--------|---|--------|
| 010102 | → | 010101 |
| 150101 | → | 150101 |

## 9.8 Centro poblado and caserío

PE\_L2\_CENTRO\_POBLADO

COLUMN NAME SECURE LOOKUP

### WHAT IT IS

The centro poblado, caserío, anexo or peasant or native community of a rural address: a few hundred people each, and the address of most rural households.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L2\_CENTRO\_POBLADO – Path

**85%** centro\_?poblado ccpp caser[íi]o anexo(\_?rural)? comunidad\_?(campesina|nativa|rural)  
comunidad aldea paraje zona\_?rural sector\_?rural hacienda fundo parcela(ci[oó]n)?  
predio\_?rural

#### HOW IT IS MASKED

Replaced by Zona rural.

**PE\_ZONA\_RURAL** **SECURE LOOKUP**

Replaces the value with one of the 1 values in pe-zona-rural.txt, chosen by a keyed hash.

#### EXAMPLES

**VALUE**

**MASKED**

Caserío El Milagro

→ Zona rural

## 9.9 Urbanización and asentamiento humano

PE\_L2\_URBANIZACION

**COLUMN NAME** **SECURE LOOKUP**

#### WHAT IT IS

The urbanización, asentamiento humano, pueblo joven, housing cooperative or residencial of residence.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L2\_URBANIZACION – Path

**80%** urbanizaci[oó]n urb asentamiento\_?humano aa\_?hh pueblo\_?joven asociaci[oó]n\_?(de\_)?  
vivienda cooperativa\_?(de\_)?vivienda conjunto\_?habitacional residencial barrio(\_?  
(residencia|domicilio))? (nom|nombre)\_?(urbanizacion|barrio|residencial) neighbou?rhood

#### HOW IT IS MASKED

Replaced by a common Peruvian one, unrelated to the original.

**PE\_URBANIZACION** **SECURE LOOKUP**

Replaces the value with one of the 104 values in pe-urbanizaciones.txt, chosen by a keyed hash.

#### EXAMPLES

**VALUE**

**MASKED**

Urb. Santa Patricia

→ AA.HH. Villa El Salvador

## 9.10 Postal code

PE\_L2\_CODIGO\_POSTAL

**COLUMN NAME** **DATA TYPE** **REGEX DECOMPOSE**

#### WHAT IT IS

The five-digit postal code in use since 2011: the first two are the departamento.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L2\_CODIGO\_POSTAL – Path

**85%** c[oó]digo\_?postal cod\_?postal codpostal zip(\_?code)? postal\_?code cp

**DATA TYPE** PE\_L2\_CODIGO\_POSTAL – Type

Accepts text (length 4 to 5) or number (length up to 5); any other type rules the domain out.

#### HOW IT IS MASKED

The first two digits stay and the others become zeros: 15001 becomes 15000 .

PE\_CODIGO\_POSTAL **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

#### EXAMPLES

| VALUE | MASKED  |
|-------|---------|
| 15001 | → 15000 |

## 9.11 Marital status

PE\_L2\_ESTADO\_CIVIL

**COLUMN NAME**

**KNOWN VALUES**

**REGEX DECOMPOSE**

#### WHAT IT IS

Single, married, conviviente — the union of fact the Civil Code recognizes —, separated, divorced or widowed, in words or as a code.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_L2\_ESTADO\_CIVIL – Path

85% estado\_?civil est\_?civil (cod|tipo|id)\_?estado\_?civil situaci[ó]n\_?conyugal uni[ó]n\_?de\_?hecho marital\_?status civil\_?status

**KNOWN VALUES** PE\_L2\_ESTADO\_CIVIL – List

| LIST                         | VALUES | STRENGTH | FIRST VALUES                                                        |
|------------------------------|--------|----------|---------------------------------------------------------------------|
| pe-detectar-estado-civil.txt | 20     | 70%      | Soltero/a, Casado/a, Conviviente, Separado/a, Divorciado/a, Viudo/a |

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

Replaced by another status from the list; a code stays a code.

PE\_ESTADO\_CIVIL **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_ESTADO\_CIVIL\_LISTA .

PE\_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_BANDERA\_SI\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.

PE\_BANDERA\_TRUE\_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.

PE\_BANDERA\_YES\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.

PE\_BANDERA\_S\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.

PE\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.

PE\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

**PE\_ESTADO\_CIVIL\_CODIGO** **SECURE LOOKUP**

Replaces the value with one of the 6 values in `pe-estado-civil-codigos.txt`, chosen by a keyed hash.

**PE\_ESTADO\_CIVIL\_LISTA** **SECURE LOOKUP**

Replaces the value with one of the 6 values in `pe-estado-civil.txt`, chosen by a keyed hash.

#### EXAMPLES

VALUE

MASKED

Unión de hecho

→ **Soltero/a**

## 9.12 Occupation

PE\_L2\_OCUPACION

**COLUMN NAME**

**REGEX DECOMPOSE**

#### WHAT IT IS

Occupation, profession or trade, in words or by the national occupation classification code. A rare occupation in a small distrito identifies on its own.

#### HOW IT IS FOUND

**COLUMN NAME**

PE\_L2\_OCUPACION - Path

80%

ocupaci[oó]n profes[ió]n oficio cno (cod|codigo)?(ocupacion|profesion|cno) puesto?  
(de\_)?trabajo categor[ií]a?ocupacional cargo?(actual|empleado) occupation profession  
job\_?title

50%

cargo puesto

#### HOW IT IS MASKED

Text is replaced by a common occupation; a code is mapped digit by digit.

**PE\_OCUPACION\_O\_CODIGO** **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: **PE\_OCUPACION**.

**PE\_CM\_DIGITOS** **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

**PE\_OCUPACION** **SECURE LOOKUP**

Replaces the value with one of the 39 values in `pe-ocupaciones.txt`, chosen by a keyed hash.

#### EXAMPLES

VALUE

MASKED

Gerente de operaciones

→ **Programador**

2221

→ **1980**

## 9.13 Employer

PE\_L2\_EMPLEADOR

**COLUMN NAME**

**SECURE LOOKUP**

#### WHAT IT IS

The name of the company or institution where the person works.

HOW IT IS FOUND

**COLUMN NAME** PE\_L2\_EMPLEADOR – Path

**90%** empleador(\_?(nombre|raz[oó]n?social|ruc))? (nombre|nom)?(empleador|empresa?donde?trabaja|patrono) empresa?(donde?)(trabaja|labora) lugar?(de?)?trabajo centro?(de?)?(trabajo|labores) entidad?empleadora employer(\_?name)? workplace

HOW IT IS MASKED

Replaced by a fictitious company.

PE\_EMPLEADOR **SECURE LOOKUP**

Replaces the value with one of the 24 values in `pe-empleadores.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE                                |   | MASKED                       |
|--------------------------------------|---|------------------------------|
| Corporación Comercial Peruana S.A.C. | → | <b>Panadería Pan de Casa</b> |

9.14 Education

PE\_L2\_NIVEL\_EDUCATIVO

**COLUMN NAME** **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

The education level attained — from none to a doctorate — in the categories the census uses, in words or as a code.

HOW IT IS FOUND

**COLUMN NAME** PE\_L2\_NIVEL\_EDUCATIVO – Path

**80%** escolaridad nivel?(educativo|de?estudios|acad[eé]mico|alcanzado|de?instrucci[oó]n) grado?(de?)?(estudio|instrucci[oó]n|escolaridad) m[aá]ximo?nivel a[nñ]os?(de?)? estudio education(\_?level)?

**KNOWN VALUES** PE\_L2\_NIVEL\_EDUCATIVO – List

| LIST                            | VALUES | STRENGTH | FIRST VALUES                                                                                           |
|---------------------------------|--------|----------|--------------------------------------------------------------------------------------------------------|
| pe-detectar-nivel-educativo.txt | 26     | 60%      | Sin nivel, Inicial, Primaria incompleta, Primaria completa, Secundaria incompleta, Secundaria completa |

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by another level from the list; a code stays a code.

PE\_NIVEL\_EDUCATIVO **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_NIVEL\_EDUCATIVO\_LISTA .

PE\_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_BANDERA\_SI\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

PE\_BANDERA\_TRUE\_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

PE\_BANDERA\_YES\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

#### PE\_BANDERA\_S\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt` , chosen by a keyed hash.

#### PE\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt` , chosen by a keyed hash.

#### PE\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt` , chosen by a keyed hash.

#### PE\_NIVEL\_EDUCATIVO\_CODIGO **SECURE LOOKUP**

Replaces the value with one of the 11 values in `pe-nivel-educativo-codigos.txt` , chosen by a keyed hash.

#### PE\_NIVEL\_EDUCATIVO\_LISTA **SECURE LOOKUP**

Replaces the value with one of the 11 values in `pe-nivel-educativo.txt` , chosen by a keyed hash.

#### EXAMPLES

##### VALUE

##### MASKED

Educación superior en curso

→

**Maestría o doctorado**

## 9.15 School

PE\_L2\_INSTITUCION\_EDUCATIVA

#### COLUMN NAME

#### SECURE LOOKUP

#### WHAT IT IS

The institución educativa, CETPRO, institute or university, by name or by the Ministry of Education modular code.

#### HOW IT IS FOUND

##### COLUMN NAME

PE\_L2\_INSTITUCION\_EDUCATIVA - Path

75%

```
instituci[oó]n?educativa i_?e(_?p)? centro?(educativo|de?estudios) cetpro
(nombre|nom)?(institucion?educativa|escuela|colegio|universidad) escuela colegio
instituto universidad c[oó]digo?modular c[oó]digo?(de_)?local?escolar school(_?
name)? university
```

#### HOW IT IS MASKED

Replaced by a fictitious school.

#### PE\_INSTITUCION\_EDUCATIVA **SECURE LOOKUP**

Replaces the value with one of the 12 values in `pe-instituciones-educativas.txt` , chosen by a keyed hash.

#### EXAMPLES

##### VALUE

##### MASKED

Universidad Nacional de Ingeniería

→

**Instituto Superior Tecnológico del Norte**

## 9.16 Dependants and children

PE\_L2\_PERSONAS\_A\_CARGO

#### COLUMN NAME

#### DATA TYPE

#### REGEX DECOMPOSE

#### WHAT IT IS

Number of dependants, children, derechohabientes or household members.

HOW IT IS FOUND

**COLUMN NAME** PE\_L2\_PERSONAS\_A\_CARGO – Path

**80%** personas\_?a\_?cargo n(o|ro|um|umero)?(de\_?)(hijos|dependientes|derechohabientes|personas\_?  
a\_?cargo) cant(idad)??(hijos|dependientes|personas\_?hogar|miembros) hijos dependientes  
derechohabientes personas\_?(en\_?el\_?)?hogar tama[ñ]o\_?(de\_l\_?)?hogar miembros\_?(de\_l\_?)?  
hogar number\_?of\_?children dependents

**DATA TYPE** PE\_L2\_PERSONAS\_A\_CARGO – Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the domain out.

HOW IT IS MASKED

Values from 0 to 5 stay; above 5 they become 5. Very large households are rare, and a rare count identifies.

PE\_CONTEO\_LIMITADO **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

| VALUE |   | MASKED |
|-------|---|--------|
| 9     | → | 5      |

# Sensitive data (art. 2, 5)

The list of article 2, numeral 5 is short and blunt: biometric data that identifies on its own, racial and ethnic origin, economic income, political, religious, philosophical and moral opinions or convictions, trade union membership, and information on health or sexual life. Income lives in the financial group, where the money is, and health in a group of its own. Here are the rest, together with the genetic data, the gender identity, the indigenous people and language, the nationality and the victim status that other Peruvian statutes protect and that re-identify just as fast.

## 10.1 Ethnic origin

PE\_L3\_ORIGEN\_ETNICO

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Self-identification as quechua, aimara, native of the Amazon, afroperuano, white, mestizo, nikkei or tusán – the categories the 2017 census asked everyone from twelve up.

HOW IT IS FOUND

COLUMN NAME

PE\_L3\_ORIGEN\_ETNICO – Path

95%

etnia origen\_?([eé]tnico|racial) grupo\_?[eé]tnico pertenencia\_?[eé]tnica autoidenticaci[oó]n(?[eé]tnica)? autorreconocimiento (cod|codigo|tipo|id)\_? (etnia|origen\_?etnico) raza ind[ií]lgena afroperuano afrodescendiente ethnicity race

KNOWN VALUES

PE\_L3\_ORIGEN\_ETNICO – List

| LIST                             | VALUES | STRENGTH | FIRST VALUES                                                                                                                                                                              |
|----------------------------------|--------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pe-detectar-origen-etnico.txt    | 29     | 90%      | Quechua, Aimara, Nativo o indígena de la Amazonía, Nativo o indígena de la Amazonia, Perteneciente a otro pueblo indígena u originario, Perteneciente a otro pueblo indígena u originario |
| pe-detectar-codigos-banderas.txt | 11     | 30%      | s, n, sí, si, no, 1                                                                                                                                                                       |

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A flag stays a flag, a code is replaced by another code and any other value by a category from the list.

PE\_ORIGEN\_ETNICO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_ORIGEN\_ETNICO\_LISTA .

PE\_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.

PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.

PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.

PE\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.

#### PE\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

#### PE\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

#### PE\_ORIGEN\_ETNICO\_CODIGO **SECURE LOOKUP**

Replaces the value with one of the 10 values in `pe-origen-etnico-codigos.txt`, chosen by a keyed hash.

#### PE\_ORIGEN\_ETNICO\_LISTA **SECURE LOOKUP**

Replaces the value with one of the 10 values in `pe-origen-etnico.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE    |   | MASKED                                            |
|----------|---|---------------------------------------------------|
| Indígena | → | Perteneciente a otro pueblo indígena u originario |
| S        | → | S                                                 |

## 10.2 Indigenous people

PE\_L3\_PUEBLO\_INDIGENA

#### COLUMN NAME

#### KNOWN VALUES

#### REGEX DECOMPOSE

### WHAT IT IS

The people of the Base de Datos Oficial de Pueblos Indígenas u Originarios the person belongs to, and the peasant or native community. Naming the people is naming the ethnic origin.

### HOW IT IS FOUND

#### COLUMN NAME

PE\_L3\_PUEBLO\_INDIGENA - Path

90%

pueblo\_?(ind[ií]gena|originario) comunidad\_?(nativa|campesina)\_?(nombre|pertenencia)  
pertenece\_?(a\_?)?(pueblo|comunidad) bdpi base\_?de\_?datos\_?(de\_?)?pueblos etnia\_  
amaz[oó]nica indigenous\_?people

#### KNOWN VALUES

PE\_L3\_PUEBLO\_INDIGENA - List

#### LIST

#### VALUES

#### STRENGTH

#### FIRST VALUES

pe-detectar-pueblos.txt

45

85%

Quechuas, Aimara, Asháninka, Ashaninka, Awajún, Awajun

A value that matches none counts 40% against the domain.

### HOW IT IS MASKED

Replaced by another people from the list; a flag stays a flag.

#### PE\_PUEBLO\_INDIGENA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

#### PE\_PUEBLO\_INDIGENA\_LISTA .

#### PE\_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

#### PE\_BANDERA\_SI\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

#### PE\_BANDERA\_TRUE\_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

#### PE\_BANDERA\_YES\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

#### PE\_BANDERA\_S\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

#### PE\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

#### PE\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

#### PE\_CM\_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### PE\_PUEBLO\_INDIGENA\_LISTA **SECURE LOOKUP**

Replaces the value with one of the 21 values in `pe-pueblos-indigenas.txt`, chosen by a keyed hash.

#### EXAMPLES

##### VALUE

##### MASKED

Pueblo Ese Eja

→ **Ticuna**

## 10.3 Language

PE\_L3 LENGUA

#### COLUMN NAME

#### KNOWN VALUES

#### REGEX DECOMPOSE

#### WHAT IT IS

The mother or indigenous language. Peru has forty-eight beside Spanish, and the Ley 29735 makes them official where they prevail: the language says where a person comes from.

#### HOW IT IS FOUND

##### COLUMN NAME PE\_L3 LENGUA - Path

90%

```
lengua(_?(ind[ií]gena|materna|nativa|originaria|de_?se[ñ]as))? idioma(_?
(ind[ií]gena|materno|nativo|originario))? habla_?(lengua|idioma) lengua_?que_?aprendi[oó]
mother_?tongue native_?language
```

##### KNOWN VALUES PE\_L3 LENGUA - List

##### LIST

##### VALUES

##### STRENGTH

##### FIRST VALUES

pe-detectar-lenguas.txt

37

80%

Castellano, Quechua, Aimara, Asháninka, Ashaninka, Awajún

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

Replaced by a language from the list; a flag stays a flag.

#### PE LENGUA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE LENGUA LISTA`.

##### PE\_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

##### PE\_BANDERA\_SI\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

##### PE\_BANDERA\_TRUE\_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

##### PE\_BANDERA\_YES\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

##### PE\_BANDERA\_S\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

##### PE\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

#### PE\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

#### PE\_CM\_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### PE LENGUA\_LISTA **SECURE LOOKUP**

Replaces the value with one of the 18 values in `pe-lenguas.txt`, chosen by a keyed hash.

#### EXAMPLES

##### VALUE

##### MASKED

Quechua chanka

→ Shipibo-Konibo

## 10.4 Nationality

PE\_L3\_NACIONALIDAD

#### COLUMN NAME

#### KNOWN VALUES

#### SECURE LOOKUP

#### WHAT IT IS

Nationality, country of origin and migratory status. The law does not list it, but it sits beside the racial and ethnic origin it names, and in Peru it separates the nationals from the Venezuelan community.

#### HOW IT IS FOUND

##### COLUMN NAME

PE\_L3\_NACIONALIDAD - Path

85%

nacionalidad pa[ií]s\_?(de\_)?(nacimiento|origen|nacionalidad|procedencia) ciudadan[ií]a  
nationality citizenship country\_of\_birth condici[oó]n\_?(de\_)?(migratoria|refugiado)  
persona\_refugiada migrante

##### KNOWN VALUES

PE\_L3\_NACIONALIDAD - List

##### LIST

##### VALUES

##### STRENGTH

##### FIRST VALUES

pe-detectar-

41

70%

nacionalidades.txt

Peruana, Venezolana, Colombiana, Ecuatoriana, Boliviana, Chilena

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

Replaced by a nationality from the list: the Peruvian one and the largest foreign communities.

#### PE\_NACIONALIDAD **SECURE LOOKUP**

Replaces the value with one of the 14 values in `pe-nacionalidades.txt`, chosen by a keyed hash.

#### EXAMPLES

##### VALUE

##### MASKED

Panameña

→ Japonesa

## 10.5 Religion

PE\_L3\_RELIGION

#### COLUMN NAME

#### KNOWN VALUES

#### REGEX DECOMPOSE

#### WHAT IT IS

The religion, church or congregation the person belongs to.

HOW IT IS FOUND

COLUMN NAME PE\_L3\_RELIGION - Path

95% religi[oó]n creencia\_?(religiosa|espiritual) convicci[oó]n\_?(religiosa|espiritual) credo  
confesi[oó]n\_?religiosa culto iglesia(\_?(a\_?la\_?que\_?pertenece|nombre))?  
denominaci[oó]n\_?religiosa (cod|codigo|tipo)\_?religion religion church

KNOWN VALUES PE\_L3\_RELIGION - List

| LIST                       | VALUES | STRENGTH | FIRST VALUES                                                      |
|----------------------------|--------|----------|-------------------------------------------------------------------|
| pe-detectar-religiones.txt | 37     | 90%      | Católica, Catolica, Evangélica, Evangelica, Cristiana, Adventista |

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced by a religion from the list.

PE\_RELIGION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_RELIGION\_LISTA .

PE\_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.

PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.

PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.

PE\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.

PE\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.

PE\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-0-1.txt , chosen by a keyed hash.

PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PE\_RELIGION\_LISTA SECURE LOOKUP

Replaces the value with one of the 11 values in pe-religiones.txt , chosen by a keyed hash.

EXAMPLES

| VALUE    | MASKED      |
|----------|-------------|
| Ortodoxa | → Cristiana |

10.6 Philosophical and moral convictions

PE\_L3\_CONVICCION\_FILOSOFICA

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

The philosophical and moral convictions article 2, 5 names beside the religious ones, conscientious objection and belonging to a lodge.

HOW IT IS FOUND

| COLUMN NAME | PE_L3_CONVICCION_FILOSOFICA - Path                                                                                                                                                                                                                                              |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 90%         | convicci[oó]n(es)?_(filos[oó]ficas? morales? espirituales? personales?) creencias?_<br>(filos[oó]ficas? morales? personales?) opini[oó]n?(filos[oó]fica moral) objeci[oó]n?<br>(de_)?conciencia objetor?(de_)?conciencia logia masoner[ii]a beliefs?<br>philosophical_?beliefs? |

HOW IT IS MASKED

The values share no vocabulary: they are suppressed to Sin información . Flags and codes keep their shape.

PE\_CATEGORIA\_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE\_SUPRIMIR .

| PE_BANDERA REGEX DECOMPOSE                                                                          |
|-----------------------------------------------------------------------------------------------------|
| Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep. |
| PE_BANDERA_SI_NO SECURE LOOKUP                                                                      |
| Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.       |
| PE_BANDERA_TRUE_FALSE SECURE LOOKUP                                                                 |
| Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.  |
| PE_BANDERA_YES_NO SECURE LOOKUP                                                                     |
| Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.      |
| PE_BANDERA_S_N SECURE LOOKUP                                                                        |
| Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.         |
| PE_BANDERA_Y_N SECURE LOOKUP                                                                        |
| Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.         |
| PE_BANDERA_0_1 SECURE LOOKUP                                                                        |
| Replaces the value with one of the 2 values in pe-bandera-0-1.txt , chosen by a keyed hash.         |
| PE_CM_ALFANUM CHARACTER MAPPING                                                                     |
| Maps each character to another of its group (7 groups), keeping length and format.                  |
| PE_SUPRIMIR SECURE LOOKUP                                                                           |
| Replaces the value with one of the 1 values in pe-sin-informacion.txt , chosen by a keyed hash.     |

EXAMPLES

| VALUE                    | MASKED            |
|--------------------------|-------------------|
| Se niega a transfusiones | → Sin información |

10.7 Political opinion

PE\_L3\_OPINION\_POLITICA

| COLUMN NAME | KNOWN VALUES | REGEX DECOMPOSE |
|-------------|--------------|-----------------|
|-------------|--------------|-----------------|

WHAT IT IS

The political opinion article 2, 5 names: leaning, sympathy and voting intention, from surveys and records.

HOW IT IS FOUND

| COLUMN NAME | PE_L3_OPINION_POLITICA - Path                                                                                                                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 90%         | ideolog[ii]a(_?pol[ii]tica)? opini[oó]n?pol[ii]tica orientaci[oó]n?pol[ii]tica<br>tendencia?pol[ii]tica preferencia?(pol[ii]tica partidaria electoral) intenci[oó]n?<br>(de_)?voto simpat[ii]a?pol[ii]tica political?(opinion orientation view) voting_<br>intention |

KNOWN VALUES PE\_L3\_OPINION\_POLITICA - List

| LIST                                 | VALUES | STRENGTH | FIRST VALUES                                                              |
|--------------------------------------|--------|----------|---------------------------------------------------------------------------|
| pe-detectar-ideologias-politicas.txt | 16     | 70%      | Izquierda, Centroizquierda, Centro, Centroderecha, Derecha, Independiente |

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by a position from the list — left to right, independent.

PE\_IDEOLOGIA\_POLITICA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_IDEOLOGIA\_POLITICA\_LISTA .

PE\_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.

PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.

PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.

PE\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.

PE\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.

PE\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-0-1.txt , chosen by a keyed hash.

PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PE\_IDEOLOGIA\_POLITICA\_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in pe-ideologias-politicas.txt , chosen by a keyed hash.

EXAMPLES

| VALUE       | MASKED   |
|-------------|----------|
| Oficialista | → Centro |

## 10.8 Party membership

PE\_L3\_AFILIACION\_PARTIDARIA

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

The political party the person belongs to, from the Registro de Organizaciones Políticas of the JNE.

HOW IT IS FOUND

COLUMN NAME PE\_L3\_AFILIACION\_PARTIDARIA - Path

95% partido(?:pol[í]tico)?(?:afiliaci[oó]n|nombre)? organizaci[oó]n?pol[í]tica  
afiliaci[oó]n?(pol[í]tica|partidaria|partido) afiliado?partido militancia(?:  
pol[í]tica)? militante (cod|codigo|nombre|nom)?partido political?party party?  
membership

## KNOWN VALUES PE\_L3\_AFILIACION\_PARTIDARIA - List

| LIST                     | VALUES | STRENGTH | FIRST VALUES                                                                                     |
|--------------------------|--------|----------|--------------------------------------------------------------------------------------------------|
| pe-detectar-partidos.txt | 26     | 80%      | Fuerza Popular, Perú Libre, Peru Libre, Acción Popular, Accion Popular, Alianza para el Progreso |

A value that matches none counts 40% against the domain.

### HOW IT IS MASKED

Replaced by another party from the list; a flag stays a flag.

#### PE\_PARTIDO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE\_PARTIDO\_LISTA .

##### PE\_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

###### PE\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.

###### PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.

###### PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.

###### PE\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.

###### PE\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.

###### PE\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-0-1.txt , chosen by a keyed hash.

##### PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

##### PE\_PARTIDO\_LISTA SECURE LOOKUP

Replaces the value with one of the 13 values in pe-partidos.txt , chosen by a keyed hash.

### EXAMPLES

| VALUE         |   | MASKED     |
|---------------|---|------------|
| Frente Amplio | → | Perú Libre |
| S             | → | S          |

## 10.9 Trade union membership

PE\_L3\_AFILIACION\_SINDICAL

### COLUMN NAME

### KNOWN VALUES

### REGEX DECOMPOSE

### WHAT IT IS

The union, membership, the union dues deducted on the payroll and union protection. Article 2, 5 names it outright.

### HOW IT IS FOUND

#### COLUMN NAME PE\_L3\_AFILIACION\_SINDICAL - Path

```
95% sindicato(_?(nombre|afiliado|afiliaci[oó]n|codigo))? sindicalizado afiliado?sindicato
afiliaci[oó]n?sindical cuota?sindical descuento?sindical aporte?sindical
asociaci[oó]n?sindical convenio?colectivo fuero?sindical dirigente?sindical trade?
union union_?member(ship)?
```

#### KNOWN VALUES PE\_L3\_AFILIACION\_SINDICAL – List

| LIST                       | VALUES | STRENGTH | FIRST VALUES                               |
|----------------------------|--------|----------|--------------------------------------------|
| pe-detectar-sindicatos.txt | 13     | 60%      | CGTP, CUT Perú, CUT Peru, CTP, CATP, SUTEP |

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

A flag stays a flag; any other value becomes Sin información .

#### PE\_CATEGORIA\_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE\_SUPRIMIR .

##### PE\_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

##### PE\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.

##### PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.

##### PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.

##### PE\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.

##### PE\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.

##### PE\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-0-1.txt , chosen by a keyed hash.

##### PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

##### PE\_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in pe-sin-informacion.txt , chosen by a keyed hash.

#### EXAMPLES

| VALUE |   | MASKED          |
|-------|---|-----------------|
| SUTEP | → | Sin información |
| S     | → | S               |

## 10.10 Biometric data

PE\_L3\_BIOMETRICO

#### COLUMN NAME DATA TYPE REGEX DECOMPOSE

#### WHAT IT IS

Fingerprint templates — the DNI's fingerprint is the country's identification pattern —, face, iris, voice and digitized signature. Article 2, 5 makes sensitive the biometric data that **by itself** can identify the subject.

HOW IT IS FOUND

COLUMN NAME PE\_L3\_BIOMETRICO - Path

90% biometr[ií]a[a-z]\* biom[eé]trico[a-z]\* huella(\_?(dactilar|digital))?( \_?(template|plantilla|imagen|hash))? huellas dactilar template?(facial|huella|biometrico) plantilla?(facial|biom[eé]trica) rostro?(hash|template) reconocimiento?facial firma?(biom[eé]trica|digitalizada|electr[oó]nica) iris voz?(biom[eé]trica|template) fingerprints? face?(template|hash|encoding) biometric[a-z]\*

DATA TYPE PE\_L3\_BIOMETRICO - Type

Accepts text; any other type rules the domain out.

HOW IT IS MASKED

Every character becomes x . A masked template authenticates nobody.

PE\_REDACTAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

| VALUE        |   | MASKED     |
|--------------|---|------------|
| A1B2C3D4E5F6 | → | XXXXXXXXXX |

10.11 Genetic data

PE\_L3\_GENETICO

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Results of genetic and DNA tests, paternity tests, genetic profiles, mutations, karyotype and newborn screening.

HOW IT IS FOUND

COLUMN NAME PE\_L3\_GENETICO - Path

90% adn dna gen[eé]tic[oa]s? datos?\_?gen[eé]ticos? genoma genotip[a-z]\* prueba?(gen[eé]tica|de?adn|de?paternidad|de?filiaci[oó]n) perfil?gen[eé]tico mutaci[oó]n brca[12]? cariotipo tamizaje?neonatal genetic[a-z]\*

HOW IT IS MASKED

Every character becomes x .

PE\_REDACTAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

| VALUE          |   | MASKED       |
|----------------|---|--------------|
| BRCA1 positivo | → | XXXXXXXXXXXX |

COLUMN NAME    KNOWN VALUES    REGEX DECOMPOSE

WHAT IT IS

Self-declared sexual orientation or preference, in health and diversity records. It belongs to the *vida sexual* article 2, 5 protects.

HOW IT IS FOUND

COLUMN NAME

PE\_L3\_ORIENTACION\_SEXUAL - Path

95%

orientaci[oó]n\_?sexual

preferencias?\_?sexuales?

(cod|codigo|tipo)\_?orientacion\_?sexual

sexual\_?orientation

KNOWN VALUES

PE\_L3\_ORIENTACION\_SEXUAL - List

| LIST                                   | VALUES | STRENGTH | FIRST VALUES                                              |
|----------------------------------------|--------|----------|-----------------------------------------------------------|
| pe-detectar-orientaciones-sexuales.txt | 13     | 80%      | Heterosexual, Gay, Lesbiana, Bisexual, Pansexual, Asexual |

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by a category from the list.

PE\_ORIENTACION\_SEXUAL    REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_ORIENTACION\_SEXUAL\_LISTA .

PE\_BANDERA    REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_BANDERA\_SI\_NO    SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt` , chosen by a keyed hash.

PE\_BANDERA\_TRUE\_FALSE    SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt` , chosen by a keyed hash.

PE\_BANDERA\_YES\_NO    SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt` , chosen by a keyed hash.

PE\_BANDERA\_S\_N    SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt` , chosen by a keyed hash.

PE\_BANDERA\_Y\_N    SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt` , chosen by a keyed hash.

PE\_BANDERA\_0\_1    SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt` , chosen by a keyed hash.

PE\_CM\_ALFANUM    CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PE\_ORIENTACION\_SEXUAL\_LISTA    SECURE LOOKUP

Replaces the value with one of the 8 values in `pe-orientaciones-sexuales.txt` , chosen by a keyed hash.

EXAMPLES

| VALUE      |   | MASKED    |
|------------|---|-----------|
| Homosexual | → | Pansexual |

## 10.13 Sexual life

PE\_L3\_VIDA\_SEXUAL

COLUMN NAME

REGEX DECOMPOSE

### WHAT IT IS

Sexual activity and behaviour, number of partners, age at first intercourse — named word for word in article 2, 5.

### HOW IT IS FOUND

COLUMN NAME

PE\_L3\_VIDA\_SEXUAL – Path

90%

vida\_?sexual actividad\_?sexual conducta\_?sexual pr[aá]cticas\_?sexuales? parejas\_?sexuales? n(o|ro|um|umero)\_?parejas sexualmente\_?activ[oa] inicio\_?(de\_?)(la\_?)?vida\_?sexual relaciones\_?sexuales sex(ual)?\_?life sexual?(activity|behavio(u)?r|partners)

### HOW IT IS MASKED

A flag stays a flag; any other value becomes Sin información .

PE\_CATEGORIA\_SUPRIMIDA

REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE\_SUPRIMIR .

PE\_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.

PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.

PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.

PE\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.

PE\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.

PE\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-0-1.txt , chosen by a keyed hash.

PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PE\_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in pe-sin-informacion.txt , chosen by a keyed hash.

### EXAMPLES

VALUE

MASKED

Activa

→

Sin información

## 10.14 Gender identity

PE\_L3\_IDENTIDAD\_GENERO

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

### WHAT IT IS

Self-declared gender identity and the social name. The law does not name it, but it travels with the sexual life it does name.

HOW IT IS FOUND

COLUMN NAME

PE\_L3\_IDENTIDAD\_GENERO – Path

95%

identidad\_(de\_)?g[eé]nero (cod|codigo|tipo)\_?identidad\_?genero g[eé]nero\_?  
(autopercebido|identitario) expresi[oó]n\_?de\_?g[eé]nero transg[eé]nero persona\_?trans  
nombre\_?social pronombres? gender\_?identity

KNOWN VALUES

PE\_L3\_IDENTIDAD\_GENERO – List

| LIST                                   | VALUES | STRENGTH | FIRST VALUES                                                                                       |
|----------------------------------------|--------|----------|----------------------------------------------------------------------------------------------------|
| pe-detectar-identidades-<br>genero.txt | 19     | 70%      | Mujer cisgénero, Mujer cisgenero, Hombre cisgénero, Hombre<br>cisgenero, Mujer trans, Hombre trans |

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by a category from the list.

PE\_IDENTIDAD\_GENERO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_IDENTIDAD\_GENERO\_LISTA .

PE\_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.

PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.

PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.

PE\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.

PE\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.

PE\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-0-1.txt , chosen by a keyed hash.

PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PE\_IDENTIDAD\_GENERO\_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in pe-identidades-genero.txt , chosen by a keyed hash.

EXAMPLES

| VALUE       | MASKED         |
|-------------|----------------|
| Transgénero | → Hombre trans |

10.15 Victims

PE\_L3\_VICTIMA

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

The Registro Único de Víctimas of the Ley 28592 records the people harmed by the violence of 1980 to 2000; the Ley 30364 protects the identity of the victims of violence against women and members of the family group.

HOW IT IS FOUND

| COLUMN NAME | PE_L3_VICTIMA – Path                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 90%         | v[í]ctima(_?(de_)?(violencia(_?(de_)?<br>g[eé]nero familiar sexual econ[ó]mica psicol[ó]gica))?<br> delito trata desplazamiento terrorismo abuso))?) ruv registro_[uú]nico_(de_)?<br>v[í]ctimas violencia_(de_)?g[eé]nero violencia_<br>(familiar dom[eé]stica sexual psicol[ó]gica) tipo_(de_)?violencia medida_(de_)?<br>protecci[ó]n ficha_de_valoraci[ó]n_(de_)?riesgo desplazamiento_forzado trata_<br>(de_)?personas denunciante_violencia cem |

HOW IT IS MASKED

A flag stays a flag; any other value becomes Sin información .

PE\_CATEGORIA\_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE\_SUPRIMIR .

|                                                                                                     |
|-----------------------------------------------------------------------------------------------------|
| PE_BANDERA REGEX DECOMPOSE                                                                          |
| Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep. |
| PE_BANDERA_SI_NO SECURE LOOKUP                                                                      |
| Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.       |
| PE_BANDERA_TRUE_FALSE SECURE LOOKUP                                                                 |
| Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.  |
| PE_BANDERA_YES_NO SECURE LOOKUP                                                                     |
| Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.      |
| PE_BANDERA_S_N SECURE LOOKUP                                                                        |
| Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.         |
| PE_BANDERA_Y_N SECURE LOOKUP                                                                        |
| Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.         |
| PE_BANDERA_0_1 SECURE LOOKUP                                                                        |
| Replaces the value with one of the 2 values in pe-bandera-0-1.txt , chosen by a keyed hash.         |
| PE_CM_ALFANUM CHARACTER MAPPING                                                                     |
| Maps each character to another of its group (7 groups), keeping length and format.                  |
| PE_SUPRIMIR SECURE LOOKUP                                                                           |
| Replaces the value with one of the 1 values in pe-sin-informacion.txt , chosen by a keyed hash.     |

EXAMPLES

| VALUE              |   | MASKED          |
|--------------------|---|-----------------|
| Violencia familiar | → | Sin información |
| S                  | → | S               |

# Health data

Article 2, 5 makes *information related to health* sensitive without qualification, and the Ley 26842 General de Salud adds medical secrecy on top. The group keeps the clinical record apart from the rest so that a health database can be masked whole.

## 11.1 Health identifiers

PE\_SALUD\_IDENTIFICADOR

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Clinical record, admission, care, appointment, prescription and FUA numbers, and the RENHICE identifier of the electronic record.

HOW IT IS FOUND

| COLUMN NAME | PE_SALUD_IDENTIFICADOR – Path                                                                                                                                                                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 85%         | <code>n(o ro um umero)?(historia(_?cl[ií]nica)? hc hce expediente(_?cl[ií]nico)? ingreso atenci[oó]n consulta cita referencia_?m[eé]dica receta fua) historia_?cl[ií]nica hc_?(n(o ro um umero))? hce renhice n(o ro um umero)?(asegurado afiliado_?(sis essalud)) c[oó]digo?(paciente historia atenci[oó]n) id_?paciente</code> |
| 55%         | <code>expediente atenci[oó]n ingreso consulta</code>                                                                                                                                                                                                                                                                             |

HOW IT IS MASKED

Character by character, keeping length and format.

PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE         | MASKED          |
|---------------|-----------------|
| HC-2024-00123 | → TW-3389-01317 |

## 11.2 Health insurance

PE\_SALUD\_COBERTURA

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

EsSalud, the SIS, an EPS, the health service of the armed forces or the police, or a private insurer, by name or code. Which one a person has says a great deal about them.

HOW IT IS FOUND

| COLUMN NAME | PE_SALUD_COBERTURA – Path                                                                                                                                                                                                                                                                                                    |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 85%         | <code>cobertura(_?(m[eé]dica de_?salud social))? seguro?(m[eé]dico salud integral) r[eé]gimen(_?(de_)?salud)? tipo?(de_)?seguro proveedor?(de_)?salud essalud(_?(afiliado cobertura))? sis(_?(afiliado tipo))? eps iafas sanidad?(pnp ffaa militar policial) aseguradora?(m[eé]dica salud) plan?(m[eé]dico de_?salud)</code> |

#### KNOWN VALUES PE\_SALUD\_COBERTURA – List

| LIST                      | VALUES | STRENGTH | FIRST VALUES                                                                                                                                 |
|---------------------------|--------|----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| pe-detectar-cobertura.txt | 16     | 80%      | EsSalud, SIS (Seguro Integral de Salud), EPS, Sanidad de las Fuerzas Armadas, Sanidad de la Policía Nacional, Sanidad de la Policía Nacional |

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

A name is replaced by another cover from the list; a code is mapped digit by digit; a flag stays a flag.

#### PE\_COBERTURA\_SALUD REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE\_COBERTURA .

##### PE\_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

##### PE\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.

##### PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.

##### PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.

##### PE\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.

##### PE\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.

##### PE\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-0-1.txt , chosen by a keyed hash.

##### PE\_CM\_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

##### PE\_COBERTURA SECURE LOOKUP

Replaces the value with one of the 8 values in pe-cobertura-salud.txt , chosen by a keyed hash.

#### EXAMPLES

| VALUE   |   | MASKED                  |
|---------|---|-------------------------|
| EsSalud | → | Seguro privado de salud |
| 12345   | → | 31564                   |

## 11.3 Diagnosis and ICD-10

PE\_SALUD\_DIAGNOSTICO

#### COLUMN NAME

#### VALUE FORMAT

#### REGEX DECOMPOSE

#### WHAT IT IS

Diagnoses in words or by ICD-10 code, with a dot ( F33.1 ) or without ( F331 ); causes of death, of sick leave and of admission; allergies and comorbidities.

HOW IT IS FOUND

COLUMN NAME PE\_SALUD\_DIAGNOSTICO – Path

85%     `diagn[oó]stico[a-z]*   dx(?:principal|secundario|ingreso|egreso)[0-9]?   cie?10   cie(?:`  
`(10|11|principal))?`   `(cod|codigo)?(diagnostico|dx|cie)`   `impresi[oó]n?diagn[oó]stica`  
`enfermedad(es)?[a-z]*   patolog[ií]a   comorbilidad(es)?   antecedentes?_?`  
`(m[eé]dicos|patol[oó]gicos|cl[ií]nicos|personales)`   `causa?(de_)?`  
`(muerte|defunci[oó]n|incapacidad|hospitalizaci[oó]n|consulta)`   `motivo?(de_)?`  
`(consulta|descanso|hospitalizaci[oó]n)`   `alergias?   diagnos(is|es)`   `icd?(10|11)?`

VALUE FORMAT PE\_SALUD\_DIAGNOSTICO – Regex

| PATTERN                                           | STRENGTH | NOTES |
|---------------------------------------------------|----------|-------|
| <code>[A-TV-Za-tv-z]\d{2}(\.\d{1,2} \d X)?</code> |          | 50%   |

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

An ICD-10 code becomes another common code in the same format, and text a common diagnosis: an HIV or mental disorder diagnosis does not survive as such.

PE\_DIAGNOSTICO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches:

PE\_DIAGNOSTICO\_TEXT0 .

PE\_CIE10\_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in `pe-cie10-decimal.txt` , chosen by a keyed hash.

PE\_CIE10 SECURE LOOKUP

Replaces the value with one of the 37 values in `pe-cie10.txt` , chosen by a keyed hash.

PE\_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt` , chosen by a keyed hash.

PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt` , chosen by a keyed hash.

PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt` , chosen by a keyed hash.

PE\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt` , chosen by a keyed hash.

PE\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt` , chosen by a keyed hash.

PE\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt` , chosen by a keyed hash.

PE\_DIAGNOSTICO\_TEXT0 SECURE LOOKUP

Replaces the value with one of the 30 values in `pe-diagnostics.txt` , chosen by a keyed hash.

EXAMPLES

| VALUE |   | MASKED |
|-------|---|--------|
| F33.1 | → | K21.9  |
| B24X  | → | R51    |

# 11.4 Procedures

PE\_SALUD\_PROCEDIMIENTO

COLUMN NAME    REGEX DECOMPOSE

## WHAT IT IS

Procedures, tests and surgeries, in words or by code. A viral load or a pregnancy test reveals the diagnosis.

## HOW IT IS FOUND

COLUMN NAME    PE\_SALUD\_PROCEDIMIENTO - Path

85%    procedimiento(?:m[eé]dico|quir[uú]rgico|realizado|autorizado))?    (cod|codigo)?

          (procedimiento|prestaci[oó]n|servicio\_?salud|cpt)    cpt    prestaci[oó]n?(m[eé]dica|de\_?salud)

          examen?(ordenado|realizado)    cirug[ií]a(?:realizada)?    intervenci[oó]n\_?quir[uú]rgica

          medical\_?procedure

## HOW IT IS MASKED

A code is mapped digit by digit; text is replaced by a common procedure.

PE\_PROCEDIMIENTO    REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches:

PE\_PROCEDIMIENTO\_TEXTO .

PE\_CM\_DIGITOS    CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

PE\_PROCEDIMIENTO\_TEXTO    SECURE LOOKUP

Replaces the value with one of the 10 values in pe-procedimientos.txt , chosen by a keyed hash.

## EXAMPLES

| VALUE  |   | MASKED |
|--------|---|--------|
| 901234 | → | 928120 |

# 11.5 Medicines

PE\_SALUD\_MEDICAMENTO

COLUMN NAME    KNOWN VALUES    REGEX DECOMPOSE

## WHAT IT IS

Prescribed or dispensed medicines, by name or active ingredient. An antiretroviral, an antipsychotic or a contraceptive says what the diagnosis column was masked for.

## HOW IT IS FOUND

COLUMN NAME    PE\_SALUD\_MEDICAMENTO - Path

85%    medicamentos(?:recetado|prescrito|entregado|despachado|nombre|uso))?    f[aá]rmacos?

          principio\_?activo    denominaci[oó]n\_?com[uú]n\_?internacional    dci    receta(?:m[eé]dica)?

          prescripci[oó]n(?:medicamento)?    posolog[ií]a    tratamiento\_?farmacol[oó]gico    (cod|codigo)?

          medicamento    petitorio?(nacional\_?)[uú]nico    medications?    drugs?\_?prescribed

KNOWN VALUES    PE\_SALUD\_MEDICAMENTO - List

| LIST                             | VALUES | STRENGTH | FIRST VALUES                                                              |
|----------------------------------|--------|----------|---------------------------------------------------------------------------|
| pe-detectar-<br>medicamentos.txt | 50     | 70%      | Paracetamol, Ibuprofeno, Diclofenaco, Metformina, Glibenclamida, Losartán |

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Replaced by a common medicine from the list; a flag stays a flag.

PE\_MEDICAMENTO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_MEDICAMENTO\_LISTA .

PE\_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt` , chosen by a keyed hash.

PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt` , chosen by a keyed hash.

PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt` , chosen by a keyed hash.

PE\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt` , chosen by a keyed hash.

PE\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt` , chosen by a keyed hash.

PE\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt` , chosen by a keyed hash.

PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PE\_MEDICAMENTO\_LISTA SECURE LOOKUP

Replaces the value with one of the 22 values in `pe-medicamentos.txt` , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Sertralina 50 mg

→ Complejo B

11.6 Clinical text

PE\_SALUD\_TEXT0\_CLINICO

COLUMN NAME

DATA TYPE

FREE TEXT REDACTION

WHAT IT IS

History, present illness, progress notes, physical examination, treatment plan, discharge summaries and nursing notes.

HOW IT IS FOUND

COLUMN NAME

PE\_SALUD\_TEXT0\_CLINICO - Path

85%

anamnesis evoluci[oó]n(\_?(m[eé]dica|cl[ií]nica|enfermer[ií]a))? enfermedad\_?actual  
examen\_?f[ií]sico plan\_?(de\_)?(manejo|tratamiento) indicaciones\_?m[eé]dicas conducta(\_?  
m[eé]dica)? epicrisis resumen\_?(de\_)?(alta|egreso|atenci[oó]n) nota\_?(m[eé]dica|de\_?  
enfermer[ií]a|cl[ií]nica) notas?\_?(m[eé]dicas|cl[ií]nicas) triaje(\_?texto)? informe\_?  
(patolog[ií]a|radiolog[ií]a|m[eé]dico) clinical\_?notes? discharge\_?summary

DATA TYPE

PE\_SALUD\_TEXT0\_CLINICO - Type

Accepts text (length 30 or more); any other type rules the domain out.

#### HOW IT IS MASKED

The same Free Text Redaction as free text: documents, contacts and names from the list are redacted. The clinical content stays, because that is what makes the copy useful.

#### PE\_TEXTO\_LIBRE FREE TEXT REDACTION

Replaces the text matching 8 patterns with [DATO], and the 1,906 terms of pe-texto-nombres.txt with [NOMBRE].

#### EXAMPLES

| VALUE                                                |   | MASKED                                               |
|------------------------------------------------------|---|------------------------------------------------------|
| Paciente Quispe, DNI 45678901, consulta por cefalea. | → | Paciente [NOMBRE], DNI [DATO], consulta por cefalea. |

## 11.7 Test results

PE\_SALUD\_RESULTADO\_EXAMEN

#### COLUMN NAME CHARACTER MAPPING

#### WHAT IT IS

Laboratory and test results — glucose, pregnancy, toxicology, blood alcohol —, also as numeric values.

#### HOW IT IS FOUND

#### COLUMN NAME PE\_SALUD\_RESULTADO\_EXAMEN - Path

|     |                                                                                                                                                                                                                                                                                                                                                |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 80% | resultado_(del_)?<br>(examen prueba laboratorio pcr serolog[ií]a biopsia glucosa citolog[ií]a papanicolaou)<br>(examen prueba)_?(resultado laboratorio) glucemia hemoglobina(_?glicosilada)? hba1c<br>colesterol imc presi[oó]n_?arterial prueba_(de_)?embarazo toxicol[oó]gico dosaje_<br>et[ií]lico alcoholemia lab_?results? test_?results? |
| 50% | resultado prueba                                                                                                                                                                                                                                                                                                                               |

#### HOW IT IS MASKED

Character by character, keeping the format: a number stays a number.

#### PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE    |   | MASKED   |
|----------|---|----------|
| Reactivo | → | Quilqaju |

## 11.8 HIV and sexually transmitted infections

PE\_SALUD\_VIH

#### COLUMN NAME REGEX DECOMPOSE

#### WHAT IT IS

Serological status, viral load and antiretroviral treatment, hepatitis, tuberculosis and sexually transmitted infections. The Ley 26626 orders the result of an HIV test to be confidential and forbids demanding it to get or keep a job.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_SALUD\_VIH – Path

**90%** vih(\_?(estado|resultado|prueba|diagn[ó]stico|positivo))? hiv sida aids serolog[ií]a?  
(vih|hiv) estado?serol[ó]gico carga?viral cd4 targa tarv antirretroviral(es)?  
its ets infecci[ó]n(es)?\_(de\_)?transmisi[ó]n?sexual s[ií]filis hepatitis?[bc]  
tuberculosis tbc

#### HOW IT IS MASKED

A flag stays a flag; any other value becomes Sin información .

**PE\_CATEGORIA\_SUPRIMIDA** **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE\_SUPRIMIR .

**PE\_BANDERA** **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

**PE\_BANDERA\_SI\_NO** **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.

**PE\_BANDERA\_TRUE\_FALSE** **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.

**PE\_BANDERA\_YES\_NO** **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.

**PE\_BANDERA\_S\_N** **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.

**PE\_BANDERA\_Y\_N** **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.

**PE\_BANDERA\_0\_1** **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-0-1.txt , chosen by a keyed hash.

**PE\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

**PE\_SUPRIMIR** **SECURE LOOKUP**

Replaces the value with one of the 1 values in pe-sin-informacion.txt , chosen by a keyed hash.

#### EXAMPLES

| VALUE    |   | MASKED          |
|----------|---|-----------------|
| Reactivo | → | Sin información |
| S        | → | S               |

## 11.9 Blood group and Rh

PE\_SALUD\_GRUPO\_SANGUINEO

**COLUMN NAME** **KNOWN VALUES** **SECURE LOOKUP**

#### WHAT IT IS

Blood group and Rh factor.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_SALUD\_GRUPO\_SANGUINEO – Path

**90%** grupo?sangu[ií]neo tipo?(de\_)?sangre rh factor?rh gs?rh grupo?rh abo(\_?rh)?  
blood?(type|group)

**KNOWN VALUES** PE\_SALUD\_GRUPO\_SANGUINEO - List

| LIST                              | VALUES | STRENGTH | FIRST VALUES             |
|-----------------------------------|--------|----------|--------------------------|
| pe-detectar-grupos-sanguineos.txt | 18     | 90%      | A+, A-, B+, B-, AB+, AB- |

A value that matches none counts 40% against the domain.

#### HOW IT IS MASKED

Replaced by another group from the list.

**PE\_GRUPO\_SANGUINEO** **SECURE LOOKUP**

Replaces the value with one of the 8 values in `pe-grupos-sanguineos.txt`, chosen by a keyed hash.

#### EXAMPLES

| VALUE | MASKED |
|-------|--------|
| 0+    | → A-   |

## 11.10 Disability

PE\_SALUD\_DISCAPACIDAD

**COLUMN NAME**

**KNOWN VALUES**

**REGEX DECOMPOSE**

#### WHAT IT IS

Disability status and type, the CONADIS registration and the disability certificate, under the Ley 29973.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_SALUD\_DISCAPACIDAD - Path

**90%** `discapacidad[a-z]* (tipo|cod|codigo|categoria)?discapacidad persona?con?discapacidad pcd conadis carn[eé]?(de_)?(conadis|discapacidad) certificado?(de_)?discapacidad condici[oó]n?(de_)?discapacidad necesidades?(educativas_)?especiales movilidad? reducida disabilit(y|ies)`

**KNOWN VALUES** PE\_SALUD\_DISCAPACIDAD - List

| LIST                           | VALUES | STRENGTH | FIRST VALUES                                                                  |
|--------------------------------|--------|----------|-------------------------------------------------------------------------------|
| pe-detectar-discapacidades.txt | 26     | 60%      | Física o motora, Fisica o motora, Auditiva, Visual, Sordoceguera, Intelectual |

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

A flag stays a flag and any other value is replaced by a category from the list.

**PE\_DISCAPACIDAD** **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

**PE\_DISCAPACIDAD\_LISTA**.

**PE\_BANDERA** **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

**PE\_BANDERA\_SI\_NO** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

**PE\_BANDERA\_TRUE\_FALSE** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

**PE\_BANDERA\_YES\_NO** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

**PE\_BANDERA\_S\_N** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt` , chosen by a keyed hash.

**PE\_BANDERA\_Y\_N** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt` , chosen by a keyed hash.

**PE\_BANDERA\_0\_1** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt` , chosen by a keyed hash.

**PE\_CM\_ALFANUM** **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

**PE\_DISCAPACIDAD\_LISTA** **SECURE LOOKUP**

Replaces the value with one of the 8 values in `pe-discapacidades.txt` , chosen by a keyed hash.

#### EXAMPLES

| VALUE                          | MASKED    |
|--------------------------------|-----------|
| Trastorno del espectro autista | → Ninguna |

## 11.11 Occupational health

PE\_SALUD\_OCUPACIONAL

**COLUMN NAME**

**KNOWN VALUES**

**REGEX DECOMPOSE**

#### WHAT IT IS

The fitness opinion of occupational medical exams (EMO), work accidents and occupational diseases, the SCTR, sick leave and absenteeism.

#### HOW IT IS FOUND

**COLUMN NAME**

PE\_SALUD\_OCUPACIONAL - Path

85%

```
aptitud_?(laboral|m[eé]dica)  concepto_?(de_)?aptitud  examen_?(m[eé]dico_)?
(ocupacional|preocupacional|de_ingreso|de_retiro|peri[oó]dico)  emo  accidente_?(de_)?
trabajo  enfermedad_?(laboral|profesional|ocupacional)  riesgo_?(profesional|laboral)  sctr
incapacidad(_?(m[eé]dica|laboral|d[ií]as))?  descanso_?m[eé]dico  d[ií]as_?(de_)?
(incapacidad|descanso)  licencia_?(m[eé]dica|por_?enfermedad|de_?maternidad)  ausentismo
comit[eé]_?(de_)?seguridad
```

**KNOWN VALUES**

PE\_SALUD\_OCUPACIONAL - List

**LIST**

**VALUES** **STRENGTH** **FIRST VALUES**

|                                 |   |     |                                                                            |
|---------------------------------|---|-----|----------------------------------------------------------------------------|
| pe-detectar-aptitud-laboral.txt | 5 | 60% | apto, no apto, apto con restricciones, pendiente, apto con recomendaciones |
|---------------------------------|---|-----|----------------------------------------------------------------------------|

A value that matches none counts 30% against the domain.

#### HOW IT IS MASKED

An opinion is replaced by another from the list — fit, unfit —; a flag stays a flag; the diagnosis falls under its own domain.

**PE\_APTITUD\_LABORAL** **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

**PE\_APTITUD\_LABORAL\_LISTA** .

**PE\_BANDERA** **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

**PE\_BANDERA\_SI\_NO** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt` , chosen by a keyed hash.

**PE\_BANDERA\_TRUE\_FALSE** **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt` , chosen by a keyed hash.

#### PE\_BANDERA\_YES\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera=yes-no.txt` , chosen by a keyed hash.

#### PE\_BANDERA\_S\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera=s-n.txt` , chosen by a keyed hash.

#### PE\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera=y-n.txt` , chosen by a keyed hash.

#### PE\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera=0-1.txt` , chosen by a keyed hash.

#### PE\_CM\_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### PE\_APTITUD\_LABORAL\_LISTA **SECURE LOOKUP**

Replaces the value with one of the 4 values in `pe-aptitud-laboral.txt` , chosen by a keyed hash.

#### EXAMPLES

##### VALUE

##### MASKED

No apto temporalmente

→ **Pendiente**

## 11.12 Sexual and reproductive health

PE\_SALUD\_REPRODUCTIVA

#### COLUMN NAME

#### REGEX DECOMPOSE

#### WHAT IT IS

Pregnancy, last menstrual period, prenatal care, childbirth, family planning, fertility.

#### HOW IT IS FOUND

##### COLUMN NAME

PE\_SALUD\_REPRODUCTIVA - Path

90%

```
embarazo embarazada gestante gestaci[oó]n edad_?gestacional semanas_?(de_)?  
gestaci[oó]n fum fecha_?[uú]ltima_?(regla|menstruaci[oó]n) control_?prenatal prenatal  
parto(_?tipo)? ces[aá]rea aborto m[eé]todo_?(de_)?planificaci[oó]n planificaci[oó]n_  
familiar anticoncepci[oó]n anticonceptivo fertilidad salud_?(sexual|reproductiva)  
pregnan(t|cy) contracepti(on|ve)
```

#### HOW IT IS MASKED

Flags keep their shape; any other value becomes `Sin información` .

#### PE\_CATEGORIA\_SUPRIMIDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_SUPRIMIR` .

##### PE\_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

##### PE\_BANDERA\_SI\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera=si-no.txt` , chosen by a keyed hash.

##### PE\_BANDERA\_TRUE\_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera=true-false.txt` , chosen by a keyed hash.

##### PE\_BANDERA\_YES\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera=yes-no.txt` , chosen by a keyed hash.

##### PE\_BANDERA\_S\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera=s-n.txt` , chosen by a keyed hash.

##### PE\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera=y-n.txt` , chosen by a keyed hash.

#### PE\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt` , chosen by a keyed hash.

#### PE\_CM\_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### PE\_SUPRIMIR **SECURE LOOKUP**

Replaces the value with one of the 1 values in `pe-sin-informacion.txt` , chosen by a keyed hash.

#### EXAMPLES

| VALUE                   |   | MASKED          |
|-------------------------|---|-----------------|
| 12 semanas de gestación | → | Sin información |
| S                       | → | S               |

## 11.13 Mental health

PE\_SALUD\_MENTAL

#### COLUMN NAME

#### REGEX DECOMPOSE

#### WHAT IT IS

Mental disorders, psychiatric or psychological care, suicide attempts and substance use.

#### HOW IT IS FOUND

#### COLUMN NAME

PE\_SALUD\_MENTAL - Path

90%

salud\_?mental trastorno[a-z]\* psiqui[aa]tri[a-z]\* psicol[oo]gi[a-z]\* depresi[oo]n  
ansiedad suicid[a-z]\* intento\_?(de\_)?suicidio autolesi[oo]n consumo\_?(de\_)?  
(sustancias|alcohol|drogas) adicci[oo]n(es)? alcoholismo tabaquismo mental\_?health

#### HOW IT IS MASKED

Flags keep their shape; any other value becomes `Sin información` .

#### PE\_CATEGORIA\_SUPRIMIDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_SUPRIMIR` .

#### PE\_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: `keep`.

#### PE\_BANDERA\_SI\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt` , chosen by a keyed hash.

#### PE\_BANDERA\_TRUE\_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt` , chosen by a keyed hash.

#### PE\_BANDERA\_YES\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt` , chosen by a keyed hash.

#### PE\_BANDERA\_S\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt` , chosen by a keyed hash.

#### PE\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt` , chosen by a keyed hash.

#### PE\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt` , chosen by a keyed hash.

#### PE\_CM\_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

#### PE\_SUPRIMIR **SECURE LOOKUP**

Replaces the value with one of the 1 values in `pe-sin-informacion.txt` , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Trastorno bipolar

→

Sin información

# Financial and socioeconomic data

**Economic income is sensitive data in Peru** (art. 2, 5) — no neighbouring law says so. The rest of the group is not sensitive by that article, but it identifies: a debt, an asset, a pension, a social programme or a SISFOH classification narrows a person down as fast as a postcode.

12.1

Income and pay

PE\_FIN\_INGRESOS

COLUMN NAME

DATA TYPE

NUMERIC MAPPING

WHAT IT IS

Salary, monthly or household income, fees and rents of every category. **Article 2, 5 lists the economic income among the sensitive data:** in Peru this column weighs the same as a diagnosis. The minimum wage was 1,130 soles a month from January 2025.

HOW IT IS FOUND

COLUMN NAME

PE\_FIN\_INGRESOS – Path

90%

sueldo(\_?(b[aá]sico|mensual|bruto|neto))? salario(\_?(b[aá]sico|mensual|bruto|neto))? remuneraci[oó]n(\_?(b[aá]sica|mensual|computable))? ingresos?(\_?(mensuales?|totales?|familiares?|declarados?|netos?|econ[oó]micos?))? haberes honorarios renta(\_?(bruta|neta|mensual|anual|de\_(primera|segunda|tercera|cuarta|quinta)\_?categor[ií]a))? ingreso?(\_?(base|familiar|percapita) salary income wages?

DATA TYPE

PE\_FIN\_INGRESOS – Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to a value between 1,025 and 25,000 soles, unrelated to the real one.

PE\_INGRESOS NUMERIC MAPPING

Maps the number to another between 1,025 and 25,000.

EXAMPLES

| VALUE |   | MASKED |
|-------|---|--------|
| 2800  | → | 3433   |

12.2

Credit history

PE\_FIN\_HISTORIAL\_CREDITICIO

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

The risk bureau report, the score, days in arrears, the SBS classification of the debtor — normal, con problema potencial, deficiente, dudoso, pérdida — and legal collection. The Ley 27489 governs the bureaus.

## HOW IT IS FOUND

**COLUMN NAME** PE\_FIN\_HISTORIAL\_CREDITICIO – Path

90% historial\_?(crediticio|de\_?cr[eé]dito) central\_?(de\_?)?riesgo infocorp sentinel  
equifax experian xchange score(\_?(crediticio|interno))? puntaje\_?(de\_?)?cr[eé]dito  
clasificaci[oó]n\_?(sbs|crediticia|del\_?deudor) calificaci[oó]n\_?(de\_?)?  
(riesgo|cr[eé]dito|cartera) categor[ií]a\_?(de\_?)?riesgo d[ií]as\_?(de\_?)?(mora|atraso)  
estado\_?(de\_?)?(cr[eé]dito|cartera|obligaci[oó]n) morosidad moroso cobranza\_?judicial  
credit\_?score credit\_?rating

## HOW IT IS MASKED

A score gets other digits of the same length; a flag stays a flag; a status is replaced by another from the list.

PE\_HISTORIAL\_CREDITICIO **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_ESTADO\_CREDITO .

PE\_CM\_DIGITOS **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

PE\_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_BANDERA\_SI\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.

PE\_BANDERA\_TRUE\_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.

PE\_BANDERA\_YES\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.

PE\_BANDERA\_S\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.

PE\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.

PE\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in pe-bandera-0-1.txt , chosen by a keyed hash.

PE\_ESTADO\_CREDITO **SECURE LOOKUP**

Replaces the value with one of the 11 values in pe-estados-credito.txt , chosen by a keyed hash.

## EXAMPLES

| VALUE                             |   | MASKED  |
|-----------------------------------|---|---------|
| Reportado en la central de riesgo | → | Pérdida |
| 745                               | → | 471     |

## 12.3 Debt

PE\_FIN\_DEUDA

**COLUMN NAME** **DATA TYPE** **NUMERIC MAPPING**

## WHAT IT IS

Credit and portfolio balances, amounts in arrears, instalments, disbursed amounts and limits.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_FIN\_DEUDA – Path

**85%** saldo\_?(deuda|cr[eé]dito|capital|obligaci[oó]n|cartera|en\_?mora|adeudado) monto\_?(de\_?la\_?)?  
(deuda|cuota|obligaci[oó]n|cr[eé]dito|pr[eé]stamo|mora|desembolsado) deuda(\_?total)?  
l[ií]mite\_?(de\_?)?(cr[eé]dito|tarjeta) l[ií]nea\_?(de\_?)?cr[eé]dito cuota\_?mensual loan\_  
(amount|balance) outstanding\_?balance debt

**DATA TYPE** PE\_FIN\_DEUDA – Type

Accepts number; any other type rules the domain out.

#### HOW IT IS MASKED

Mapped to a value between 100 and 500,000 soles, unrelated to the real one.

**PE\_VALOR\_DEUDA** **NUMERIC MAPPING**

Maps the number to another between 100 and 500,000.

#### EXAMPLES

| VALUE | MASKED  |
|-------|---------|
| 18400 | → 69493 |

## 12.4 Assets

PE\_FIN\_PATRIMONIO

**COLUMN NAME** **DATA TYPE** **NUMERIC MAPPING**

#### WHAT IT IS

Net worth, assets, valuations and self-assessments, account, CTS and fund balances, term deposits and the sworn declaration of assets.

#### HOW IT IS FOUND

**COLUMN NAME** PE\_FIN\_PATRIMONIO – Path

**85%** patrimonio(\_?(neto|l[ií]quido))? activos\_?(totales)? valor\_?(del\_?)?  
(inmueble|predio|veh[ií]culo|bienes|aval[uú]o|comercial) autoval[uú]o aval[uú]o saldo\_  
(cuenta|ahorro|dep[oó]sito|inversi[oó]n|cts|fondo) dep[oó]sito\_?a\_?plazo inversiones  
declaraci[oó]n\_?jurada\_?(de\_?)?(bienes|ingresos) net\_?worth account\_?balance

**DATA TYPE** PE\_FIN\_PATRIMONIO – Type

Accepts number; any other type rules the domain out.

#### HOW IT IS MASKED

Mapped to a value between 1,000 and 900,000 soles.

**PE\_PATRIMONIO** **NUMERIC MAPPING**

Maps the number to another between 1,000 and 900,000.

#### EXAMPLES

| VALUE  | MASKED   |
|--------|----------|
| 148000 | → 814723 |

## 12.5 Pension and benefits

PE\_FIN\_PENSION

COLUMN NAME DATA TYPE NUMERIC MAPPING

### WHAT IT IS

The amount of the pension, subsidy or benefit, and child support.

### HOW IT IS FOUND

COLUMN NAME PE\_FIN\_PENSION - Path

90% pensi[oó]n(\_?(monto|mensual|valor))? monto?(pensi[oó]n|jubilaci[oó]n|beneficio|subsidio)  
valor?(pensi[oó]n|subsidio|beneficio) haber?(pensionable|jubilatorio) pensi[oó]n\_  
(de\_)?alimentos cuota?alimentaria pension?amount

DATA TYPE PE\_FIN\_PENSION - Type

Accepts number; any other type rules the domain out.

### HOW IT IS MASKED

Mapped to a value between 300 and 8,000 soles.

PE\_VALOR\_PENSION NUMERIC MAPPING

Maps the number to another between 300 and 8,000.

### EXAMPLES

VALUE

MASKED

1050

→

4687

## 12.6 Social programmes

PE\_FIN\_PROGRAMA\_SOCIAL

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

### WHAT IT IS

Juntos, Pensión 65, Contigo, Cuna Más, Qali Warma, Beca 18, the FISE gas voucher and other programmes, by name or as a flag. Being in one says the household is poor.

### HOW IT IS FOUND

COLUMN NAME PE\_FIN\_PROGRAMA\_SOCIAL - Path

90% programa?social beneficiario?(programa|subsidio|bono|pensi[oó]n?65) juntos  
pensi[oó]n?65 cuna?m[aá]s qali?warma beca?18 contigo trabaja?per[uú] fise  
techo?propio bono(\_?(familiar|universal|yanapay|demogr[aá]fico))? subsidio(\_?  
(gas|energ[ií]a|transporte|tipo))? transferencia?monetaria ayuda?(social|estatal)  
social?programme

KNOWN VALUES PE\_FIN\_PROGRAMA\_SOCIAL - List

LIST

VALUES

STRENGTH

FIRST VALUES

pe-detectar-programas-  
sociales.txt

24

80%

Juntos, Pensión 65, Pension 65, Contigo, Cuna Más, Cuna Mas

pe-detectar-codigos-banderas.txt

11

30%

s, n, sí, si, no, 1

A value that matches none counts 40% against the domain.

### HOW IT IS MASKED

A flag stays a flag and any other value is replaced by a programme from the list.

## PE\_PROGRAMA\_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

### PE\_PROGRAMA\_SOCIAL\_LISTA

#### PE\_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

##### PE\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

##### PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

##### PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

##### PE\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

##### PE\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

##### PE\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

#### PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

### PE\_PROGRAMA\_SOCIAL\_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in `pe-programas-sociales.txt`, chosen by a keyed hash.

## EXAMPLES

### VALUE

### MASKED

Bono Yanapay

→

Cuna Más

## 12.7 SISFOH classification

PE\_FIN\_CLASIFICACION\_SOCIOECONOMICA

### COLUMN NAME

### KNOWN VALUES

### REGEX DECOMPOSE

## WHAT IT IS

The socioeconomic classification of the Padrón General de Hogares — extremely poor, poor, not poor — that decides who gets a subsidy, and the socioeconomic level or quintile.

## HOW IT IS FOUND

### COLUMN NAME

PE\_FIN\_CLASIFICACION\_SOCIOECONOMICA - Path

90%

clasificaci[oó]n?socioecon[oó]mica csc sisfoh padr[oó]n?general?(de?)?hogares  
nivel?socioecon[oó]mico nse condici[oó]n?(de?)?pobreza pobreza quintil(\_?(de?)?  
(ingreso|riqueza))? estrato?socioecon[oó]mico

### KNOWN VALUES

PE\_FIN\_CLASIFICACION\_SOCIOECONOMICA - List

### LIST

### VALUES

### STRENGTH

### FIRST VALUES

pe-detectar-clasificacion-  
socioeconomica.txt

13

60%

Pobre extremo, Pobre, No pobre, Sin clasificación, Sin  
clasificacion, pobre

A value that matches none counts 40% against the domain.

## HOW IT IS MASKED

A flag stays a flag, a code is replaced by another code and any other value by a class from the list.

## PE\_CLASIFICACION\_SOCIOECONOMICA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_CLASIFICACION\_SOCIOECONOMICA\_LISTA .

PE\_BANDERA **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

PE\_BANDERA\_SI\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt` , chosen by a keyed hash.

PE\_BANDERA\_TRUE\_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt` , chosen by a keyed hash.

PE\_BANDERA\_YES\_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt` , chosen by a keyed hash.

PE\_BANDERA\_S\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt` , chosen by a keyed hash.

PE\_BANDERA\_Y\_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt` , chosen by a keyed hash.

PE\_BANDERA\_0\_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt` , chosen by a keyed hash.

PE\_CLASIFICACION\_SOCIOECONOMICA\_CODIGO **SECURE LOOKUP**

Replaces the value with one of the 4 values in `pe-clasificacion-socioeconomica-codigos.txt` , chosen by a keyed hash.

PE\_CLASIFICACION\_SOCIOECONOMICA\_LISTA **SECURE LOOKUP**

Replaces the value with one of the 4 values in `pe-clasificacion-socioeconomica.txt` , chosen by a keyed hash.

EXAMPLES

| VALUE         |   | MASKED            |
|---------------|---|-------------------|
| Pobre extremo | → | Sin clasificación |
| 2             | → | 4                 |

12.8 Housing tenure

PE\_FIN\_VIVIENDA

**COLUMN NAME** **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

Housing tenure in the categories the census uses: owned outright or being paid for, rented, lent by the employer or another household, owned by invasion.

HOW IT IS FOUND

**COLUMN NAME** PE\_FIN\_VIVIENDA - Path

90% `tenencia_(de_)?(la_)?vivienda` `tipo_(de_)?tenencia` `vivienda_(propia|alquilada|tipo_?tenencia)` `condici[oó]n_(de_)?(la_)?vivienda` `ocupaci[oó]n_vivienda` `housing_?tenure`

**KNOWN VALUES** PE\_FIN\_VIVIENDA - List

| LIST                              | VALUES | STRENGTH | FIRST VALUES                                                                                                                                                 |
|-----------------------------------|--------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pe-detectar-tenencia-vivienda.txt | 17     | 60%      | Propia totalmente pagada, Propia pagándola a plazos, Propia pagandola a plazos, Alquilada, Cedida por el centro de trabajo u otro hogar, Propia por invasión |

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Replaced by another tenure from the list.

#### PE\_TENENCIA\_VIVIENDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

#### PE\_TENENCIA\_VIVIENDA\_LISTA .

##### PE\_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

##### PE\_BANDERA\_SI\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt` , chosen by a keyed hash.

##### PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt` , chosen by a keyed hash.

##### PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt` , chosen by a keyed hash.

##### PE\_BANDERA\_S\_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt` , chosen by a keyed hash.

##### PE\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt` , chosen by a keyed hash.

##### PE\_BANDERA\_0\_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt` , chosen by a keyed hash.

##### PE\_CM\_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

##### PE\_TENENCIA\_VIVIENDA\_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in `pe-tenencia-vivienda.txt` , chosen by a keyed hash.

#### EXAMPLES

##### VALUE

##### MASKED

Ocupación de hecho

→

Propia por invasión

# Criminal records

The law does not list criminal records among the sensitive data, but the Constitution and the Código Procesal Penal treat them as reserved, and a record certificate in a test copy is a liability with no upside.

## 13.1 Criminal records

PE\_PENAL\_ANTECEDENTES

COLUMN NAME    REGEX DECOMPOSE

WHAT IT IS

The criminal, police and judicial record certificate, the type of offence, the procedural situation, the requisitoria, being deprived of liberty, the INPE record and the REDAM register of maintenance debtors.

HOW IT IS FOUND

| COLUMN NAME | PE_PENAL_ANTECEDENTES – Path                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 95%         | <code>antecedentes?(?_?(penales policiales judiciales))?</code> <code>certificado_(de_)?antecedentes</code><br><code>record_?policial</code> <code>reincidencia</code> <code>condena_(?_(penal tipo))?</code> <code>delito_(?_(tipo cometido))?</code> <code>tipo_?</code><br><code>(de_)?delito</code> <code>situaci[oó]n_?jur[ií]dica</code> <code>requisitoria</code> <code>orden_(de_)?captura</code> <code>privad[oa]_?</code><br><code>(de_)?libertad</code> <code>interno_(?_(penal penitenciario))</code> <code>inpe</code> <code>reo</code> <code>deudor_?alimentario_?moroso</code><br><code>redam</code> <code>criminal_?record</code> |
| 55%         | <code>estado_(del_)?proceso</code> <code>sentencia</code> <code>fallo</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

HOW IT IS MASKED

Replaced by a generic status from the list — no record, sentenced, acquitted —, whatever the outcome; a flag stays a flag.

PE\_ANTECEDENTES    REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_ANTECEDENTES\_LISTA .

|                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|
| PE_BANDERA    REGEX DECOMPOSE                                                                                   |
| Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.             |
| PE_BANDERA_SI_NO    SECURE LOOKUP                                                                               |
| Replaces the value with one of the 2 values in <code>pe-bandera-si-no.txt</code> , chosen by a keyed hash.      |
| PE_BANDERA_TRUE_FALSE    SECURE LOOKUP                                                                          |
| Replaces the value with one of the 2 values in <code>pe-bandera-true-false.txt</code> , chosen by a keyed hash. |
| PE_BANDERA_YES_NO    SECURE LOOKUP                                                                              |
| Replaces the value with one of the 2 values in <code>pe-bandera-yes-no.txt</code> , chosen by a keyed hash.     |
| PE_BANDERA_S_N    SECURE LOOKUP                                                                                 |
| Replaces the value with one of the 2 values in <code>pe-bandera-s-n.txt</code> , chosen by a keyed hash.        |
| PE_BANDERA_Y_N    SECURE LOOKUP                                                                                 |
| Replaces the value with one of the 2 values in <code>pe-bandera-y-n.txt</code> , chosen by a keyed hash.        |
| PE_BANDERA_0_1    SECURE LOOKUP                                                                                 |
| Replaces the value with one of the 2 values in <code>pe-bandera-0-1.txt</code> , chosen by a keyed hash.        |
| PE_CM_ALFANUM    CHARACTER MAPPING                                                                              |
| Maps each character to another of its group (7 groups), keeping length and format.                              |
| PE_ANTECEDENTES_LISTA    SECURE LOOKUP                                                                          |
| Replaces the value with one of the 7 values in <code>pe-antecedentes.txt</code> , chosen by a keyed hash.       |

EXAMPLES

VALUE

MASKED

Sentenciado por hurto en 2019

→ Absuelto

13.2 Court or prosecution file

PE\_PENAL\_EXPEDIENTE

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

The number the Poder Judicial gives a case — correlative, year, judicial district and specialty, 00123-2024-0-1801-JR-CI-01 — and the carpeta fiscal of the Ministerio Público.

HOW IT IS FOUND

COLUMN NAME

PE\_PENAL\_EXPEDIENTE - Path

90%

n(o|ro|um|umero)?(de\_)?(expediente?(judicial|penal|fiscal)|causa|proceso?(judicial|penal)|denuncia|carpeta?fiscal|referencia?(judicial|fiscal)) expediente?(judicial|penal|fiscal) carpeta?fiscal referencia?(judicial|fiscal) causa?penal n(o|ro|um|umero)?[uú]nico?(de\_)?expediente cui?judicial

50%

expediente causa proceso

VALUE FORMAT

PE\_PENAL\_EXPEDIENTE - Regex

PATTERN

STRENGTH

NOTES

\d{4,6}-\d{4}-\d-\d{4}-[A-Z-]{2,12}-\d{2}

90%

\d{1,6}-[A-Z]{2,10}-\d{4}

60%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The correlative and the court number are mapped digit by digit; year, judicial district and specialty stay. Other numbers are masked character by character.

PE\_EXPEDIENTE\_JUDICIAL

REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PE\_CM\_ALFANUM .

PE\_CM\_DIGITOS

CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

PE\_CM\_ALFANUM

CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

00123-2024-0-1801-JR-CI-01

→ 47916-2024-0-1801-JR-CI-80

1234/2023

→ 9707/2023

## The DNI and RUC check digits

Both rules are modulus 11, and both end where the Masking Engine's Check Digit algorithm cannot follow them on its own.

- **DNI:** weights 3, 2, 7, 6, 5, 4, 3 and 2 from the left; eleven minus the remainder (zero stays zero) gives a position from 0 to 10, which indexes two series — the digits 6 7 8 9 0 1 1 2 3 4 5 and the letters K A B C D E F G H I J. The letters belong to the documents issued up to 14 August 2007.
- **RUC:** weights 5, 4, 3, 2, 7, 6, 5, 4, 3 and 2 over the first ten digits; the digit is eleven minus the remainder, written 0 when that is ten and **1 when it is eleven**.

Check Digit writes *modulus minus remainder* and writes ten as 0. Both rules agree with it everywhere except in the one place where that written zero is ambiguous: it stands for a remainder of 0 and for a remainder of 1 at once, and the two need different digits — 6 and 5 for the DNI, 1 and 0 for the RUC.

The set resolves it with an **auxiliary Check Digit whose weights are the real ones doubled, modulo 11**. Doubling maps a remainder of 0 to 0 and a remainder of 1 to 2, so the auxiliary writes 0 for the first and 9 for the second — the two cases are told apart. A **Regex Decompose** then reads the pair of digits, writes the true one and drops the auxiliary. Four steps in a chain, no lookup table, and the whole body masked digit by digit before the digit is computed.

Because a natural person's RUC is the number 10 followed by the DNI, and both keep those eight digits under the same mapping, the masked DNI and the masked RUC of the same person still agree.

## Where people live: distritos, UBIGEO and postal code

With the population the 2017 census counted and the location the INEI publishes, the set knows the size, provincia, departamento and position of the 1,892 distritos and the 196 provincias.

- A **distrito** with fewer than 20,000 inhabitants becomes the nearest distrito of its **provincia** with at least 20,000; when its provincia has none, the nearest of its departamento. There are 1,633 of 1,892, with 7.4 million inhabitants.
- The **33 provincias** under 20,000 — Purús has 2,860 — follow the same rule inside their departamento. The 25 departamentos stay.
- Distance is measured between the census centroids, in a straight line. The eighteen distritos created after the census have no population or location of their own and take the centre of their provincia.
- Names repeat a great deal — San Juan, Santa Rosa, San José, Independencia, Progreso. A name is generalized only when every place bearing it is small, and then follows the most populous of them; when that bearer is a provincia, the replacement is a provincia. Written with the provincia or the departamento (Balsas – Amazonas, Balsas (Amazonas)) the same rule applies inside it.
- The **UBIGEO** follows the same table, so a person's distrito name and code still agree.
- The **centro poblado** and the **caserío** become `Zona rural`, the **urbanización** is replaced by another and the **postal code** keeps only the departamento.

Distrito names that are also given names or surnames, and departamento names, are left out of the detection list, so that a surname column is not taken for distritos.

## Limits

- **Binary columns** — DNI photos, fingerprints, validation selfies, clinical PDFs in BLOBs — cannot be masked by these algorithms. Inventory them and report them as residual risk.
- **A verification character in a column of its own** is not recomputed: it is masked and no longer matches the masked DNI.
- **Departamento, the bank code and the phone area code stay.**
- **Data Cleansing leaves unknown values as they are:** a distrito spelled in a way that is not in the table — abbreviated, misspelled — passes unchanged.
- **Free text** is redacted word by word: a phone written with spaces inside a sentence may not be found, and a name that is also a common word ( Rosa , Cruz , Flores ) is redacted wherever it appears.
- **Children and adolescents** — a birth date moves at most 60 days, so few people cross their 14th or 18th birthday, but some do. The regulation makes that line matter (art. 25): where it does, split the rule by age group with Multi-Column Condition.
- **Addresses** are replaced as a whole line. Street, urbanización, distrito and postal code in separate columns are masked independently and no longer form a coherent address.
- **Masked bitcoin addresses** no longer pass their checksum: they serve to test the system, not to transact with it.
- **The engine key** links masked values back to production for whoever holds both: the copy is **dissociated** in the sense of article 41 of the regulation — a reversible procedure —, not anonymized, and remains personal data. Keep the key separate and safeguarded.

## Verification

Before publication, the set was checked outside a Masking Engine:

- The DNI rule was checked against the worked example RENIEC's own routine is published with, and the RUC rule against RUCs published by public bodies.
- Every classifier configuration was reviewed against what a Masking Engine accepts.
- Profiling was simulated on 145 columns with the whole set at its assignment threshold — well named, badly named and 29 columns that are not personal data — and again with only the classifiers of the essential pack.
- The table that generalizes places was audited for all 1,892 distritos and 196 provincias: by name, by name with the provincia and the departamento, and by UBIGEO.
- The algorithms, run with the Delphix algorithm plugin, masked each domain's sample values and placeholders such as N/A , checking every output against its vocabulary; 150 DNIs of the digit series and 60 of the letter series, 200 RUCs of the four kinds of taxpayer for validity and collisions, and 20 DNI and RUC pairs to confirm they still agree. Every algorithm also masked its own sample value.

Every column was assigned as expected and every value masked as expected. Before using the set on real data, run a profiling job and a masking job on a representative copy.

# Sources

- Ley 29733, Ley de Protección de Datos Personales (3 July 2011). Reglamento, Decreto Supremo 016-2024-JUS (El Peruano, 30 November 2024, in force 30 March 2025). Autoridad Nacional de Protección de Datos Personales, Ministerio de Justicia y Derechos Humanos.
- Ley 26842 General de Salud. Ley 30024 (Registro Nacional de Historias Clínicas Electrónicas). Ley 26626 (CONTRASIDA). Ley 29973 (persons with disabilities). Ley 27489 (risk bureaus). Ley 28592 (Registro Único de Víctimas). Ley 30364 (violence against women and the family group). Ley 29735 (indigenous languages). Ley 29785 (prior consultation).
- Instituto Nacional de Estadística e Informática: Censos Nacionales 2017 (XII de Población, VII de Vivienda y III de Comunidades Indígenas), Directorio Nacional de Centros Poblados, and the UBIGEO with the area and location of every distrito.
- RENIEC (DNI and the verification character). SUNAT (RUC and its check digit). SBS (debtor classification). MIDIS (SISFOH and the social programmes). Ministerio de Cultura (Base de Datos Oficial de Pueblos Indígenas u Originarios). JNE (Registro de Organizaciones Políticas). MTPE (minimum wage in force since January 2025). MTC (Placa Única Nacional de Rodaje). SERPOST (five-digit postal code).

# Algorithms

Every algorithm of the set in alphabetical order, with its configuration. The one a domain names is the top of its tree; the others are building blocks.

**PE\_ANIO** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PE\_L2\_ANIO\_NACIMIENTO

PATTERNS, TRIED IN ORDER

|             |                      |
|-------------|----------------------|
| (\d{3})(\d) | keep · PE_CM_DIGITOS |
| otherwise   | keep                 |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_ANTECEDENTES** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_ANTECEDENTES\_LISTA .

Used by: PE\_PENAL\_ANTECEDENTES

PATTERNS, TRIED IN ORDER

|                                        |                       |
|----------------------------------------|-----------------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA            |
| (\d{1,6})                              | PE_CM_ALFANUM         |
| otherwise                              | PE_ANTECEDENTES_LISTA |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_ANTECEDENTES\_LISTA** SECURE LOOKUP

Replaces the value with one of the 7 values in pe-antecedentes.txt , chosen by a keyed hash.

Used by: PE\_ANTECEDENTES

CONFIGURATION

|                            |                     |
|----------------------------|---------------------|
| lookupFile                 | pe-antecedentes.txt |
| hashMethod                 | SHA256              |
| maskedValueCase            | PRESERVE_INPUT      |
| inputCaseSensitive         | no                  |
| trimWhitespaceFromInput    | yes                 |
| trimWhitespaceInLookupFile | yes                 |

**PE\_APELLIDO** NAME

Replaces the name with one of the 422 names in pe-apellidos.txt .

Used by: PE\_PALABRA\_APELLIDO

**CONFIGURATION**

|                    |                  |
|--------------------|------------------|
| lookupFile         | pe-apellidos.txt |
| maskedValueCase    | PRESERVE_INPUT   |
| filterAccent       | yes              |
| inputCaseSensitive | no               |

**PE\_APELLIDOS** REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PE\_CM\_ALFANUM .

Used by: PE\_L1\_APELLIDO , PE\_NOMBRE\_COMPLETO

**PATTERNS, TRIED IN ORDER**

|                               |                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------|
| (\S+)                         | PE_PALABRA_APELLIDO                                                                   |
| (\S+)\s+(\S+)                 | PE_PALABRA_APELLIDO · PE_PALABRA_APELLIDO                                             |
| (\S+)\s+(\S+)\s+(\S+)         | PE_PALABRA_APELLIDO · PE_PALABRA_APELLIDO · PE_PALABRA_APELLIDO                       |
| (\S+)\s+(\S+)\s+(\S+)\s+(\S+) | PE_PALABRA_APELLIDO · PE_PALABRA_APELLIDO · PE_PALABRA_APELLIDO · PE_PALABRA_APELLIDO |
| otherwise                     | PE_CM_ALFANUM                                                                         |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_APTITUD\_LABORAL** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_APTITUD\_LABORAL\_LISTA .

Used by: PE\_SALUD\_OCUPACIONAL

**PATTERNS, TRIED IN ORDER**

|                                        |                          |
|----------------------------------------|--------------------------|
| (?i)(s i j no true false yes [snyx01]) | PE_BANDERA               |
| (\d{1,6})                              | PE_CM_ALFANUM            |
| otherwise                              | PE_APTITUD_LABORAL_LISTA |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_APTITUD\_LABORAL\_LISTA** SECURE LOOKUP

Replaces the value with one of the 4 values in pe-aptitud-laboral.txt , chosen by a keyed hash.

Used by: PE\_APTITUD\_LABORAL

**CONFIGURATION**

|                            |                        |
|----------------------------|------------------------|
| lookupFile                 | pe-aptitud-laboral.txt |
| hashMethod                 | SHA256                 |
| maskedValueCase            | PRESERVE_INPUT         |
| inputCaseSensitive         | no                     |
| trimWhitespaceFromInput    | yes                    |
| trimWhitespaceInLookupFile | yes                    |

**PE\_BANDERA** REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PE\_CATEGORIA\_SUPRIMIDA , PE\_ESTADO\_CIVIL , PE\_NIVEL\_EDUCATIVO , PE\_ORIGEN\_ETNICO , PE\_PUEBLO\_INDIGENA , PE\_LENGUA , PE\_RELIGION , PE\_IDEOLOGIA\_POLITICA , PE\_PARTIDO , PE\_ORIENTACION\_SEXUAL , PE\_IDENTIDAD\_GENERO , PE\_COBERTURA\_SALUD , PE\_DIAGNOSTICO , PE\_MEDICAMENTO , PE\_DISCAPACIDAD , PE\_APTITUD\_LABORAL , PE\_HISTORIAL\_CREDITICIO , PE\_PROGRAMA\_SOCIAL , PE\_CLASIFICACION\_SOCIOECONOMICA , PE\_TENENCIA\_VIVIENDA , PE\_ANTECEDENTES

PATTERNS, TRIED IN ORDER

|                  |                       |
|------------------|-----------------------|
| (?i)(s[ií] no)   | PE_BANDERA_SI_NO      |
| (?i)(true false) | PE_BANDERA_TRUE_FALSE |
| (?i)(yes)        | PE_BANDERA_YES_NO     |
| (?i)([sn])       | PE_BANDERA_S_N        |
| (?i)([yx])       | PE_BANDERA_Y_N        |
| ([01])           | PE_BANDERA_0_1        |
| otherwise        | keep                  |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_BANDERA\_0\_1** SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-0-1.txt , chosen by a keyed hash.

Used by: PE\_BANDERA

CONFIGURATION

|                            |                    |
|----------------------------|--------------------|
| lookupFile                 | pe-bandera-0-1.txt |
| hashMethod                 | SHA256             |
| maskedValueCase            | PRESERVE_INPUT     |
| inputCaseSensitive         | no                 |
| trimWhitespaceFromInput    | yes                |
| trimWhitespaceInLookupFile | yes                |

**PE\_BANDERA\_S\_N** SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-s-n.txt , chosen by a keyed hash.

Used by: PE\_BANDERA

CONFIGURATION

|                            |                    |
|----------------------------|--------------------|
| lookupFile                 | pe-bandera-s-n.txt |
| hashMethod                 | SHA256             |
| maskedValueCase            | PRESERVE_INPUT     |
| inputCaseSensitive         | no                 |
| trimWhitespaceFromInput    | yes                |
| trimWhitespaceInLookupFile | yes                |

**PE\_BANDERA\_SI\_NO** SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-si-no.txt , chosen by a keyed hash.

Used by: PE\_BANDERA

CONFIGURATION

|                            |                      |
|----------------------------|----------------------|
| lookupFile                 | pe-bandera-si-no.txt |
| hashMethod                 | SHA256               |
| maskedValueCase            | PRESERVE_INPUT       |
| inputCaseSensitive         | no                   |
| trimWhitespaceFromInput    | yes                  |
| trimWhitespaceInLookupFile | yes                  |

PE\_BANDERA\_TRUE\_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-true-false.txt , chosen by a keyed hash.

Used by: PE\_BANDERA

CONFIGURATION

|                            |                           |
|----------------------------|---------------------------|
| lookupFile                 | pe-bandera-true-false.txt |
| hashMethod                 | SHA256                    |
| maskedValueCase            | PRESERVE_INPUT            |
| inputCaseSensitive         | no                        |
| trimWhitespaceFromInput    | yes                       |
| trimWhitespaceInLookupFile | yes                       |

PE\_BANDERA\_Y\_N SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-y-n.txt , chosen by a keyed hash.

Used by: PE\_BANDERA

CONFIGURATION

|                            |                    |
|----------------------------|--------------------|
| lookupFile                 | pe-bandera-y-n.txt |
| hashMethod                 | SHA256             |
| maskedValueCase            | PRESERVE_INPUT     |
| inputCaseSensitive         | no                 |
| trimWhitespaceFromInput    | yes                |
| trimWhitespaceInLookupFile | yes                |

PE\_BANDERA\_YES\_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pe-bandera-yes-no.txt , chosen by a keyed hash.

Used by: PE\_BANDERA

CONFIGURATION

|                            |                       |
|----------------------------|-----------------------|
| lookupFile                 | pe-bandera-yes-no.txt |
| hashMethod                 | SHA256                |
| maskedValueCase            | PRESERVE_INPUT        |
| inputCaseSensitive         | no                    |
| trimWhitespaceFromInput    | yes                   |
| trimWhitespaceInLookupFile | yes                   |

PE\_CATEGORIA\_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE\_SUPRIMIR .

Used by: PE\_L3\_CONVICCION\_FILOSOFICA , PE\_L3\_AFILIACION\_SINDICAL , PE\_L3\_VIDA\_SEXUAL , PE\_L3\_VICTIMA , PE\_SALUD\_VIH , PE\_SALUD\_REPRODUCTIVA , PE\_SALUD\_MENTAL

PATTERNS, TRIED IN ORDER

|                                        |               |
|----------------------------------------|---------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA    |
| (\d{1,6})                              | PE_CM_ALFANUM |
| otherwise                              | PE_SUPRIMIR   |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_CIE10 SECURE LOOKUP

Replaces the value with one of the 37 values in `pe-cie10.txt` , chosen by a keyed hash.

Used by: `PE_DIAGNOSTICO`

CONFIGURATION

|                            |                      |
|----------------------------|----------------------|
| lookupFile                 | pe-cie10.txt         |
| hashMethod                 | SHA256               |
| maskedValueCase            | PRESERVE_LOOKUP_FILE |
| inputCaseSensitive         | no                   |
| trimWhitespaceFromInput    | yes                  |
| trimWhitespaceInLookupFile | yes                  |

PE\_CIE10\_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in `pe-cie10-decimal.txt` , chosen by a keyed hash.

Used by: `PE_DIAGNOSTICO`

CONFIGURATION

|                            |                      |
|----------------------------|----------------------|
| lookupFile                 | pe-cie10-decimal.txt |
| hashMethod                 | SHA256               |
| maskedValueCase            | PRESERVE_LOOKUP_FILE |
| inputCaseSensitive         | no                   |
| trimWhitespaceFromInput    | yes                  |
| trimWhitespaceInLookupFile | yes                  |

PE\_CLASIFICACION\_SOCIOECONOMICA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PE_CLASIFICACION_SOCIOECONOMICA_LISTA` .

Used by: `PE_FIN_CLASIFICACION_SOCIOECONOMICA`

PATTERNS, TRIED IN ORDER

|                                        |                                        |
|----------------------------------------|----------------------------------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA                             |
| (\d{1,6})                              | PE_CLASIFICACION_SOCIOECONOMICA_CODIGO |
| otherwise                              | PE_CLASIFICACION_SOCIOECONOMICA_LISTA  |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_CLASIFICACION\_SOCIOECONOMICA\_CODIGO** SECURE LOOKUP

Replaces the value with one of the 4 values in `pe-clasificacion-socioeconomica-codigos.txt` , chosen by a keyed hash.

Used by: `PE_CLASIFICACION_SOCIOECONOMICA`

**CONFIGURATION**

|                            |                                             |
|----------------------------|---------------------------------------------|
| lookupFile                 | pe-clasificacion-socioeconomica-codigos.txt |
| hashMethod                 | SHA256                                      |
| maskedValueCase            | PRESERVE_INPUT                              |
| inputCaseSensitive         | no                                          |
| trimWhitespaceFromInput    | yes                                         |
| trimWhitespaceInLookupFile | yes                                         |

**PE\_CLASIFICACION\_SOCIOECONOMICA\_LISTA** SECURE LOOKUP

Replaces the value with one of the 4 values in `pe-clasificacion-socioeconomica.txt` , chosen by a keyed hash.

Used by: `PE_CLASIFICACION_SOCIOECONOMICA`

**CONFIGURATION**

|                            |                                     |
|----------------------------|-------------------------------------|
| lookupFile                 | pe-clasificacion-socioeconomica.txt |
| hashMethod                 | SHA256                              |
| maskedValueCase            | PRESERVE_INPUT                      |
| inputCaseSensitive         | no                                  |
| trimWhitespaceFromInput    | yes                                 |
| trimWhitespaceInLookupFile | yes                                 |

**PE\_CM\_ALFANUM** CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

Used by: `PE_L1_CARNE_EXTRANJERIA` , `PE_L1_PASAPORTE` , `PE_L1_ACTA` , `PE_L1_SEGURO_SOCIAL` , `PE_L1_CODIGO_ESTUDIANTE` , `PE_L1_COLEGIATURA` , `PE_L1_DIRECCION_COMPLEMENTO` , `PE_L1_BILLETERA` , `PE_L1_COOKIE` , `PE_L1_VEHICULO` , `PE_L1_INMUEBLE` , `PE_L1_CONTRATO` , `PE_L1_USUARIO` , `PE_SALUD_IDENTIFICADOR` , `PE_SALUD_RESULTADO_EXAMEN` , `PE_CATEGORIA_SUPRIMIDA` , `PE_DOCUMENTO` , `PE_NOMBRES` , `PE_APELLIDOS` , `PE_NOMBRE_COMPLETO` , `PE_EMAIL` , `PE_PUEBLO_INDIGENA` , `PE LENGUA` , `PE_RELIGION` , `PE_IDEOLOGIA_POLITICA` , `PE_PARTIDO` , `PE_ORIENTACION_SEXUAL` , `PE_IDENTIDAD_GENERO` , `PE_MEDICAMENTO` , `PE_DISCAPACIDAD` , `PE_APTITUD_LABORAL` , `PE_PROGRAMA_SOCIAL` , `PE_TENENCIA_VIVIENDA` , `PE_ANTECEDENTES` , `PE_EXPEDIENTE_JUDICIAL`

**CONFIGURATION**

|                    |                                                                                      |
|--------------------|--------------------------------------------------------------------------------------|
| characterGroups    | 0123456789 AEIOU BCDFGHJKLMNPQRSTVWXYZ aeiou<br>bcd fghjklmnpqrstvwxyz ÁÉÍÓÚú áéíóúü |
| caseSensitive      | yes                                                                                  |
| minMaskedPositions | 0                                                                                    |

**PE\_CM\_DIGITOS** CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

Used by: `PE_L1_CUENTA_BANCARIA` , `PE_DNI_CD` , `PE_RUC_CUERPO` , `PE_DOCUMENTO` , `PE_TELEFONO` , `PE_COORDENADA` , `PE_ANIO` , `PE_EDAD` , `PE_OCUPACION_O_CODIGO` , `PE_COBERTURA_SALUD` , `PE_PROCEDIMIENTO` , `PE_HISTORIAL_CREDITICIO` , `PE_EXPEDIENTE_JUDICIAL`

**CONFIGURATION**

|                 |            |
|-----------------|------------|
| characterGroups | 0123456789 |
| caseSensitive   | yes        |

|                    |   |
|--------------------|---|
| minMaskedPositions | 0 |
|--------------------|---|

**PE\_CM\_DIGITOS\_LETRAS** CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

Used by: PE\_L1\_DIGITO\_VERIFICACION , PE\_L1\_PLACA

**CONFIGURATION**

|                    |                                       |
|--------------------|---------------------------------------|
| characterGroups    | 0123456789 ABCDEFGHIJKLMNOPQRSTUVWXYZ |
| caseSensitive      | yes                                   |
| minMaskedPositions | 0                                     |

**PE\_CM\_HEX** CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

Used by: PE\_L1\_DISPOSITIVO , PE\_IP

**CONFIGURATION**

|                    |                                                                       |
|--------------------|-----------------------------------------------------------------------|
| characterGroups    | 0123456789 abcdef ABCDEF ghijklmnopqrstuvwxyz<br>GHIJKLMNOPQRSTUVWXYZ |
| caseSensitive      | yes                                                                   |
| minMaskedPositions | 0                                                                     |

**PE\_COBERTURA** SECURE LOOKUP

Replaces the value with one of the 8 values in pe-cobertura-salud.txt , chosen by a keyed hash.

Used by: PE\_COBERTURA\_SALUD

**CONFIGURATION**

|                            |                        |
|----------------------------|------------------------|
| lookupFile                 | pe-cobertura-salud.txt |
| hashMethod                 | SHA256                 |
| maskedValueCase            | PRESERVE_LOOKUP_FILE   |
| inputCaseSensitive         | no                     |
| trimWhitespaceFromInput    | yes                    |
| trimWhitespaceInLookupFile | yes                    |

**PE\_COBERTURA\_SALUD** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE\_COBERTURA .

Used by: PE\_SALUD\_COBERTURA

**PATTERNS, TRIED IN ORDER**

|                                        |               |
|----------------------------------------|---------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA    |
| (\d{1,8})                              | PE_CM_DIGITOS |
| otherwise                              | PE_COBERTURA  |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_CODIGO\_POSTAL** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PE\_L2\_CODIGO\_POSTAL

PATTERNS, TRIED IN ORDER

|                |                         |
|----------------|-------------------------|
| (\d{2})(\d{3}) | keep · replace with 000 |
| (\d{2})(\d{2}) | keep · replace with 00  |
| otherwise      | keep                    |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_CONTEO\_LIMITADO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PE\_L2\_PERSONAS\_A\_CARGO

PATTERNS, TRIED IN ORDER

|           |                |
|-----------|----------------|
| ([0-5])   | keep           |
| (\d+)     | replace with 5 |
| otherwise | keep           |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_COORDENADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PE\_L1\_GEOLOCALIZACION

PATTERNS, TRIED IN ORDER

|                                                          |                                                    |
|----------------------------------------------------------|----------------------------------------------------|
| (-?\d{1,3}[.,]\d)(\d+)\s*([,;])\s*(-?\d{1,3}[.,]\d)(\d+) | keep · PE_CM_DIGITOS · keep · keep · PE_CM_DIGITOS |
| (-?\d{1,3}[.,]\d)(\d+)                                   | keep · PE_CM_DIGITOS                               |
| otherwise                                                | keep                                               |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_DIAGNOSTICO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches:

PE\_DIAGNOSTICO\_TEXTO .

Used by: PE\_SALUD\_DIAGNOSTICO

PATTERNS, TRIED IN ORDER

|                                        |                      |
|----------------------------------------|----------------------|
| ([A-TV-Za-tv-z]\d{2}\.\d{1,2})         | PE_CIE10_DECIMAL     |
| ([A-TV-Za-tv-z]\d{2,3}X?)              | PE_CIE10             |
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA           |
| otherwise                              | PE_DIAGNOSTICO_TEXTO |

CONFIGURATION

|           |     |
|-----------|-----|
| trimInput | yes |
|-----------|-----|

|             |    |
|-------------|----|
| requireMask | no |
|-------------|----|

**PE\_DIAGNOSTICO\_TEXTO** SECURE LOOKUP

Replaces the value with one of the 30 values in `pe-diagnostics.txt` , chosen by a keyed hash.

Used by: `PE_DIAGNOSTICO`

**CONFIGURATION**

|                            |                                 |
|----------------------------|---------------------------------|
| lookupFile                 | <code>pe-diagnostics.txt</code> |
| hashMethod                 | <code>SHA256</code>             |
| maskedValueCase            | <code>PRESERVE_INPUT</code>     |
| inputCaseSensitive         | no                              |
| trimWhitespaceFromInput    | yes                             |
| trimWhitespaceInLookupFile | yes                             |

**PE\_DIRECCION** SECURE LOOKUP

Replaces the value with one of the 4,000 values in `pe-direcciones.txt` , chosen by a keyed hash.

Used by: `PE_L1_DIRECCION`

**CONFIGURATION**

|                            |                                   |
|----------------------------|-----------------------------------|
| lookupFile                 | <code>pe-direcciones.txt</code>   |
| hashMethod                 | <code>SHA256</code>               |
| maskedValueCase            | <code>PRESERVE_LOOKUP_FILE</code> |
| inputCaseSensitive         | no                                |
| trimWhitespaceFromInput    | yes                               |
| trimWhitespaceInLookupFile | yes                               |

**PE\_DISCAPACIDAD** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PE_DISCAPACIDAD_LISTA` .

Used by: `PE_SALUD_DISCAPACIDAD`

**PATTERNS, TRIED IN ORDER**

|                                                    |                                    |
|----------------------------------------------------|------------------------------------|
| <code>(?i)([ií] no true false yes [snyx01])</code> | <code>PE_BANDERA</code>            |
| <code>(\d{1,6})</code>                             | <code>PE_CM_ALFANUM</code>         |
| otherwise                                          | <code>PE_DISCAPACIDAD_LISTA</code> |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_DISCAPACIDAD\_LISTA** SECURE LOOKUP

Replaces the value with one of the 8 values in `pe-discapacidades.txt` , chosen by a keyed hash.

Used by: `PE_DISCAPACIDAD`

**CONFIGURATION**

|                    |                                    |
|--------------------|------------------------------------|
| lookupFile         | <code>pe-discapacidades.txt</code> |
| hashMethod         | <code>SHA256</code>                |
| maskedValueCase    | <code>PRESERVE_INPUT</code>        |
| inputCaseSensitive | no                                 |

|                            |     |
|----------------------------|-----|
| trimWhitespaceFromInput    | yes |
| trimWhitespaceInLookupFile | yes |

#### PE\_DISTRITO DATA CLEANSING

Replaces each value found in `pe-distritos-generalizados.txt` (44,952 pairs) by its counterpart; values not in the file stay as they are.

Used by: `PE_L2_DISTRITO`

##### CONFIGURATION

|                |                                             |
|----------------|---------------------------------------------|
| lookupFile     | <code>pe-distritos-generalizados.txt</code> |
| delimiter      | <code> </code>                              |
| caseSensitive  | yes                                         |
| trimWhitespace | yes                                         |

#### PE\_DNI\_AUX CHECK DIGIT

Masks the digits with `PE_IDENTIDAD` and recomputes a modulus 11 check digit with weights `0 4 6 8 10 1 3 4 6` from the right.

Used by: `PE_DNI_CON_DIGITO`, `PE_DNI_CON_LETRA`

##### CONFIGURATION

|                                          |                                 |
|------------------------------------------|---------------------------------|
| weightList                               | <code>0 4 6 8 10 1 3 4 6</code> |
| modulusNumber                            | <code>11</code>                 |
| checkDigitIndex                          | <code>9</code>                  |
| numDigitsForCheckdigitCalculation        | <code>9</code>                  |
| calculateChecksumRightToLeft             | yes                             |
| numericAlgorithm                         | <code>PE_IDENTIDAD</code>       |
| alphaNumericAlgorithm                    | <code>PE_IDENTIDAD</code>       |
| preserveRegex                            | <code>[.\-\s]</code>            |
| inputHandlingConfig.characterHandling    | <code>STANDARD</code>           |
| inputHandlingConfig.invalidInputHandling | <code>ERROR</code>              |
| inputHandlingConfig.shortInputHandling   | <code>FALLBACK</code>           |
| inputHandlingConfig.padCharacter         | <code>0</code>                  |
| inputHandlingConfig.trimWhitespace       | yes                             |

#### PE\_DNI\_CD CHECK DIGIT

Masks the digits with `PE_CM_DIGITOS` and recomputes a modulus 11 check digit with weights `2 3 4 5 6 7 2 3` from the right.

Used by: `PE_DNI_CON_DIGITO`, `PE_DNI_CON_LETRA`

##### CONFIGURATION

|                                   |                              |
|-----------------------------------|------------------------------|
| weightList                        | <code>2 3 4 5 6 7 2 3</code> |
| modulusNumber                     | <code>11</code>              |
| checkDigitIndex                   | <code>8</code>               |
| numDigitsForCheckdigitCalculation | <code>8</code>               |
| calculateChecksumRightToLeft      | yes                          |
| numericAlgorithm                  | <code>PE_CM_DIGITOS</code>   |
| alphaNumericAlgorithm             | <code>PE_CM_DIGITOS</code>   |
| preserveRegex                     | <code>[.\-\s]</code>         |

|                                          |          |
|------------------------------------------|----------|
| inputHandlingConfig.characterHandling    | STANDARD |
| inputHandlingConfig.invalidInputHandling | ERROR    |
| inputHandlingConfig.shortInputHandling   | FALLBACK |
| inputHandlingConfig.padCharacter         | 0        |
| inputHandlingConfig.trimWhitespace       | yes      |

**PE\_DNI\_CON\_DIGITO** STRING ALGORITHM CHAIN

Applies PE\_DNI\_CD → PE\_DNI\_ESPACIO → PE\_DNI\_AUX → PE\_DNI\_VERIFICADOR , in this order.

Used by: PE\_DOCUMENTO

**CONFIGURATION**

|                        |                    |
|------------------------|--------------------|
| algorithmReferences[1] | PE_DNI_CD          |
| algorithmReferences[2] | PE_DNI_ESPACIO     |
| algorithmReferences[3] | PE_DNI_AUX         |
| algorithmReferences[4] | PE_DNI_VERIFICADOR |

**PE\_DNI\_CON\_LETRA** STRING ALGORITHM CHAIN

Applies PE\_DNI\_CD → PE\_DNI\_ESPACIO → PE\_DNI\_AUX → PE\_DNI\_LETRA , in this order.

Used by: PE\_DOCUMENTO

**CONFIGURATION**

|                        |                |
|------------------------|----------------|
| algorithmReferences[1] | PE_DNI_CD      |
| algorithmReferences[2] | PE_DNI_ESPACIO |
| algorithmReferences[3] | PE_DNI_AUX     |
| algorithmReferences[4] | PE_DNI_LETRA   |

**PE\_DNI\_ESPACIO** REGEX DECOMPOSE

Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PE\_DNI\_CON\_DIGITO , PE\_DNI\_CON\_LETRA

**PATTERNS, TRIED IN ORDER**

|                   |                        |
|-------------------|------------------------|
| (\d{8}[\s\-]?)(0) | keep · replace with 00 |
| (\d{8}[\s\-]?)(1) | keep · replace with 10 |
| (\d{8}[\s\-]?)(2) | keep · replace with 20 |
| (\d{8}[\s\-]?)(3) | keep · replace with 30 |
| (\d{8}[\s\-]?)(4) | keep · replace with 40 |
| (\d{8}[\s\-]?)(5) | keep · replace with 50 |
| (\d{8}[\s\-]?)(6) | keep · replace with 60 |
| (\d{8}[\s\-]?)(7) | keep · replace with 70 |
| (\d{8}[\s\-]?)(8) | keep · replace with 80 |
| (\d{8}[\s\-]?)(9) | keep · replace with 90 |
| otherwise         | keep                   |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_DNI\_LETRA REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PE\_DNI\_CON\_LETRA

PATTERNS, TRIED IN ORDER

|                       |                       |
|-----------------------|-----------------------|
| (\d{8}[\s\-\-]?)(00)  | keep · replace with K |
| (\d{8}[\s\-\-]?)(0\d) | keep · replace with J |
| (\d{8}[\s\-\-]?)(1\d) | keep · replace with A |
| (\d{8}[\s\-\-]?)(2\d) | keep · replace with B |
| (\d{8}[\s\-\-]?)(3\d) | keep · replace with C |
| (\d{8}[\s\-\-]?)(4\d) | keep · replace with D |
| (\d{8}[\s\-\-]?)(5\d) | keep · replace with E |
| (\d{8}[\s\-\-]?)(6\d) | keep · replace with F |
| (\d{8}[\s\-\-]?)(7\d) | keep · replace with G |
| (\d{8}[\s\-\-]?)(8\d) | keep · replace with H |
| (\d{8}[\s\-\-]?)(9\d) | keep · replace with I |
| otherwise             | keep                  |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_DNI\_VERIFICADOR REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PE\_DNI\_CON\_DIGITO

PATTERNS, TRIED IN ORDER

|                       |                       |
|-----------------------|-----------------------|
| (\d{8}[\s\-\-]?)(00)  | keep · replace with 6 |
| (\d{8}[\s\-\-]?)(0\d) | keep · replace with 5 |
| (\d{8}[\s\-\-]?)(1\d) | keep · replace with 7 |
| (\d{8}[\s\-\-]?)(2\d) | keep · replace with 8 |
| (\d{8}[\s\-\-]?)(3\d) | keep · replace with 9 |
| (\d{8}[\s\-\-]?)(4\d) | keep · replace with 0 |
| (\d{8}[\s\-\-]?)(5\d) | keep · replace with 1 |
| (\d{8}[\s\-\-]?)(6\d) | keep · replace with 1 |
| (\d{8}[\s\-\-]?)(7\d) | keep · replace with 2 |
| (\d{8}[\s\-\-]?)(8\d) | keep · replace with 3 |
| (\d{8}[\s\-\-]?)(9\d) | keep · replace with 4 |
| otherwise             | keep                  |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_DOCUMENTO REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PE\_CM\_ALFANUM .

Used by: PE\_L1\_DNI

PATTERNS, TRIED IN ORDER

|                          |                   |
|--------------------------|-------------------|
| ((?:10 15 17 20)\d{9})   | PE_RUC            |
| (\d{8}[\s\~]?[0-9])      | PE_DNI_CON_DIGITO |
| (\d{8}[\s\~]?[A-JKa-jk]) | PE_DNI_CON_LETRA  |
| (\d{8})                  | PE_CM_DIGITOS     |
| otherwise                | PE_CM_ALFANUM     |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_EDAD REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PE\_L2\_EDAD

PATTERNS, TRIED IN ORDER

|                      |                             |
|----------------------|-----------------------------|
| (\d{3,})             | replace with 90             |
| ([1-9])(\d)([.,]\d+) | keep · PE_CM_DIGITOS · keep |
| ([1-9])(\d)          | keep · PE_CM_DIGITOS        |
| (\d)                 | PE_CM_DIGITOS               |
| otherwise            | keep                        |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_EMAIL REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PE\_CM\_ALFANUM .

Used by: PE\_L1\_EMAIL

PATTERNS, TRIED IN ORDER

|                                                        |                                                   |
|--------------------------------------------------------|---------------------------------------------------|
| ((^@\s+)+)(^@\s+)\.([A-Za-z]{2,24})(?:\.[A-Za-z]{2})?) | PE_CM_ALFANUM · PE_CM_ALFANUM · replace with test |
| otherwise                                              | PE_CM_ALFANUM                                     |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_EMPLEADOR SECURE LOOKUP

Replaces the value with one of the 24 values in pe-empleadores.txt , chosen by a keyed hash.

Used by: PE\_L2\_EMPLEADOR

CONFIGURATION

|                         |                      |
|-------------------------|----------------------|
| lookupFile              | pe-empleadores.txt   |
| hashMethod              | SHA256               |
| maskedValueCase         | PRESERVE_LOOKUP_FILE |
| inputCaseSensitive      | no                   |
| trimWhitespaceFromInput | yes                  |

|                            |     |
|----------------------------|-----|
| trimWhitespaceInLookupFile | yes |
|----------------------------|-----|

**PE\_ESTADO\_CIVIL** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_ESTADO\_CIVIL\_LISTA .

Used by: PE\_L2\_ESTADO\_CIVIL

**PATTERNS, TRIED IN ORDER**

|                                        |                        |
|----------------------------------------|------------------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA             |
| (\d{1,6})                              | PE_ESTADO_CIVIL_CODIGO |
| otherwise                              | PE_ESTADO_CIVIL_LISTA  |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_ESTADO\_CIVIL\_CODIGO** SECURE LOOKUP

Replaces the value with one of the 6 values in pe-estado-civil-codigos.txt , chosen by a keyed hash.

Used by: PE\_ESTADO\_CIVIL

**CONFIGURATION**

|                            |                             |
|----------------------------|-----------------------------|
| lookupFile                 | pe-estado-civil-codigos.txt |
| hashMethod                 | SHA256                      |
| maskedValueCase            | PRESERVE_INPUT              |
| inputCaseSensitive         | no                          |
| trimWhitespaceFromInput    | yes                         |
| trimWhitespaceInLookupFile | yes                         |

**PE\_ESTADO\_CIVIL\_LISTA** SECURE LOOKUP

Replaces the value with one of the 6 values in pe-estado-civil.txt , chosen by a keyed hash.

Used by: PE\_ESTADO\_CIVIL

**CONFIGURATION**

|                            |                     |
|----------------------------|---------------------|
| lookupFile                 | pe-estado-civil.txt |
| hashMethod                 | SHA256              |
| maskedValueCase            | PRESERVE_INPUT      |
| inputCaseSensitive         | no                  |
| trimWhitespaceFromInput    | yes                 |
| trimWhitespaceInLookupFile | yes                 |

**PE\_ESTADO\_CREDITO** SECURE LOOKUP

Replaces the value with one of the 11 values in pe-estados-credito.txt , chosen by a keyed hash.

Used by: PE\_HISTORIAL\_CREDITICIO

**CONFIGURATION**

|                    |                        |
|--------------------|------------------------|
| lookupFile         | pe-estados-credito.txt |
| hashMethod         | SHA256                 |
| maskedValueCase    | PRESERVE_INPUT         |
| inputCaseSensitive | no                     |

|                            |     |
|----------------------------|-----|
| trimWhitespaceFromInput    | yes |
| trimWhitespaceInLookupFile | yes |
|                            |     |

**PE\_EXPEDIENTE\_JUDICIAL** REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PE\_CM\_ALFANUM .

Used by: PE\_PENAL\_EXPEDIENTE

**PATTERNS, TRIED IN ORDER**

|                                                                    |                                                                                                 |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| (\d{4,6})(-)(\d{4})(-)(\d)(-)(\d{4})(-)([A-Za-z-]{2,12})(-)(\d{2}) | PE_CM_DIGITOS · keep · keep · keep · keep · keep · keep · keep · keep · keep ·<br>PE_CM_DIGITOS |
| (\d{1,6})([\\s\\-\\/])([A-Za-z]{2,10})([\\s\\-\\/])(\d{4})         | PE_CM_DIGITOS · keep · keep · keep · keep                                                       |
| ([A-Za-z]{1,6}[\\s\\-\\?])(\d{1,6})([\\s\\-\\/])(\d{4})            | keep · PE_CM_DIGITOS · keep · keep                                                              |
| (\d{1,6})([\\s\\-\\/])(\d{4})                                      | PE_CM_DIGITOS · keep · keep                                                                     |
| otherwise                                                          | PE_CM_ALFANUM                                                                                   |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |
|             |     |

**PE\_FECHA\_EVENTO** DATE SHIFT

Moves the date by -30 to 30 days.

Used by: PE\_L2\_FECHA\_EVENTO

**CONFIGURATION**

|          |      |
|----------|------|
| minRange | -30  |
| maxRange | 30   |
| unit     | DAYS |
| roll     | no   |
|          |      |

**PE\_FECHA\_NACIMIENTO** DATE SHIFT

Moves the date by -60 to 60 days.

Used by: PE\_L2\_FECHA\_NACIMIENTO

**CONFIGURATION**

|          |      |
|----------|------|
| minRange | -60  |
| maxRange | 60   |
| unit     | DAYS |
| roll     | no   |
|          |      |

**PE\_GRUPO\_SANGUINEO** SECURE LOOKUP

Replaces the value with one of the 8 values in pe-grupos-sanguineos.txt , chosen by a keyed hash.

Used by: PE\_SALUD\_GRUPO\_SANGUINEO

**CONFIGURATION**

|                    |                          |
|--------------------|--------------------------|
| lookupFile         | pe-grupos-sanguineos.txt |
| hashMethod         | SHA256                   |
| maskedValueCase    | PRESERVE_LOOKUP_FILE     |
| inputCaseSensitive | no                       |
|                    |                          |

|                            |     |
|----------------------------|-----|
| trimWhitespaceFromInput    | yes |
| trimWhitespaceInLookupFile | yes |

**PE\_HISTORIAL\_CREDITICIO** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE\_ESTADO\_CREDITO .

Used by: PE\_FIN\_HISTORIAL\_CREDITICIO

**PATTERNS, TRIED IN ORDER**

|                                        |                   |
|----------------------------------------|-------------------|
| (\d{1,4})                              | PE_CM_DIGITOS     |
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA        |
| otherwise                              | PE_ESTADO_CREDITO |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_IDENTIDAD** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PE\_DNI\_AUX , PE\_RUC\_AUX

**PATTERNS, TRIED IN ORDER**

|           |      |
|-----------|------|
| (.*)      | keep |
| otherwise | keep |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_IDENTIDAD\_GENERO** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_IDENTIDAD\_GENERO\_LISTA .

Used by: PE\_L3\_IDENTIDAD\_GENERO

**PATTERNS, TRIED IN ORDER**

|                                        |                           |
|----------------------------------------|---------------------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA                |
| (\d{1,6})                              | PE_CM_ALFANUM             |
| otherwise                              | PE_IDENTIDAD_GENERO_LISTA |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_IDENTIDAD\_GENERO\_LISTA** SECURE LOOKUP

Replaces the value with one of the 7 values in pe-identidades-genero.txt , chosen by a keyed hash.

Used by: PE\_IDENTIDAD\_GENERO

**CONFIGURATION**

|                 |                           |
|-----------------|---------------------------|
| lookupFile      | pe-identidades-genero.txt |
| hashMethod      | SHA256                    |
| maskedValueCase | PRESERVE_INPUT            |

|                            |     |
|----------------------------|-----|
| inputCaseSensitive         | no  |
| trimWhitespaceFromInput    | yes |
| trimWhitespaceInLookupFile | yes |

#### PE\_IDEOLOGIA\_POLITICA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_IDEOLOGIA\_POLITICA\_LISTA .

Used by: PE\_L3\_OPINION\_POLITICA

#### PATTERNS, TRIED IN ORDER

|                                        |                             |
|----------------------------------------|-----------------------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA                  |
| (\d{1,6})                              | PE_CM_ALFANUM               |
| otherwise                              | PE_IDEOLOGIA_POLITICA_LISTA |

#### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PE\_IDEOLOGIA\_POLITICA\_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in pe-ideologias-politicas.txt , chosen by a keyed hash.

Used by: PE\_IDEOLOGIA\_POLITICA

#### CONFIGURATION

|                            |                             |
|----------------------------|-----------------------------|
| lookupFile                 | pe-ideologias-politicas.txt |
| hashMethod                 | SHA256                      |
| maskedValueCase            | PRESERVE_INPUT              |
| inputCaseSensitive         | no                          |
| trimWhitespaceFromInput    | yes                         |
| trimWhitespaceInLookupFile | yes                         |

#### PE\_INGRESOS NUMERIC MAPPING

Maps the number to another between 1,025 and 25,000.

Used by: PE\_FIN\_INGRESOS

#### CONFIGURATION

|          |       |
|----------|-------|
| minValue | 1025  |
| maxValue | 25000 |

#### PE\_INSTITUCION\_EDUCATIVA SECURE LOOKUP

Replaces the value with one of the 12 values in pe-instituciones-educativas.txt , chosen by a keyed hash.

Used by: PE\_L2\_INSTITUCION\_EDUCATIVA

#### CONFIGURATION

|                         |                                 |
|-------------------------|---------------------------------|
| lookupFile              | pe-instituciones-educativas.txt |
| hashMethod              | SHA256                          |
| maskedValueCase         | PRESERVE_INPUT                  |
| inputCaseSensitive      | no                              |
| trimWhitespaceFromInput | yes                             |

|                            |     |
|----------------------------|-----|
| trimWhitespaceInLookupFile | yes |
|----------------------------|-----|

**PE\_IP** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PE\_CM\_HEX .

Used by: PE\_L1\_IP

**PATTERNS, TRIED IN ORDER**

|                                               |                                               |
|-----------------------------------------------|-----------------------------------------------|
| (\d{1,3})\.\(\d{1,3})\.\(\d{1,3})\.\(\d{1,3}) | PE_OCTETO · PE_OCTETO · PE_OCTETO · PE_OCTETO |
| otherwise                                     | PE_CM_HEX                                     |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE LENGUA** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE LENGUA LISTA .

Used by: PE\_L3 LENGUA

**PATTERNS, TRIED IN ORDER**

|                                        |                 |
|----------------------------------------|-----------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE BANDERA      |
| (\d{1,6})                              | PE_CM_ALFANUM   |
| otherwise                              | PE LENGUA LISTA |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE LENGUA LISTA** SECURE LOOKUP

Replaces the value with one of the 18 values in pe-lenguas.txt , chosen by a keyed hash.

Used by: PE LENGUA

**CONFIGURATION**

|                            |                |
|----------------------------|----------------|
| lookupFile                 | pe-lenguas.txt |
| hashMethod                 | SHA256         |
| maskedValueCase            | PRESERVE_INPUT |
| inputCaseSensitive         | no             |
| trimWhitespaceFromInput    | yes            |
| trimWhitespaceInLookupFile | yes            |

**PE\_MEDICAMENTO** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_MEDICAMENTO LISTA .

Used by: PE\_SALUD\_MEDICAMENTO

**PATTERNS, TRIED IN ORDER**

|                                        |                      |
|----------------------------------------|----------------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE BANDERA           |
| (\d{1,6})                              | PE_CM_ALFANUM        |
| otherwise                              | PE_MEDICAMENTO LISTA |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_MEDICAMENTO\_LISTA SECURE LOOKUP

Replaces the value with one of the 22 values in `pe-medicamentos.txt` , chosen by a keyed hash.

Used by: `PE_MEDICAMENTO`

CONFIGURATION

|                            |                                  |
|----------------------------|----------------------------------|
| lookupFile                 | <code>pe-medicamentos.txt</code> |
| hashMethod                 | SHA256                           |
| maskedValueCase            | PRESERVE_INPUT                   |
| inputCaseSensitive         | no                               |
| trimWhitespaceFromInput    | yes                              |
| trimWhitespaceInLookupFile | yes                              |

PE\_NACIONALIDAD SECURE LOOKUP

Replaces the value with one of the 14 values in `pe-nacionalidades.txt` , chosen by a keyed hash.

Used by: `PE_L3_NACIONALIDAD`

CONFIGURATION

|                            |                                    |
|----------------------------|------------------------------------|
| lookupFile                 | <code>pe-nacionalidades.txt</code> |
| hashMethod                 | SHA256                             |
| maskedValueCase            | PRESERVE_INPUT                     |
| inputCaseSensitive         | no                                 |
| trimWhitespaceFromInput    | yes                                |
| trimWhitespaceInLookupFile | yes                                |

PE\_NIVEL\_EDUCATIVO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PE_NIVEL_EDUCATIVO_LISTA` .

Used by: `PE_L2_NIVEL_EDUCATIVO`

PATTERNS, TRIED IN ORDER

|                                                    |                                        |
|----------------------------------------------------|----------------------------------------|
| <code>(?i)([ií] no true false yes [snyx01])</code> | <code>PE_BANDERA</code>                |
| <code>(\d{1,6})</code>                             | <code>PE_NIVEL_EDUCATIVO_CODIGO</code> |
| otherwise                                          | <code>PE_NIVEL_EDUCATIVO_LISTA</code>  |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_NIVEL\_EDUCATIVO\_CODIGO SECURE LOOKUP

Replaces the value with one of the 11 values in `pe-nivel-educativo-codigos.txt` , chosen by a keyed hash.

Used by: `PE_NIVEL_EDUCATIVO`

CONFIGURATION

|            |                                             |
|------------|---------------------------------------------|
| lookupFile | <code>pe-nivel-educativo-codigos.txt</code> |
| hashMethod | SHA256                                      |

|                            |                |
|----------------------------|----------------|
| maskedValueCase            | PRESERVE_INPUT |
| inputCaseSensitive         | no             |
| trimWhitespaceFromInput    | yes            |
| trimWhitespaceInLookupFile | yes            |

**PE\_NIVEL\_EDUCATIVO\_LISTA** SECURE LOOKUP

Replaces the value with one of the 11 values in `pe-nivel-educativo.txt` , chosen by a keyed hash.

Used by: `PE_NIVEL_EDUCATIVO`

**CONFIGURATION**

|                            |                        |
|----------------------------|------------------------|
| lookupFile                 | pe-nivel-educativo.txt |
| hashMethod                 | SHA256                 |
| maskedValueCase            | PRESERVE_INPUT         |
| inputCaseSensitive         | no                     |
| trimWhitespaceFromInput    | yes                    |
| trimWhitespaceInLookupFile | yes                    |

**PE\_NOMBRE** NAME

Replaces the name with one of the 395 names in `pe-nombres.txt` .

Used by: `PE_PALABRA_NOMBRE`

**CONFIGURATION**

|                    |                |
|--------------------|----------------|
| lookupFile         | pe-nombres.txt |
| maskedValueCase    | PRESERVE_INPUT |
| filterAccent       | yes            |
| inputCaseSensitive | no             |

**PE\_NOMBRE\_COMPLETO** REGEX DECOMPOSE

Tries 7 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM` .

Used by: `PE_L1_NOMBRE_COMPLETO`

**PATTERNS, TRIED IN ORDER**

|                                                                                  |                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>( [^, ]+ ), \s*( .+ )</code>                                               | <code>PE_APELLIDOS</code> . <code>PE_NOMBRES</code>                                                                                                                                                       |
| <code>( \S+ )</code>                                                             | <code>PE_PALABRA_NOMBRE</code>                                                                                                                                                                            |
| <code>( \S+ ) \s+ ( \S+ )</code>                                                 | <code>PE_PALABRA_NOMBRE</code> . <code>PE_PALABRA_APELLIDO</code>                                                                                                                                         |
| <code>( \S+ ) \s+ ( \S+ ) \s+ ( \S+ )</code>                                     | <code>PE_PALABRA_NOMBRE</code> . <code>PE_PALABRA_APELLIDO</code> . <code>PE_PALABRA_APELLIDO</code>                                                                                                      |
| <code>( \S+ ) \s+ ( \S+ ) \s+ ( \S+ ) \s+ ( \S+ )</code>                         | <code>PE_PALABRA_NOMBRE</code> . <code>PE_PALABRA_NOMBRE</code> . <code>PE_PALABRA_APELLIDO</code> . <code>PE_PALABRA_APELLIDO</code>                                                                     |
| <code>( \S+ ) \s+ ( \S+ ) \s+ ( \S+ ) \s+ ( \S+ ) \s+ ( \S+ )</code>             | <code>PE_PALABRA_NOMBRE</code> . <code>PE_PALABRA_NOMBRE</code> . <code>PE_PALABRA_APELLIDO</code> . <code>PE_PALABRA_APELLIDO</code> . <code>PE_PALABRA_APELLIDO</code>                                  |
| <code>( \S+ ) \s+ ( \S+ ) \s+ ( \S+ ) \s+ ( \S+ ) \s+ ( \S+ ) \s+ ( \S+ )</code> | <code>PE_PALABRA_NOMBRE</code> . <code>PE_PALABRA_NOMBRE</code> . <code>PE_PALABRA_NOMBRE</code> . <code>PE_PALABRA_APELLIDO</code> . <code>PE_PALABRA_APELLIDO</code> . <code>PE_PALABRA_APELLIDO</code> |
| otherwise                                                                        | <code>PE_CM_ALFANUM</code>                                                                                                                                                                                |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_NOMBRES** REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PE\_CM\_ALFANUM .

Used by: PE\_L1\_NOMBRE , PE\_NOMBRE\_COMPLETO

PATTERNS, TRIED IN ORDER

|                               |                                                                               |
|-------------------------------|-------------------------------------------------------------------------------|
| (\S+)                         | PE_PALABRA_NOMBRE                                                             |
| (\S+)\s+(\S+)                 | PE_PALABRA_NOMBRE · PE_PALABRA_NOMBRE                                         |
| (\S+)\s+(\S+)\s+(\S+)         | PE_PALABRA_NOMBRE · PE_PALABRA_NOMBRE · PE_PALABRA_NOMBRE                     |
| (\S+)\s+(\S+)\s+(\S+)\s+(\S+) | PE_PALABRA_NOMBRE · PE_PALABRA_NOMBRE · PE_PALABRA_NOMBRE · PE_PALABRA_NOMBRE |
| otherwise                     | PE_CM_ALFANUM                                                                 |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_OCTETO** NUMERIC MAPPING

Maps the number to another between 1 and 254.

Used by: PE\_IP

CONFIGURATION

|          |     |
|----------|-----|
| minValue | 1   |
| maxValue | 254 |

**PE\_OCUPACION** SECURE LOOKUP

Replaces the value with one of the 39 values in pe-ocupaciones.txt , chosen by a keyed hash.

Used by: PE\_OCUPACION\_0\_CODIGO

CONFIGURATION

|                            |                    |
|----------------------------|--------------------|
| lookupFile                 | pe-ocupaciones.txt |
| hashMethod                 | SHA256             |
| maskedValueCase            | PRESERVE_INPUT     |
| inputCaseSensitive         | no                 |
| trimWhitespaceFromInput    | yes                |
| trimWhitespaceInLookupFile | yes                |

**PE\_OCUPACION\_0\_CODIGO** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PE\_OCUPACION .

Used by: PE\_L2\_OCUPACION

PATTERNS, TRIED IN ORDER

|           |               |
|-----------|---------------|
| (\d{2,5}) | PE_CM_DIGITOS |
| otherwise | PE_OCUPACION  |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_ORIENTACION\_SEXUAL** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_ORIENTACION\_SEXUAL\_LISTA .

Used by: PE\_L3\_ORIENTACION\_SEXUAL

PATTERNS, TRIED IN ORDER

|                                        |                             |
|----------------------------------------|-----------------------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA                  |
| (\d{1,6})                              | PE_CM_ALFANUM               |
| otherwise                              | PE_ORIENTACION_SEXUAL_LISTA |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_ORIENTACION\_SEXUAL\_LISTA** SECURE LOOKUP

Replaces the value with one of the 8 values in pe-orientaciones-sexuales.txt , chosen by a keyed hash.

Used by: PE\_ORIENTACION\_SEXUAL

CONFIGURATION

|                            |                               |
|----------------------------|-------------------------------|
| lookupFile                 | pe-orientaciones-sexuales.txt |
| hashMethod                 | SHA256                        |
| maskedValueCase            | PRESERVE_INPUT                |
| inputCaseSensitive         | no                            |
| trimWhitespaceFromInput    | yes                           |
| trimWhitespaceInLookupFile | yes                           |

**PE\_ORIGEN\_ETNICO** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_ORIGEN\_ETNICO\_LISTA .

Used by: PE\_L3\_ORIGEN\_ETNICO

PATTERNS, TRIED IN ORDER

|                                        |                         |
|----------------------------------------|-------------------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA              |
| (\d{1,6})                              | PE_ORIGEN_ETNICO_CODIGO |
| otherwise                              | PE_ORIGEN_ETNICO_LISTA  |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_ORIGEN\_ETNICO\_CODIGO** SECURE LOOKUP

Replaces the value with one of the 10 values in pe-origen-etnico-codigos.txt , chosen by a keyed hash.

Used by: PE\_ORIGEN\_ETNICO

CONFIGURATION

|                    |                              |
|--------------------|------------------------------|
| lookupFile         | pe-origen-etnico-codigos.txt |
| hashMethod         | SHA256                       |
| maskedValueCase    | PRESERVE_INPUT               |
| inputCaseSensitive | no                           |

|                            |     |
|----------------------------|-----|
| trimWhitespaceFromInput    | yes |
| trimWhitespaceInLookupFile | yes |

#### PE\_ORIGEN\_ETNICO\_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in `pe-origen-etnico.txt` , chosen by a keyed hash.

Used by: `PE_ORIGEN_ETNICO`

##### CONFIGURATION

|                            |                                   |
|----------------------------|-----------------------------------|
| lookupFile                 | <code>pe-origen-etnico.txt</code> |
| hashMethod                 | SHA256                            |
| maskedValueCase            | PRESERVE_INPUT                    |
| inputCaseSensitive         | no                                |
| trimWhitespaceFromInput    | yes                               |
| trimWhitespaceInLookupFile | yes                               |

#### PE\_PALABRA\_APELLIDO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_APELLIDO` .

Used by: `PE_APELLIDOS` , `PE_NOMBRE_COMPLETO`

##### PATTERNS, TRIED IN ORDER

|                                                              |                          |
|--------------------------------------------------------------|--------------------------|
| <code>(?i)</code>                                            | keep                     |
| <code>(de del la las los y e san santa van von di da)</code> |                          |
| otherwise                                                    | <code>PE_APELLIDO</code> |

##### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PE\_PALABRA\_NOMBRE REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_NOMBRE` .

Used by: `PE_NOMBRES` , `PE_NOMBRE_COMPLETO`

##### PATTERNS, TRIED IN ORDER

|                                                              |                        |
|--------------------------------------------------------------|------------------------|
| <code>(?i)</code>                                            | keep                   |
| <code>(de del la las los y e san santa van von di da)</code> |                        |
| otherwise                                                    | <code>PE_NOMBRE</code> |

##### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PE\_PARTIDO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_PARTIDO_LISTA` .

Used by: `PE_L3_AFILIACION_PARTIDARIA`

##### PATTERNS, TRIED IN ORDER

|                                                     |                            |
|-----------------------------------------------------|----------------------------|
| <code>(?i)(s[ií] no true false yes [snyx01])</code> | <code>PE_BANDERA</code>    |
| <code>(\d{1,6})</code>                              | <code>PE_CM_ALFANUM</code> |

|           |                  |
|-----------|------------------|
| otherwise | PE_PARTIDO_LISTA |
|-----------|------------------|

#### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PE\_PARTIDO\_LISTA SECURE LOOKUP

Replaces the value with one of the 13 values in `pe-partidos.txt`, chosen by a keyed hash.

Used by: `PE_PARTIDO`

#### CONFIGURATION

|                            |                 |
|----------------------------|-----------------|
| lookupFile                 | pe-partidos.txt |
| hashMethod                 | SHA256          |
| maskedValueCase            | PRESERVE_INPUT  |
| inputCaseSensitive         | no              |
| trimWhitespaceFromInput    | yes             |
| trimWhitespaceInLookupFile | yes             |

#### PE\_PATRIMONIO NUMERIC MAPPING

Maps the number to another between 1,000 and 900,000.

Used by: `PE_FIN_PATRIMONIO`

#### CONFIGURATION

|          |        |
|----------|--------|
| minValue | 1000   |
| maxValue | 900000 |

#### PE\_PROCEDIMIENTO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches:

`PE_PROCEDIMIENTO_TEXTO`.

Used by: `PE_SALUD_PROCEDIMIENTO`

#### PATTERNS, TRIED IN ORDER

|                        |                        |
|------------------------|------------------------|
| <code>(\d{4,6})</code> | PE_CM_DIGITOS          |
| otherwise              | PE_PROCEDIMIENTO_TEXTO |

#### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PE\_PROCEDIMIENTO\_TEXTO SECURE LOOKUP

Replaces the value with one of the 10 values in `pe-procedimientos.txt`, chosen by a keyed hash.

Used by: `PE_PROCEDIMIENTO`

#### CONFIGURATION

|                         |                       |
|-------------------------|-----------------------|
| lookupFile              | pe-procedimientos.txt |
| hashMethod              | SHA256                |
| maskedValueCase         | PRESERVE_INPUT        |
| inputCaseSensitive      | no                    |
| trimWhitespaceFromInput | yes                   |

|                            |     |
|----------------------------|-----|
| trimWhitespaceInLookupFile | yes |
|----------------------------|-----|

**PE\_PROGRAMA\_SOCIAL** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_PROGRAMA\_SOCIAL\_LISTA .

Used by: PE\_FIN\_PROGRAMA\_SOCIAL

**PATTERNS, TRIED IN ORDER**

|                                        |                          |
|----------------------------------------|--------------------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA               |
| (\d{1,6})                              | PE_CM_ALFANUM            |
| otherwise                              | PE_PROGRAMA_SOCIAL_LISTA |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_PROGRAMA\_SOCIAL\_LISTA** SECURE LOOKUP

Replaces the value with one of the 10 values in pe-programas-sociales.txt , chosen by a keyed hash.

Used by: PE\_PROGRAMA\_SOCIAL

**CONFIGURATION**

|                            |                           |
|----------------------------|---------------------------|
| lookupFile                 | pe-programas-sociales.txt |
| hashMethod                 | SHA256                    |
| maskedValueCase            | PRESERVE_INPUT            |
| inputCaseSensitive         | no                        |
| trimWhitespaceFromInput    | yes                       |
| trimWhitespaceInLookupFile | yes                       |

**PE\_PUEBLO\_INDIGENA** REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_PUEBLO\_INDIGENA\_LISTA .

Used by: PE\_L3\_PUEBLO\_INDIGENA

**PATTERNS, TRIED IN ORDER**

|                                        |                          |
|----------------------------------------|--------------------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA               |
| (\d{1,6})                              | PE_CM_ALFANUM            |
| otherwise                              | PE_PUEBLO_INDIGENA_LISTA |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_PUEBLO\_INDIGENA\_LISTA** SECURE LOOKUP

Replaces the value with one of the 21 values in pe-pueblos-indigenas.txt , chosen by a keyed hash.

Used by: PE\_PUEBLO\_INDIGENA

**CONFIGURATION**

|                 |                          |
|-----------------|--------------------------|
| lookupFile      | pe-pueblos-indigenas.txt |
| hashMethod      | SHA256                   |
| maskedValueCase | PRESERVE_INPUT           |

|                            |     |
|----------------------------|-----|
| inputCaseSensitive         | no  |
| trimWhitespaceFromInput    | yes |
| trimWhitespaceInLookupFile | yes |

#### PE\_RAZON\_SOCIAL SECURE LOOKUP

Replaces the value with one of the 20 values in `pe-razones-sociales.txt`, chosen by a keyed hash.

Used by: `PE_L1_RAZON_SOCIAL`

##### CONFIGURATION

|                            |                                      |
|----------------------------|--------------------------------------|
| lookupFile                 | <code>pe-razones-sociales.txt</code> |
| hashMethod                 | <code>SHA256</code>                  |
| maskedValueCase            | <code>PRESERVE_LOOKUP_FILE</code>    |
| inputCaseSensitive         | no                                   |
| trimWhitespaceFromInput    | yes                                  |
| trimWhitespaceInLookupFile | yes                                  |

#### PE\_REDACTAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `PE_L1_CREDENCIAL`, `PE_L3_BIOMETRICO`, `PE_L3_GENETICO`

##### PATTERNS, TRIED IN ORDER

|                   |                                 |
|-------------------|---------------------------------|
| <code>(.+)</code> | each character → <code>X</code> |
| otherwise         | keep                            |

##### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PE\_RELIGION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_RELIGION_LISTA`.

Used by: `PE_L3_RELIGION`

##### PATTERNS, TRIED IN ORDER

|                                                     |                                |
|-----------------------------------------------------|--------------------------------|
| <code>(?i)(s i l no true false yes [snyx01])</code> | <code>PE_BANDERA</code>        |
| <code>(\d{1,6})</code>                              | <code>PE_CM_ALFANUM</code>     |
| otherwise                                           | <code>PE_RELIGION_LISTA</code> |

##### CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

#### PE\_RELIGION\_LISTA SECURE LOOKUP

Replaces the value with one of the 11 values in `pe-religiones.txt`, chosen by a keyed hash.

Used by: `PE_RELIGION`

##### CONFIGURATION

|                 |                                |
|-----------------|--------------------------------|
| lookupFile      | <code>pe-religiones.txt</code> |
| hashMethod      | <code>SHA256</code>            |
| maskedValueCase | <code>PRESERVE_INPUT</code>    |

|                            |     |
|----------------------------|-----|
| inputCaseSensitive         | no  |
| trimWhitespaceFromInput    | yes |
| trimWhitespaceInLookupFile | yes |

**PE\_RUC** STRING ALGORITHM CHAIN

Applies PE\_RUC\_CD → PE\_RUC\_ESPACIO → PE\_RUC\_AUX → PE\_RUC\_VERIFICADOR , in this order.

Used by: PE\_L1\_RUC , PE\_DOCUMENTO

**CONFIGURATION**

|                        |                    |
|------------------------|--------------------|
| algorithmReferences[1] | PE_RUC_CD          |
| algorithmReferences[2] | PE_RUC_ESPACIO     |
| algorithmReferences[3] | PE_RUC_AUX         |
| algorithmReferences[4] | PE_RUC_VERIFICADOR |

**PE\_RUC\_AUX** CHECK DIGIT

Masks the digits with PE\_IDENTIDAD and recomputes a modulus 11 check digit with weights 0 4 6 8 10 1 3 4 6 8 10 from the right.

Used by: PE\_RUC

**CONFIGURATION**

|                                          |                         |
|------------------------------------------|-------------------------|
| weightList                               | 0 4 6 8 10 1 3 4 6 8 10 |
| modulusNumber                            | 11                      |
| checkDigitIndex                          | 11                      |
| numDigitsForCheckdigitCalculation        | 11                      |
| calculateChecksumRightToLeft             | yes                     |
| numericAlgorithm                         | PE_IDENTIDAD            |
| alphaNumericAlgorithm                    | PE_IDENTIDAD            |
| preserveRegex                            | [.\-\\s]                |
| inputHandlingConfig.characterHandling    | STANDARD                |
| inputHandlingConfig.invalidInputHandling | ERROR                   |
| inputHandlingConfig.shortInputHandling   | FALLBACK                |
| inputHandlingConfig.padCharacter         | 0                       |
| inputHandlingConfig.trimWhitespace       | yes                     |

**PE\_RUC\_CD** CHECK DIGIT

Masks the digits with PE\_RUC\_CUERPO and recomputes a modulus 11 check digit with weights 2 3 4 5 6 7 2 3 4 5 from the right.

Used by: PE\_RUC

**CONFIGURATION**

|                                   |                     |
|-----------------------------------|---------------------|
| weightList                        | 2 3 4 5 6 7 2 3 4 5 |
| modulusNumber                     | 11                  |
| checkDigitIndex                   | 10                  |
| numDigitsForCheckdigitCalculation | 10                  |
| calculateChecksumRightToLeft      | yes                 |
| numericAlgorithm                  | PE_RUC_CUERPO       |
| alphaNumericAlgorithm             | PE_RUC_CUERPO       |
| preserveRegex                     | [.\-\\s]            |

|                                          |          |
|------------------------------------------|----------|
| inputHandlingConfig.characterHandling    | STANDARD |
| inputHandlingConfig.invalidInputHandling | ERROR    |
| inputHandlingConfig.shortInputHandling   | FALLBACK |
| inputHandlingConfig.padCharacter         | 0        |
| inputHandlingConfig.trimWhitespace       | yes      |

**PE\_RUC\_CUERPO** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PE\_CM\_DIGITOS .

Used by: PE\_RUC\_CD

**PATTERNS, TRIED IN ORDER**

|                |                      |
|----------------|----------------------|
| (\d{2})(\d{8}) | keep · PE_CM_DIGITOS |
| otherwise      | PE_CM_DIGITOS        |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_RUC\_ESPACIO** REGEX DECOMPOSE

Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PE\_RUC

**PATTERNS, TRIED IN ORDER**

|             |                        |
|-------------|------------------------|
| (\d{10})(0) | keep · replace with 00 |
| (\d{10})(1) | keep · replace with 10 |
| (\d{10})(2) | keep · replace with 20 |
| (\d{10})(3) | keep · replace with 30 |
| (\d{10})(4) | keep · replace with 40 |
| (\d{10})(5) | keep · replace with 50 |
| (\d{10})(6) | keep · replace with 60 |
| (\d{10})(7) | keep · replace with 70 |
| (\d{10})(8) | keep · replace with 80 |
| (\d{10})(9) | keep · replace with 90 |
| otherwise   | keep                   |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_RUC\_VERIFICADOR** REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PE\_RUC

**PATTERNS, TRIED IN ORDER**

|               |                       |
|---------------|-----------------------|
| (\d{10})(00)  | keep · replace with 1 |
| (\d{10})(0\d) | keep · replace with 0 |
| (\d{10})(1\d) | keep · replace with 1 |
| (\d{10})(2\d) | keep · replace with 2 |

|               |                       |
|---------------|-----------------------|
| (\d{10})(3\d) | keep · replace with 3 |
| (\d{10})(4\d) | keep · replace with 4 |
| (\d{10})(5\d) | keep · replace with 5 |
| (\d{10})(6\d) | keep · replace with 6 |
| (\d{10})(7\d) | keep · replace with 7 |
| (\d{10})(8\d) | keep · replace with 8 |
| (\d{10})(9\d) | keep · replace with 9 |
| otherwise     | keep                  |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_SEX0** REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PE\_L2\_SEX0

**PATTERNS, TRIED IN ORDER**

|                          |                            |
|--------------------------|----------------------------|
| (?i)(femenino masculino) | PE_SEX0_FEMENINO_MASCULINO |
| (?i)(mujer hombre)       | PE_SEX0_MUJER_HOMBRE       |
| (?i)(female male)        | PE_SEX0_FEMALE_MALE        |
| (?i)([fm])               | PE_SEX0_F_M                |
| (?i)([hm])               | PE_SEX0_H_M                |
| ([12])                   | PE_SEX0_1_2                |
| otherwise                | keep                       |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_SEX0\_1\_2** SECURE LOOKUP

Replaces the value with one of the 2 values in pe-sexo-1-2.txt , chosen by a keyed hash.

Used by: PE\_SEX0

**CONFIGURATION**

|                            |                 |
|----------------------------|-----------------|
| lookupFile                 | pe-sexo-1-2.txt |
| hashMethod                 | SHA256          |
| maskedValueCase            | PRESERVE_INPUT  |
| inputCaseSensitive         | no              |
| trimWhitespaceFromInput    | yes             |
| trimWhitespaceInLookupFile | yes             |

**PE\_SEX0\_F\_M** SECURE LOOKUP

Replaces the value with one of the 2 values in pe-sexo-f-m.txt , chosen by a keyed hash.

Used by: PE\_SEX0

**CONFIGURATION**

|            |                 |
|------------|-----------------|
| lookupFile | pe-sexo-f-m.txt |
|------------|-----------------|

|                            |                |
|----------------------------|----------------|
| hashMethod                 | SHA256         |
| maskedValueCase            | PRESERVE_INPUT |
| inputCaseSensitive         | no             |
| trimWhitespaceFromInput    | yes            |
| trimWhitespaceInLookupFile | yes            |

#### PE\_SEXO\_FEMALE\_MALE SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-female-male.txt` , chosen by a keyed hash.

Used by: `PE_SEX0`

##### CONFIGURATION

|                            |                         |
|----------------------------|-------------------------|
| lookupFile                 | pe-sexo-female-male.txt |
| hashMethod                 | SHA256                  |
| maskedValueCase            | PRESERVE_INPUT          |
| inputCaseSensitive         | no                      |
| trimWhitespaceFromInput    | yes                     |
| trimWhitespaceInLookupFile | yes                     |

#### PE\_SEXO\_FEMENINO\_MASCULINO SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-femenino-masculino.txt` , chosen by a keyed hash.

Used by: `PE_SEX0`

##### CONFIGURATION

|                            |                                |
|----------------------------|--------------------------------|
| lookupFile                 | pe-sexo-femenino-masculino.txt |
| hashMethod                 | SHA256                         |
| maskedValueCase            | PRESERVE_INPUT                 |
| inputCaseSensitive         | no                             |
| trimWhitespaceFromInput    | yes                            |
| trimWhitespaceInLookupFile | yes                            |

#### PE\_SEXO\_H\_M SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-h-m.txt` , chosen by a keyed hash.

Used by: `PE_SEX0`

##### CONFIGURATION

|                            |                 |
|----------------------------|-----------------|
| lookupFile                 | pe-sexo-h-m.txt |
| hashMethod                 | SHA256          |
| maskedValueCase            | PRESERVE_INPUT  |
| inputCaseSensitive         | no              |
| trimWhitespaceFromInput    | yes             |
| trimWhitespaceInLookupFile | yes             |

#### PE\_SEXO\_MUJER\_HOMBRE SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-mujer-hombre.txt` , chosen by a keyed hash.

Used by: `PE_SEX0`

##### CONFIGURATION

|            |                          |
|------------|--------------------------|
| lookupFile | pe-sexo-mujer-hombre.txt |
|------------|--------------------------|

|                            |                |
|----------------------------|----------------|
| hashMethod                 | SHA256         |
| maskedValueCase            | PRESERVE_INPUT |
| inputCaseSensitive         | no             |
| trimWhitespaceFromInput    | yes            |
| trimWhitespaceInLookupFile | yes            |

**PE\_SUPRIMIR** SECURE LOOKUP

Replaces the value with one of the 1 values in `pe-sin-informacion.txt` , chosen by a keyed hash.

Used by: `PE_CATEGORIA_SUPRIMIDA`

**CONFIGURATION**

|                            |                        |
|----------------------------|------------------------|
| lookupFile                 | pe-sin-informacion.txt |
| hashMethod                 | SHA256                 |
| maskedValueCase            | PRESERVE_LOOKUP_FILE   |
| inputCaseSensitive         | no                     |
| trimWhitespaceFromInput    | yes                    |
| trimWhitespaceInLookupFile | yes                    |

**PE\_TARJETA** REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `PE_L1_TARJETA`

**PATTERNS, TRIED IN ORDER**

|                                   |                 |
|-----------------------------------|-----------------|
| <code>(\d[\d\s-]{11,22}\d)</code> | PE_TARJETA_LUHN |
| otherwise                         | keep            |

**CONFIGURATION**

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

**PE\_TARJETA\_LUHN** PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

Used by: `PE_TARJETA`

**CONFIGURATION**

|                    |   |
|--------------------|---|
| minMaskedPositions | 6 |
| preserve           | 0 |

**PE\_TELEFONO** REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_DIGITOS` .

Used by: `PE_L1_TELEFONO`

**PATTERNS, TRIED IN ORDER**

|                                                                        |                                                                                                                  |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <code>(\+?51[\s-]?)?(9)(\d{2})([\s-]?) (\d{3}) ([\s-]?) (\d{3})</code> | keep · keep · <code>PE_CM_DIGITOS</code> · keep · <code>PE_CM_DIGITOS</code> · keep · <code>PE_CM_DIGITOS</code> |
| <code>(\(?0?1\)?[\s-]?) (\d{3})([\s-]?) (\d{4})</code>                 | keep · <code>PE_CM_DIGITOS</code> · keep · <code>PE_CM_DIGITOS</code>                                            |
| <code>(\(?0?[2-8]\d\)?[\s-]?) (\d{3})([\s-]?) (\d{3})</code>           | keep · <code>PE_CM_DIGITOS</code> · keep · <code>PE_CM_DIGITOS</code>                                            |
| otherwise                                                              | <code>PE_CM_DIGITOS</code>                                                                                       |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_TENENCIA\_VIVIENDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PE\_TENENCIA\_VIVIENDA\_LISTA .

Used by: PE\_FIN\_VIVIENDA

PATTERNS, TRIED IN ORDER

|                                        |                            |
|----------------------------------------|----------------------------|
| (?i)(s[ií] no true false yes [snyx01]) | PE_BANDERA                 |
| (\d{1,6})                              | PE_CM_ALFANUM              |
| otherwise                              | PE_TENENCIA_VIVIENDA_LISTA |

CONFIGURATION

|             |     |
|-------------|-----|
| trimInput   | yes |
| requireMask | no  |

PE\_TENENCIA\_VIVIENDA\_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in pe-tenencia-vivienda.txt , chosen by a keyed hash.

Used by: PE\_TENENCIA\_VIVIENDA

CONFIGURATION

|                            |                          |
|----------------------------|--------------------------|
| lookupFile                 | pe-tenencia-vivienda.txt |
| hashMethod                 | SHA256                   |
| maskedValueCase            | PRESERVE_INPUT           |
| inputCaseSensitive         | no                       |
| trimWhitespaceFromInput    | yes                      |
| trimWhitespaceInLookupFile | yes                      |

PE\_TEXTO\_LIBRE FREE TEXT REDACTION

Replaces the text matching 8 patterns with [DATO] , and the 1,906 terms of pe-texto-nombres.txt with [NOMBRE] .

Used by: PE\_L1\_TEXTO\_LIBRE , PE\_SALUD\_TEXTO\_CLINICO

PATTERNS, TRIED IN ORDER

|                                                                                   |
|-----------------------------------------------------------------------------------|
| (?<!\d)\d{8}-[0-9A-JKa-jk](?!\\d)                                                 |
| (?<!\d)\d{8}(?!\\d)                                                               |
| (?<!\d)(10 15 17 20)\d{9}(?!\\d)                                                  |
| [\\w.+\\-]+@[\\w\\-]+(\\. [\\w\\-]+)+                                             |
| (?<![\\d\\-])(\\+?51[\\s\\-]?)?9\\d{2}[\\s\\-]?\\d{3}[\\s\\-]?\\d{3}(?! [\\d\\-]) |
| (?<!\d)\d{13,19}(?!\\d)                                                           |
| (?<![A-Za-z])([A-Z][0-9A-Z][A-Z][\\s\\-]?\\d{3})(?! [A-Za-z\\d])                  |
| (?<!\d)(\\d{1,3}\\.){3}\\d{1,3}(?!\\d)                                            |

CONFIGURATION

|                       |                      |
|-----------------------|----------------------|
| regExRedactValue      | [DATO]               |
| lookupFile            | pe-texto-nombres.txt |
| lookupFileRedactValue | [NOMBRE]             |

|            |     |
|------------|-----|
| isDenyList | yes |
|------------|-----|

#### PE\_UBIGEO DATA CLEANSING

Replaces each value found in `pe-ubigeo-generalizado.txt` (1,666 pairs) by its counterpart; values not in the file stay as they are.

Used by: `PE_L2_UBIGEO`

##### CONFIGURATION

|                |                                         |
|----------------|-----------------------------------------|
| lookupFile     | <code>pe-ubigeo-generalizado.txt</code> |
| delimiter      | <code> </code>                          |
| caseSensitive  | yes                                     |
| trimWhitespace | yes                                     |

#### PE\_URBANIZACION SECURE LOOKUP

Replaces the value with one of the 104 values in `pe-urbanizaciones.txt`, chosen by a keyed hash.

Used by: `PE_L2_URBANIZACION`

##### CONFIGURATION

|                            |                                    |
|----------------------------|------------------------------------|
| lookupFile                 | <code>pe-urbanizaciones.txt</code> |
| hashMethod                 | <code>SHA256</code>                |
| maskedValueCase            | <code>PRESERVE_LOOKUP_FILE</code>  |
| inputCaseSensitive         | no                                 |
| trimWhitespaceFromInput    | yes                                |
| trimWhitespaceInLookupFile | yes                                |

#### PE\_VALOR\_DEUDA NUMERIC MAPPING

Maps the number to another between 100 and 500,000.

Used by: `PE_FIN_DEUDA`

##### CONFIGURATION

|          |                     |
|----------|---------------------|
| minValue | <code>100</code>    |
| maxValue | <code>500000</code> |

#### PE\_VALOR\_PENSION NUMERIC MAPPING

Maps the number to another between 300 and 8,000.

Used by: `PE_FIN_PENSION`

##### CONFIGURATION

|          |                   |
|----------|-------------------|
| minValue | <code>300</code>  |
| maxValue | <code>8000</code> |

#### PE\_ZONA\_RURAL SECURE LOOKUP

Replaces the value with one of the 1 values in `pe-zona-rural.txt`, chosen by a keyed hash.

Used by: `PE_L2_CENTRO_POBLADO`

##### CONFIGURATION

|                 |                                   |
|-----------------|-----------------------------------|
| lookupFile      | <code>pe-zona-rural.txt</code>    |
| hashMethod      | <code>SHA256</code>               |
| maskedValueCase | <code>PRESERVE_LOOKUP_FILE</code> |

|                            |     |
|----------------------------|-----|
| inputCaseSensitive         | no  |
| trimWhitespaceFromInput    | yes |
| trimWhitespaceInLookupFile | yes |

## APPENDIX B

# Files

The lookup and detection files the set reads. On a Masking Engine each one is uploaded with the object that reads it — the lookup file of a Secure Lookup or Data Cleansing algorithm, the value list of a List classifier — and referenced by that object's configuration.

| FILE                                                    | LINES | USED BY                                                      | FIRST LINES                                                                        |
|---------------------------------------------------------|-------|--------------------------------------------------------------|------------------------------------------------------------------------------------|
| pe-<br>antecedentes.txt                                 | 7     | PE_ANTECEDENTES_LIST<br>A                                    | No registra antecedentes penales · Registra antecedentes · Procesado · Sentenciado |
| pe-<br>apellidos.txt                                    | 422   | PE_APELLIDO                                                  | Quispe · Mamani · Flores · Huamán                                                  |
| pe-aptitud-<br>laboral.txt                              | 4     | PE_APTITUD_LABORAL_LISTA                                     | Apto · Apto con restricciones · No apto · Pendiente                                |
| pe-bandera-0-<br>1.txt                                  | 2     | PE_BANDERA_0_1                                               | 0 · 1                                                                              |
| pe-bandera-s-<br>n.txt                                  | 2     | PE_BANDERA_S_N                                               | S · N                                                                              |
| pe-bandera-si-<br>no.txt                                | 2     | PE_BANDERA_SI_NO                                             | Sí · No                                                                            |
| pe-bandera-<br>true-false.txt                           | 2     | PE_BANDERA_TRUE_FALSE<br>E                                   | true · false                                                                       |
| pe-bandera-y-<br>n.txt                                  | 2     | PE_BANDERA_Y_N                                               | Y · N                                                                              |
| pe-bandera-yes-<br>no.txt                               | 2     | PE_BANDERA_YES_NO                                            | Yes · No                                                                           |
| pe-cie10-<br>decimal.txt                                | 33    | PE_CIE10_DECIMAL                                             | E11.9 · E78.5 · J45.9 · J06.9                                                      |
| pe-cie10.txt                                            | 37    | PE_CIE10                                                     | I10 · E11 · E78 · J45                                                              |
| pe-<br>clasificacion-<br>socioeconomica-<br>codigos.txt | 4     | PE_CLASIFICACION_SOCIOECONOMICA_CODIGO                       | 1 · 2 · 3 · 4                                                                      |
| pe-<br>clasificacion-<br>socioeconomica.t<br>xt         | 4     | PE_CLASIFICACION_SOCIOECONOMICA_LISTA                        | Pobre extremo · Pobre · No pobre · Sin clasificación                               |
| pe-cobertura-<br>salud.txt                              | 8     | PE_COBERTURA                                                 | EsSalud · SIS (Seguro Integral de Salud) · EPS · Sanidad de las Fuerzas Armadas    |
| pe-detectar-<br>apellidos.txt                           | 519   | PE_L1_APELLIDO -<br>List<br>PE_L1_NOMBRE_COMPLET<br>O - List | Quispe · Mamani · Flores · Huamán                                                  |
| pe-detectar-<br>aptitud-<br>laboral.txt                 | 5     | PE_SALUD_OCUPACIONAL<br>- List                               | apto · no apto · apto con restricciones · pendiente                                |

| FILE                                         | LINES | USED BY                                                     | FIRST LINES                                                                            |
|----------------------------------------------|-------|-------------------------------------------------------------|----------------------------------------------------------------------------------------|
| pe-detectar-clasificacion-socioeconomica.txt | 13    | PE_FIN_CLASIFICACION_SOCIOECONOMICA – List                  | Pobre extremo · Pobre · No pobre · Sin clasificación                                   |
| pe-detectar-cobertura.txt                    | 16    | PE_SALUD_COBERTURA – List                                   | EsSalud · SIS (Seguro Integral de Salud) · EPS · Sanidad de las Fuerzas Armadas        |
| pe-detectar-codigos-banderas.txt             | 11    | PE_L3_ORIGEN_ETNICO – List<br>PE_FIN_PROGRAMA_SOCIAL – List | s · n · sí · si                                                                        |
| pe-detectar-discapacidades.txt               | 26    | PE_SALUD_DISCAPACIDAD – List                                | Física o motora · Física o motora · Auditiva · Visual                                  |
| pe-detectar-districtos.txt                   | 1,981 | PE_L2_DISTRITO – List                                       | Chachapoyas · Asunción · Asuncion · Balsas                                             |
| pe-detectar-estado-civil.txt                 | 20    | PE_L2_ESTADO_CIVIL – List                                   | Soltero/a · Casado/a · Conviviente · Separado/a                                        |
| pe-detectar-grupos-sanguineos.txt            | 18    | PE_SALUD_GRUPO_SANGUINEO – List                             | A+ · A- · B+ · B-                                                                      |
| pe-detectar-identidades-genero.txt           | 19    | PE_L3_IDENTIDAD_GENERO – List                               | Mujer cisgénero · Mujer cisgenero · Hombre cisgénero · Hombre cisgenero                |
| pe-detectar-ideologias-politicas.txt         | 16    | PE_L3_OPINION_POLITICA – List                               | Izquierda · Centroizquierda · Centro · Centroderecha                                   |
| pe-detectar-lenguas.txt                      | 37    | PE_L3 LENGUA – List                                         | Castellano · Quechua · Aimara · Asháninka                                              |
| pe-detectar-medicamentos.txt                 | 50    | PE_SALUD_MEDICAMENTO – List                                 | Paracetamol · Ibuprofeno · Diclofenaco · Metformina                                    |
| pe-detectar-nacionalidades.txt               | 41    | PE_L3_NACIONALIDAD – List                                   | Peruana · Venezolana · Colombiana · Ecuatoriana                                        |
| pe-detectar-nivel-educativo.txt              | 26    | PE_L2_NIVEL_EDUCATIVO – List                                | Sin nivel · Inicial · Primaria incompleta · Primaria completa                          |
| pe-detectar-nombres.txt                      | 456   | PE_L1_NOMBRE – List<br>PE_L1_NOMBRE_COMPLETO – List         | José · Jose · María · Maria                                                            |
| pe-detectar-orientaciones-sexuales.txt       | 13    | PE_L3_ORIENTACION_SEXUAL – List                             | Heterosexual · Gay · Lesbiana · Bisexual                                               |
| pe-detectar-origen-etnico.txt                | 29    | PE_L3_ORIGEN_ETNICO – List                                  | Quechua · Aimara · Nativo o indígena de la Amazonía · Nativo o indígena de la Amazonia |
| pe-detectar-partidos.txt                     | 26    | PE_L3_AFILIACION_PARTIDARIA – List                          | Fuerza Popular · Perú Libre · Peru Libre · Acción Popular                              |

| FILE                               | LINES  | USED BY                          | FIRST LINES                                                                                                                                                                        |
|------------------------------------|--------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pe-detectar-programas-sociales.txt | 24     | PE_FIN_PROGRAMA_SOCIAL – List    | Juntos · Pensión 65 · Pension 65 · Contigo                                                                                                                                         |
| pe-detectar-pueblos.txt            | 45     | PE_L3_PUEBLO_INDIGENA – List     | Quechuas · Aimara · Asháninka · Ashaninka                                                                                                                                          |
| pe-detectar-religiones.txt         | 37     | PE_L3_RELIGION – List            | Católica · Catolica · Evangélica · Evangelica                                                                                                                                      |
| pe-detectar-sexo.txt               | 9      | PE_L2_SEX0 – List                | f · m · h · femenino                                                                                                                                                               |
| pe-detectar-sindicatos.txt         | 13     | PE_L3_AFILIACION_SINDICAL – List | CGTP · CUT Perú · CUT Peru · CTP                                                                                                                                                   |
| pe-detectar-tenencia-vivienda.txt  | 17     | PE_FIN_VIVIENDA – List           | Propia totalmente pagada · Propia pagándola a plazos · Propia pagandola a plazos · Alquilada                                                                                       |
| pe-diagnosticos.txt                | 30     | PE_DIAGNOSTICO_TEXT0             | Hipertensión arterial · Diabetes mellitus tipo 2 · Asma · Lumbalgia                                                                                                                |
| pe-direcciones.txt                 | 4,000  | PE_DIRECCION                     | Alameda Brasil 918 Int. 592 · Pasaje Pachacútec 1787, Urb. Zárate · Alameda Benavides 598, Urb. San Isidro · AA.HH. Nueva Esperanza, Jr. Brasil 787                                |
| pe-discapacidades.txt              | 8      | PE_DISCAPACIDAD_LISTA            | Física o motora · Auditiva · Visual · Sordoceguera                                                                                                                                 |
| pe-distritos-generalizados.txt     | 44,952 | PE_DISTRITO                      | Asunción Los Baños del Inca · Asunción - Chachapoyas Chachapoyas - Chachapoyas · Asunción, Chachapoyas Chachapoyas, Chachapoyas · Asunción (Chachapoyas) Chachapoyas (Chachapoyas) |
| pe-empleadores.txt                 | 24     | PE_EMPLEADOR                     | Distribuidora Andina S.A.C. · Constructora Pacífico S.A. · Transportes El Inca S.A.C. · Supermercados del Sur S.A.                                                                 |
| pe-estado-civil-codigos.txt        | 6      | PE_ESTADO_CIVIL_CODIGO           | 1 · 2 · 3 · 4                                                                                                                                                                      |
| pe-estado-civil.txt                | 6      | PE_ESTADO_CIVIL_LISTA            | Soltero/a · Casado/a · Conviviente · Separado/a                                                                                                                                    |
| pe-estados-credito.txt             | 11     | PE_ESTADO_CREDITO                | Normal · Con problema potencial · Deficiente · Dudoso                                                                                                                              |
| pe-grupos-sanguineos.txt           | 8      | PE_GRUPO_SANGUINEO               | A+ · A- · B+ · B-                                                                                                                                                                  |
| pe-identidades-genero.txt          | 7      | PE_IDENTIDAD_GENERO_LISTA        | Mujer cisgénero · Hombre cisgénero · Mujer trans · Hombre trans                                                                                                                    |
| pe-ideologias-politicas.txt        | 7      | PE_IDEOLOGIA_POLITICA_LISTA      | Izquierda · Centroizquierda · Centro · Centroderecha                                                                                                                               |
| pe-instituciones-educativas.txt    | 12     | PE_INSTITUCION_EDUCATIVA         | I.E. 1234 Nuestra Señora del Carmen · I.E. José Carlos Mariátegui · I.E.P. San Agustín · Colegio Nacional Mariscal Cáceres                                                         |
| pe-lenguas.txt                     | 18     | PE LENGUA LISTA                  | Castellano · Quechua · Aimara · Asháninka                                                                                                                                          |
| pe-medicamentos.txt                | 22     | PE_MEDICAMENTO_LISTA             | Paracetamol · Ibuprofeno · Diclofenaco · Metformina                                                                                                                                |

| FILE                           | LINES | USED BY                     | FIRST LINES                                                                                                                  |
|--------------------------------|-------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| pe-nacionalidades.txt          | 14    | PE_NACIONALIDAD             | Peruana · Venezolana · Colombiana · Ecuatoriana                                                                              |
| pe-nivel-educativo-codigos.txt | 11    | PE_NIVEL_EDUCATIVO_CODIGO   | 1 · 2 · 3 · 4                                                                                                                |
| pe-nivel-educativo.txt         | 11    | PE_NIVEL_EDUCATIVO_LISTA    | Sin nivel · Inicial · Primaria incompleta · Primaria completa                                                                |
| pe-nombres.txt                 | 395   | PE_NOMBRE                   | José · María · Juan · Carlos                                                                                                 |
| pe-ocupaciones.txt             | 39    | PE_OCUPACION                | Empleado administrativo · Vendedor · Cajero · Chofer                                                                         |
| pe-orientaciones-sexuales.txt  | 8     | PE_ORIENTACION_SEXUAL_LISTA | Heterosexual · Gay · Lesbiana · Bisexual                                                                                     |
| pe-origen-etnico-codigos.txt   | 10    | PE_ORIGEN_ETNICO_CODIGO     | 1 · 2 · 3 · 4                                                                                                                |
| pe-origen-etnico.txt           | 10    | PE_ORIGEN_ETNICO_LISTA      | Quechua · Aimara · Nativo o indígena de la Amazonía · Perteneciente a otro pueblo indígena u originario                      |
| pe-partidos.txt                | 13    | PE_PARTIDO_LISTA            | Fuerza Popular · Perú Libre · Acción Popular · Alianza para el Progreso                                                      |
| pe-procedimientos.txt          | 10    | PE_PROCEDIMIENTO_TEXTO      | Consulta médica general · Consulta de emergencia · Hemograma completo · Glucosa en sangre                                    |
| pe-programas-sociales.txt      | 10    | PE_PROGRAMA_SOCIAL_LISTA    | Juntos · Pensión 65 · Contigo · Cuna Más                                                                                     |
| pe-pueblos-indigenas.txt       | 21    | PE_PUEBLO_INDIGENA_LISTA    | Quechuas · Aimara · Asháninka · Awajún                                                                                       |
| pe-razones-sociales.txt        | 20    | PE_RAZON_SOCIAL             | Inversiones Andinas S.A.C. · Comercial Los Incas E.I.R.L. · Transportes Altiplano S.A.C. · Distribuidora del Pacífico S.R.L. |
| pe-religiones.txt              | 11    | PE_RELIGION_LISTA           | Católica · Evangélica · Cristiana · Adventista                                                                               |
| pe-sexo-1-2.txt                | 2     | PE_SEXO_1_2                 | 1 · 2                                                                                                                        |
| pe-sexo-f-m.txt                | 2     | PE_SEXO_F_M                 | F · M                                                                                                                        |
| pe-sexo-female-male.txt        | 2     | PE_SEXO_FEMALE_MALE         | Female · Male                                                                                                                |
| pe-sexo-femenino-masculino.txt | 2     | PE_SEXO_FEMENINO_MASCULINO  | Femenino · Masculino                                                                                                         |
| pe-sexo-h-m.txt                | 2     | PE_SEXO_H_M                 | H · M                                                                                                                        |
| pe-sexo-mujer-hombre.txt       | 2     | PE_SEXO_MUJER_HOMBRE        | Mujer · Hombre                                                                                                               |
| pe-sin-informacion.txt         | 1     | PE_SUPRIMIR                 | Sin información                                                                                                              |
| pe-tenencia-vivienda.txt       | 6     | PE_TENENCIA_VIVIENDA_LISTA  | Propia totalmente pagada · Propia pagándola a plazos · Alquilada · Cedida por el centro de trabajo u otro hogar              |

| FILE                       | LINES | USED BY         | FIRST LINES                                                                                |
|----------------------------|-------|-----------------|--------------------------------------------------------------------------------------------|
| pe-texto-nombres.txt       | 1,906 | PE_TEXT0_LIBRE  | José · JOSÉ · Jose · JOSE                                                                  |
| pe-ubigeo-generalizado.txt | 1,666 | PE_UBIGE0       | 010102 010101 · 010103 010101 · 010104 010101 · 010105 010101                              |
| pe-urbanizaciones.txt      | 104   | PE_URBANIZACION | Urb. San Borja Norte · Urb. Las Gardenias · Urb. Monterrico · Urb. Chacarilla del Estanque |
| pe-zona-rural.txt          | 1     | PE_ZONA_RURAL   | Zona rural                                                                                 |

## Exclusion lists

Words that keep a column-name rule from matching when they follow it — so that `NOMBRE_PRODUCTO` is not taken for a person's name.

**E1** producto prod item articulo empresa emp razon social comercial archivo calle avenida jiron pasaje urbanizacion asentamiento caserio anexo centro poblado barrio distrito provincia departamento region depto dpto pais banco sucursal agencia plan campana proyecto servicio tabla columna campo usuario host servidor dominio marca modelo categoria tipo escuela colegio institucion universidad curso materia documento doc unidad clinica hospital essalud seguro area dependencia cargo sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento diagnostico estudio programa beneficio partido sindicato religion pueblo lengua idioma lugar sede local tienda almacen proveedor convenio contrato poliza aseguradora entidad organismo organizacion parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor factura comprobante pago forma zona ruta linea actividad rubro sector file table column field user server product company category type school document department system group report role account place store supplier contract device image page module action key label code project city country street bank cliente cli paciente pac usuario afiliado asegurado beneficiario benef titular empleado trabajador colaborador alumno estudiante deudor codeudor fiador avalista solicitante propietario arrendatario inquilino conductor responsable representante apoderado persona ciudadano contribuyente madre padre conyuge conviviente contacto referencia victima denunciante imputado procesado testigo medico docente socio heredero donante tutor jubilado pensionista derechohabiente customer patient employee member holder mother father spouse contact person student driver owner completo apellidos? usuario corto largo

**E2** producto prod item articulo empresa emp razon social comercial archivo calle avenida jiron pasaje urbanizacion asentamiento caserio anexo centro poblado barrio distrito provincia departamento region depto dpto pais banco sucursal agencia plan campana proyecto servicio tabla columna campo usuario host servidor dominio marca modelo categoria tipo escuela colegio institucion universidad curso materia documento doc unidad clinica hospital essalud seguro area dependencia cargo sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento diagnostico estudio programa beneficio partido sindicato religion pueblo lengua idioma lugar sede local tienda almacen proveedor convenio contrato poliza aseguradora entidad organismo organizacion parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor factura comprobante pago forma zona ruta linea actividad rubro sector file table column field user server product company category type school document department system group report role account place store supplier contract device image page module action key label code project city country street bank cliente cli paciente pac usuario afiliado asegurado beneficiario benef titular empleado trabajador colaborador alumno estudiante deudor codeudor fiador avalista solicitante propietario arrendatario inquilino conductor responsable representante apoderado persona ciudadano contribuyente madre padre conyuge conviviente contacto referencia victima denunciante imputado procesado testigo medico docente socio heredero donante tutor jubilado pensionista derechohabiente customer patient employee member holder mother father spouse contact person student driver owner

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**E3** producto prod item articulo empresa emp razon social comercial archivo calle avenida jiron pasaje urbanizacion asentamiento caserio anexo centro poblado barrio distrito provincia departamento region depto dpto pais banco sucursal agencia plan campana proyecto servicio tabla columna campo usuario host servidor dominio marca modelo categoria tipo escuela colegio institucion universidad curso materia documento doc unidad clinica hospital essalud seguro area dependencia cargo sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento diagnostico estudio programa beneficio partido sindicato religion pueblo lengua idioma lugar sede local tienda almacen proveedor convenio contrato poliza aseguradora entidad organismo organizacion parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor factura comprobante pago forma zona ruta linea actividad rubro sector file table column field user server product company category type school document department system group report role account place store supplier contract device image page module action key label code project city country street bank cliente cli paciente pac usuario afiliado asegurado beneficiario benef titular empleado trabajador colaborador alumno estudiante deudor codeudor fiador avalista solicitante propietario arrendatario inquilino conductor responsable representante apoderado persona ciudadano contribuyente madre padre conyuge conviviente contacto referencia victima denunciante imputado procesado testigo medico docente socio heredero donante tutor jubilado pensionista derechohabiente customer patient employee member holder mother father spouse contact person student driver owner full complete last first

---

**E4** ip mac email e\_?mail correo electr[ó]nica web url tipo id general regional

---

**E5** ip mac email e\_?mail web url type id

---

**E6** id cd cod codigo tipo fecha estado num nro

---

**E7** id tipo codigo fecha estado cola log

---

**E8** cod codigo id judicial electoral fiscal escolar de\_?riego

---

# Profile set and classifiers

The profile set `PE - Ley 29733 - v1` runs these 144 classifiers, with an assignment threshold of 60%.

Identificadores directos, cuasi-identificadores, datos sensibles (art. 2, 5), datos de salud, datos financieros y antecedentes penales, conforme a la Ley 29733 de Protección de Datos Personales del Perú y su reglamento (D.S. 016-2024-JUS).

| CLASSIFIER                         | KIND         | DOMAIN                      |
|------------------------------------|--------------|-----------------------------|
| PE_L1_DNI - Path                   | COLUMN NAME  | PE_L1_DNI                   |
| PE_L1_DNI - Type                   | DATA TYPE    | PE_L1_DNI                   |
| PE_L1_DNI - Regex                  | VALUE FORMAT | PE_L1_DNI                   |
| PE_L1_RUC - Path                   | COLUMN NAME  | PE_L1_RUC                   |
| PE_L1_RUC - Regex                  | VALUE FORMAT | PE_L1_RUC                   |
| PE_L1_DIGITO_VERIFICACION - Path   | COLUMN NAME  | PE_L1_DIGITO_VERIFICACION   |
| PE_L1_DIGITO_VERIFICACION - Type   | DATA TYPE    | PE_L1_DIGITO_VERIFICACION   |
| PE_L1_CARNE_EXTRANJERIA - Path     | COLUMN NAME  | PE_L1_CARNE_EXTRANJERIA     |
| PE_L1_PASAPORTE - Path             | COLUMN NAME  | PE_L1_PASAPORTE             |
| PE_L1_PASAPORTE - Regex            | VALUE FORMAT | PE_L1_PASAPORTE             |
| PE_L1_ACTA - Path                  | COLUMN NAME  | PE_L1_ACTA                  |
| PE_L1_SEGURO_SOCIAL - Path         | COLUMN NAME  | PE_L1_SEGURO_SOCIAL         |
| PE_L1_CODIGO_ESTUDIANTE - Path     | COLUMN NAME  | PE_L1_CODIGO_ESTUDIANTE     |
| PE_L1_COLEGIATURA - Path           | COLUMN NAME  | PE_L1_COLEGIATURA           |
| PE_L1_NOMBRE - Path                | COLUMN NAME  | PE_L1_NOMBRE                |
| PE_L1_NOMBRE - List                | KNOWN VALUES | PE_L1_NOMBRE                |
| PE_L1_APELLIDO - Path              | COLUMN NAME  | PE_L1_APELLIDO              |
| PE_L1_APELLIDO - List              | KNOWN VALUES | PE_L1_APELLIDO              |
| PE_L1_NOMBRE_COMPLETO - Path       | COLUMN NAME  | PE_L1_NOMBRE_COMPLETO       |
| PE_L1_NOMBRE_COMPLETO - List       | KNOWN VALUES | PE_L1_NOMBRE_COMPLETO       |
| PE_L1_RAZON_SOCIAL - Path          | COLUMN NAME  | PE_L1_RAZON_SOCIAL          |
| PE_L1_EMAIL - Path                 | COLUMN NAME  | PE_L1_EMAIL                 |
| PE_L1_EMAIL - Type                 | DATA TYPE    | PE_L1_EMAIL                 |
| PE_L1_EMAIL - Regex                | VALUE FORMAT | PE_L1_EMAIL                 |
| PE_L1_TELEFONO - Path              | COLUMN NAME  | PE_L1_TELEFONO              |
| PE_L1_TELEFONO - Regex             | VALUE FORMAT | PE_L1_TELEFONO              |
| PE_L1_DIRECCION - Path             | COLUMN NAME  | PE_L1_DIRECCION             |
| PE_L1_DIRECCION - Type             | DATA TYPE    | PE_L1_DIRECCION             |
| PE_L1_DIRECCION - Regex            | VALUE FORMAT | PE_L1_DIRECCION             |
| PE_L1_DIRECCION_COMPLEMENTO - Path | COLUMN NAME  | PE_L1_DIRECCION_COMPLEMENTO |

| CLASSIFIER                    | KIND         | DOMAIN                 |
|-------------------------------|--------------|------------------------|
| PE_L1_CUENTA_BANCARIA – Path  | COLUMN NAME  | PE_L1_CUENTA_BANCARIA  |
| PE_L1_TARJETA – Path          | COLUMN NAME  | PE_L1_TARJETA          |
| PE_L1_TARJETA – Regex         | VALUE FORMAT | PE_L1_TARJETA          |
| PE_L1_BILLETERA – Path        | COLUMN NAME  | PE_L1_BILLETERA        |
| PE_L1_BILLETERA – Regex       | VALUE FORMAT | PE_L1_BILLETERA        |
| PE_L1_IP – Path               | COLUMN NAME  | PE_L1_IP               |
| PE_L1_IP – Regex              | VALUE FORMAT | PE_L1_IP               |
| PE_L1_DISPOSITIVO – Path      | COLUMN NAME  | PE_L1_DISPOSITIVO      |
| PE_L1_DISPOSITIVO – Regex     | VALUE FORMAT | PE_L1_DISPOSITIVO      |
| PE_L1_COOKIE – Path           | COLUMN NAME  | PE_L1_COOKIE           |
| PE_L1_PLACA – Path            | COLUMN NAME  | PE_L1_PLACA            |
| PE_L1_PLACA – Regex           | VALUE FORMAT | PE_L1_PLACA            |
| PE_L1_VEHICULO – Path         | COLUMN NAME  | PE_L1_VEHICULO         |
| PE_L1_VEHICULO – Regex        | VALUE FORMAT | PE_L1_VEHICULO         |
| PE_L1_INMUEBLE – Path         | COLUMN NAME  | PE_L1_INMUEBLE         |
| PE_L1_CONTRATO – Path         | COLUMN NAME  | PE_L1_CONTRATO         |
| PE_L1_USUARIO – Path          | COLUMN NAME  | PE_L1_USUARIO          |
| PE_L1_CREDENCIAL – Path       | COLUMN NAME  | PE_L1_CREDENCIAL       |
| PE_L1_GEOLOCALIZACION – Path  | COLUMN NAME  | PE_L1_GEOLOCALIZACION  |
| PE_L1_GEOLOCALIZACION – Regex | VALUE FORMAT | PE_L1_GEOLOCALIZACION  |
| PE_L1_TEXTO_LIBRE – Path      | COLUMN NAME  | PE_L1_TEXTO_LIBRE      |
| PE_L1_TEXTO_LIBRE – Type      | DATA TYPE    | PE_L1_TEXTO_LIBRE      |
| PE_L1_TEXTO_LIBRE – Regex     | VALUE FORMAT | PE_L1_TEXTO_LIBRE      |
| PE_L2_FECHA_NACIMIENTO – Path | COLUMN NAME  | PE_L2_FECHA_NACIMIENTO |
| PE_L2_FECHA_NACIMIENTO – Type | DATA TYPE    | PE_L2_FECHA_NACIMIENTO |
| PE_L2_ANIO_NACIMIENTO – Path  | COLUMN NAME  | PE_L2_ANIO_NACIMIENTO  |
| PE_L2_ANIO_NACIMIENTO – Type  | DATA TYPE    | PE_L2_ANIO_NACIMIENTO  |
| PE_L2_EDAD – Path             | COLUMN NAME  | PE_L2_EDAD             |
| PE_L2_EDAD – Type             | DATA TYPE    | PE_L2_EDAD             |
| PE_L2_FECHA_EVENTO – Path     | COLUMN NAME  | PE_L2_FECHA_EVENTO     |
| PE_L2_FECHA_EVENTO – Type     | DATA TYPE    | PE_L2_FECHA_EVENTO     |
| PE_L2_SEX0 – Path             | COLUMN NAME  | PE_L2_SEX0             |
| PE_L2_SEX0 – List             | KNOWN VALUES | PE_L2_SEX0             |
| PE_L2_DISTRITO – Path         | COLUMN NAME  | PE_L2_DISTRITO         |
| PE_L2_DISTRITO – List         | KNOWN VALUES | PE_L2_DISTRITO         |
| PE_L2_UBIGE0 – Path           | COLUMN NAME  | PE_L2_UBIGE0           |
| PE_L2_UBIGE0 – Regex          | VALUE FORMAT | PE_L2_UBIGE0           |

| CLASSIFIER                         | KIND         | DOMAIN                      |
|------------------------------------|--------------|-----------------------------|
| PE_L2_CENTRO_POBLADO – Path        | COLUMN NAME  | PE_L2_CENTRO_POBLADO        |
| PE_L2_URBANIZACION – Path          | COLUMN NAME  | PE_L2_URBANIZACION          |
| PE_L2_CODIGO_POSTAL – Path         | COLUMN NAME  | PE_L2_CODIGO_POSTAL         |
| PE_L2_CODIGO_POSTAL – Type         | DATA TYPE    | PE_L2_CODIGO_POSTAL         |
| PE_L2_ESTADO_CIVIL – Path          | COLUMN NAME  | PE_L2_ESTADO_CIVIL          |
| PE_L2_ESTADO_CIVIL – List          | KNOWN VALUES | PE_L2_ESTADO_CIVIL          |
| PE_L2_OCUPACION – Path             | COLUMN NAME  | PE_L2_OCUPACION             |
| PE_L2_EMPLEADOR – Path             | COLUMN NAME  | PE_L2_EMPLEADOR             |
| PE_L2_NIVEL_EDUCATIVO – Path       | COLUMN NAME  | PE_L2_NIVEL_EDUCATIVO       |
| PE_L2_NIVEL_EDUCATIVO – List       | KNOWN VALUES | PE_L2_NIVEL_EDUCATIVO       |
| PE_L2_INSTITUCION_EDUCATIVA – Path | COLUMN NAME  | PE_L2_INSTITUCION_EDUCATIVA |
| PE_L2_PERSONAS_A_CARGO – Path      | COLUMN NAME  | PE_L2_PERSONAS_A_CARGO      |
| PE_L2_PERSONAS_A_CARGO – Type      | DATA TYPE    | PE_L2_PERSONAS_A_CARGO      |
| PE_L3_ORIGEN_ETNICO – Path         | COLUMN NAME  | PE_L3_ORIGEN_ETNICO         |
| PE_L3_ORIGEN_ETNICO – List         | KNOWN VALUES | PE_L3_ORIGEN_ETNICO         |
| PE_L3_PUEBLO_INDIGENA – Path       | COLUMN NAME  | PE_L3_PUEBLO_INDIGENA       |
| PE_L3_PUEBLO_INDIGENA – List       | KNOWN VALUES | PE_L3_PUEBLO_INDIGENA       |
| PE_L3 LENGUA – Path                | COLUMN NAME  | PE_L3 LENGUA                |
| PE_L3 LENGUA – List                | KNOWN VALUES | PE_L3 LENGUA                |
| PE_L3_NACIONALIDAD – Path          | COLUMN NAME  | PE_L3_NACIONALIDAD          |
| PE_L3_NACIONALIDAD – List          | KNOWN VALUES | PE_L3_NACIONALIDAD          |
| PE_L3_RELIGION – Path              | COLUMN NAME  | PE_L3_RELIGION              |
| PE_L3_RELIGION – List              | KNOWN VALUES | PE_L3_RELIGION              |
| PE_L3_CONVICCION_FILOSOFICA – Path | COLUMN NAME  | PE_L3_CONVICCION_FILOSOFICA |
| PE_L3_OPINION_POLITICA – Path      | COLUMN NAME  | PE_L3_OPINION_POLITICA      |
| PE_L3_OPINION_POLITICA – List      | KNOWN VALUES | PE_L3_OPINION_POLITICA      |
| PE_L3_AFILIACION_PARTIDARIA – Path | COLUMN NAME  | PE_L3_AFILIACION_PARTIDARIA |
| PE_L3_AFILIACION_PARTIDARIA – List | KNOWN VALUES | PE_L3_AFILIACION_PARTIDARIA |
| PE_L3_AFILIACION_SINDICAL – Path   | COLUMN NAME  | PE_L3_AFILIACION_SINDICAL   |
| PE_L3_AFILIACION_SINDICAL – List   | KNOWN VALUES | PE_L3_AFILIACION_SINDICAL   |
| PE_L3_BIOMETRICO – Path            | COLUMN NAME  | PE_L3_BIOMETRICO            |
| PE_L3_BIOMETRICO – Type            | DATA TYPE    | PE_L3_BIOMETRICO            |
| PE_L3_GENETICO – Path              | COLUMN NAME  | PE_L3_GENETICO              |
| PE_L3_ORIENTACION_SEXUAL – Path    | COLUMN NAME  | PE_L3_ORIENTACION_SEXUAL    |
| PE_L3_ORIENTACION_SEXUAL – List    | KNOWN VALUES | PE_L3_ORIENTACION_SEXUAL    |
| PE_L3_VIDA_SEXUAL – Path           | COLUMN NAME  | PE_L3_VIDA_SEXUAL           |
| PE_L3_IDENTIDAD_GENERO – Path      | COLUMN NAME  | PE_L3_IDENTIDAD_GENERO      |

| CLASSIFIER                                 | KIND         | DOMAIN                              |
|--------------------------------------------|--------------|-------------------------------------|
| PE_L3_IDENTIDAD_GENERO - List              | KNOWN VALUES | PE_L3_IDENTIDAD_GENERO              |
| PE_L3_VICTIMA - Path                       | COLUMN NAME  | PE_L3_VICTIMA                       |
| PE_SALUD_IDENTIFICADOR - Path              | COLUMN NAME  | PE_SALUD_IDENTIFICADOR              |
| PE_SALUD_COBERTURA - Path                  | COLUMN NAME  | PE_SALUD_COBERTURA                  |
| PE_SALUD_COBERTURA - List                  | KNOWN VALUES | PE_SALUD_COBERTURA                  |
| PE_SALUD_DIAGNOSTICO - Path                | COLUMN NAME  | PE_SALUD_DIAGNOSTICO                |
| PE_SALUD_DIAGNOSTICO - Regex               | VALUE FORMAT | PE_SALUD_DIAGNOSTICO                |
| PE_SALUD_PROCEDIMIENTO - Path              | COLUMN NAME  | PE_SALUD_PROCEDIMIENTO              |
| PE_SALUD_MEDICAMENTO - Path                | COLUMN NAME  | PE_SALUD_MEDICAMENTO                |
| PE_SALUD_MEDICAMENTO - List                | KNOWN VALUES | PE_SALUD_MEDICAMENTO                |
| PE_SALUD_TEXTO_CLINICO - Path              | COLUMN NAME  | PE_SALUD_TEXTO_CLINICO              |
| PE_SALUD_TEXTO_CLINICO - Type              | DATA TYPE    | PE_SALUD_TEXTO_CLINICO              |
| PE_SALUD_RESULTADO_EXAMEN - Path           | COLUMN NAME  | PE_SALUD_RESULTADO_EXAMEN           |
| PE_SALUD_VIH - Path                        | COLUMN NAME  | PE_SALUD_VIH                        |
| PE_SALUD_GRUPO_SANGUINEO - Path            | COLUMN NAME  | PE_SALUD_GRUPO_SANGUINEO            |
| PE_SALUD_GRUPO_SANGUINEO - List            | KNOWN VALUES | PE_SALUD_GRUPO_SANGUINEO            |
| PE_SALUD_DISCAPACIDAD - Path               | COLUMN NAME  | PE_SALUD_DISCAPACIDAD               |
| PE_SALUD_DISCAPACIDAD - List               | KNOWN VALUES | PE_SALUD_DISCAPACIDAD               |
| PE_SALUD_OCUPACIONAL - Path                | COLUMN NAME  | PE_SALUD_OCUPACIONAL                |
| PE_SALUD_OCUPACIONAL - List                | KNOWN VALUES | PE_SALUD_OCUPACIONAL                |
| PE_SALUD_REPRODUCTIVA - Path               | COLUMN NAME  | PE_SALUD_REPRODUCTIVA               |
| PE_SALUD_MENTAL - Path                     | COLUMN NAME  | PE_SALUD_MENTAL                     |
| PE_FIN_INGRESOS - Path                     | COLUMN NAME  | PE_FIN_INGRESOS                     |
| PE_FIN_INGRESOS - Type                     | DATA TYPE    | PE_FIN_INGRESOS                     |
| PE_FIN_HISTORIAL_CREDITICIO - Path         | COLUMN NAME  | PE_FIN_HISTORIAL_CREDITICIO         |
| PE_FIN_DEUDA - Path                        | COLUMN NAME  | PE_FIN_DEUDA                        |
| PE_FIN_DEUDA - Type                        | DATA TYPE    | PE_FIN_DEUDA                        |
| PE_FIN_PATRIMONIO - Path                   | COLUMN NAME  | PE_FIN_PATRIMONIO                   |
| PE_FIN_PATRIMONIO - Type                   | DATA TYPE    | PE_FIN_PATRIMONIO                   |
| PE_FIN_PENSION - Path                      | COLUMN NAME  | PE_FIN_PENSION                      |
| PE_FIN_PENSION - Type                      | DATA TYPE    | PE_FIN_PENSION                      |
| PE_FIN_PROGRAMA_SOCIAL - Path              | COLUMN NAME  | PE_FIN_PROGRAMA_SOCIAL              |
| PE_FIN_PROGRAMA_SOCIAL - List              | KNOWN VALUES | PE_FIN_PROGRAMA_SOCIAL              |
| PE_FIN_CLASIFICACION_SOCIOECONOMICA - Path | COLUMN NAME  | PE_FIN_CLASIFICACION_SOCIOECONOMICA |
| PE_FIN_CLASIFICACION_SOCIOECONOMICA - List | KNOWN VALUES | PE_FIN_CLASIFICACION_SOCIOECONOMICA |
| PE_FIN_VIVIENDA - Path                     | COLUMN NAME  | PE_FIN_VIVIENDA                     |
| PE_FIN_VIVIENDA - List                     | KNOWN VALUES | PE_FIN_VIVIENDA                     |

| CLASSIFIER                   | KIND         | DOMAIN                |
|------------------------------|--------------|-----------------------|
| PE_PENAL_ANTECEDENTES – Path | COLUMN NAME  | PE_PENAL_ANTECEDENTES |
| PE_PENAL_EXPEDIENTE – Path   | COLUMN NAME  | PE_PENAL_EXPEDIENTE   |
| PE_PENAL_EXPEDIENTE – Regex  | VALUE FORMAT | PE_PENAL_EXPEDIENTE   |